#1 · Allergy and Immunology

A patient develops hives, lip swelling, wheezing, and hypotension minutes after a bee sting. What is the first-line treatment and why?

Answer

Intramuscular epinephrine in the anterolateral thigh is the first-line treatment for anaphylaxis and should be given immediately, not delayed for antihistamines or steroids. It reverses vasodilation, bronchospasm, and mucosal edema through alpha and beta adrenergic effects. Antihistamines and glucocorticoids are adjuncts only and do not treat the airway or hemodynamic collapse. Patients should be observed for several hours afterward because biphasic reactions can occur.

#2 · Allergy and Immunology

How do you distinguish allergic asthma phenotype and what test confirms an allergic trigger?

Answer

Allergic asthma typically starts in childhood, is associated with eczema and allergic rhinitis, and worsens with exposure to specific triggers like pollen, dust mites, or pet dander. Skin prick testing or serum specific IgE testing confirms sensitization to suspected allergens. Elevated total IgE and blood eosinophilia often support the diagnosis. Identifying the trigger allows for allergen avoidance and consideration of allergen immunotherapy or biologic therapy such as omalizumab.

#3 · Allergy and Immunology

A young woman has year-round nasal congestion, clear rhinorrhea, itchy watery eyes, and a nasal crease from chronic rubbing. What is the diagnosis and first-line therapy?

Answer

This presentation is classic for allergic rhinoconjunctivitis, often perennial due to indoor allergens like dust mites or pet dander. Intranasal corticosteroids are first-line therapy and are more effective than oral antihistamines for nasal symptoms. Ocular symptoms can be treated with topical antihistamine or mast cell stabilizer eye drops. Allergen avoidance and immunotherapy are options for refractory or severe cases.

#4 · Allergy and Immunology

A child has an immediate reaction of hives and vomiting after eating peanuts. What testing confirms IgE-mediated food allergy, and what is the emergency management plan?

Answer

Skin prick testing or serum specific IgE to peanut supports the diagnosis, though a positive test alone does not confirm clinical allergy without a consistent history. Oral food challenge remains the gold standard when the diagnosis is uncertain. Patients should carry an epinephrine autoinjector and receive an anaphylaxis action plan, since accidental exposure can cause life-threatening reactions. Strict avoidance of the trigger food is the mainstay of chronic management.

#5 · Allergy and Immunology

A patient has recurrent episodes of angioedema without urticaria, and antihistamines and steroids do not help. What diagnosis should be considered and how is it confirmed?

Answer

Hereditary angioedema should be suspected when angioedema occurs without hives and fails to respond to antihistamines, steroids, or epinephrine. It is caused by C1 esterase inhibitor deficiency or dysfunction, leading to excess bradykinin. Diagnosis is confirmed with low C4 level plus low C1 inhibitor level or function. Acute attacks are treated with C1 inhibitor concentrate, icatibant, or ecallantide rather than standard allergy medications.

#6 · Allergy and Immunology

A patient developed a diffuse maculopapular rash on day 8 of amoxicillin without airway or hemodynamic compromise. Is this a true IgE-mediated allergy, and what should be done?

Answer

This delayed morbilliform rash pattern is more consistent with a non-IgE mediated, T-cell mediated hypersensitivity reaction rather than immediate anaphylaxis. It does not necessarily preclude future penicillin use the way true IgE-mediated urticaria or anaphylaxis would. Management involves stopping the drug and supportive care, and referral for allergy evaluation or graded challenge can clarify future antibiotic options. Warning signs of a more severe drug reaction include mucosal involvement, blistering, fever, and organ involvement, which suggest SJS/TEN or DRESS.

#7 · Allergy and Immunology

A young man has recurrent sinopulmonary infections, low immunoglobulin levels, and poor response to vaccines. What primary immunodeficiency should be suspected and how is it confirmed?

Answer

Common variable immunodeficiency should be suspected in adults with recurrent bacterial sinopulmonary infections and low IgG, often with low IgA or IgM as well. Diagnosis requires demonstrating impaired specific antibody response to vaccines, such as pneumococcal polysaccharide vaccine, after excluding other causes of hypogammaglobulinemia. Treatment is immunoglobulin replacement therapy, given intravenously or subcutaneously, to reduce infection risk. Patients also have increased risk of autoimmune disease and lymphoma, so long-term monitoring is needed.

#8 · Allergy and Immunology

A middle-aged man presents with painless swelling of the salivary and lacrimal glands, elevated serum IgG4, and pancreatic enlargement on imaging. What is the diagnosis and how is it confirmed?

Answer

This presentation is consistent with IgG4-related disease, a fibroinflammatory condition that can affect the pancreas, biliary tree, salivary glands, orbits, and other organs. Diagnosis is supported by elevated serum IgG4 and tissue biopsy showing dense lymphoplasmacytic infiltrate with IgG4-positive plasma cells, storiform fibrosis, and obliterative phlebitis. It can mimic malignancy such as pancreatic cancer, so biopsy is important before assuming a benign diagnosis. Glucocorticoids are first-line treatment and usually produce a rapid response.

#9 · Allergy and Immunology

A patient has recurrent flushing, abdominal pain, and diarrhea, and develops anaphylaxis-like symptoms with minor triggers like heat or exercise. What condition should be considered and what test helps confirm it?

Answer

This picture suggests mastocytosis, a disorder of mast cell proliferation and mediator release triggered by heat, exercise, alcohol, or certain medications. An elevated serum tryptase level, especially during or after a flare, supports the diagnosis, and a persistently elevated baseline tryptase raises suspicion for systemic disease. Bone marrow biopsy showing abnormal mast cell infiltrates confirms systemic mastocytosis. Patients should avoid known triggers and carry epinephrine given their risk of severe anaphylactic reactions.

#10 · Allergy and Immunology

A young man with a history of asthma and allergic rhinitis develops dysphagia and food impaction, with esophageal biopsy showing more than 15 eosinophils per high-power field. What is the diagnosis and treatment?

Answer

This is eosinophilic esophagitis, an antigen-driven eosinophilic inflammatory disease of the esophagus, often seen in patients with atopic conditions. Treatment includes proton pump inhibitors, topical swallowed corticosteroids, and dietary elimination of trigger foods such as dairy, wheat, egg, and soy. Esophageal dilation may be needed for strictures that cause food impaction. This is distinct from eosinophilic granulomatosis with polyangiitis, which presents with asthma, sinusitis, eosinophilia, and vasculitis affecting nerves, skin, and other organs, and from hypersensitivity pneumonitis, which follows inhaled organic antigen exposure and causes interstitial lung disease.

#11 · Cardiovascular Disease

First-line drug classes for essential hypertension

Answer

Thiazide diuretics, ACE inhibitors, ARBs, and long-acting calcium channel blockers are all acceptable first-line choices. In Black patients without heart failure or CKD, a thiazide or calcium channel blocker usually works better than an ACE inhibitor or ARB alone. Comorbidities steer the choice, for example an ACE inhibitor or ARB is preferred with diabetic proteinuria or heart failure with reduced ejection fraction.

#12 · Cardiovascular Disease

Young patient with resistant hypertension, hypokalemia, and metabolic alkalosis

Answer

This picture points to primary hyperaldosteronism. Screen with a plasma aldosterone to renin ratio drawn off spironolactone, and confirm with a saline suppression or captopril challenge test. If confirmed, CT adrenal imaging and adrenal vein sampling in patients over 35 help distinguish a unilateral adenoma treated with surgery from bilateral hyperplasia treated with a mineralocorticoid receptor antagonist.

#13 · Cardiovascular Disease

Renal artery stenosis in a young woman versus an older smoker

Answer

Fibromuscular dysplasia causes renal artery stenosis in young women, typically affecting the mid to distal renal artery with a string of beads appearance on angiography. Atherosclerotic renal artery stenosis affects older patients with vascular risk factors and involves the proximal artery and ostium. Both can present with resistant hypertension, but fibromuscular dysplasia responds well to angioplasty while atherosclerotic disease is usually managed medically first.

#14 · Cardiovascular Disease

Clue that hypertension is renovascular in origin

Answer

Suspect renovascular hypertension when there is an abrupt rise in creatinine after starting an ACE inhibitor or ARB, an abdominal bruit, recurrent flash pulmonary edema, or onset of severe hypertension before age 30 or after age 55. A significant creatinine rise after starting an ACE inhibitor suggests bilateral renal artery stenosis or stenosis in a solitary kidney. Duplex ultrasound or CT/MR angiography confirms the diagnosis.

#15 · Cardiovascular Disease

Hypertensive urgency versus hypertensive emergency

Answer

Hypertensive urgency is severely elevated blood pressure, generally above 180/120, without acute end-organ damage, and it is managed with oral medication and close outpatient follow-up rather than rapid lowering. Hypertensive emergency is severe elevation with evidence of acute end-organ injury such as encephalopathy, papilledema, acute kidney injury, pulmonary edema, or aortic dissection. Emergencies require IV antihypertensives with a goal of lowering mean arterial pressure by about 25 percent in the first hour, except in aortic dissection or ischemic stroke where targets differ.

#16 · Cardiovascular Disease

Preferred IV agent for hypertensive emergency with aortic dissection

Answer

Esmolol or labetalol is used first to control heart rate and blood pressure together, since lowering pressure alone without controlling shear stress can worsen the dissection. A vasodilator like nitroprusside may be added only after adequate beta blockade. The systolic blood pressure goal is under 120 and heart rate under 60, achieved within minutes.

#17 · Cardiovascular Disease

Coarctation of the aorta: classic exam findings

Answer

Look for upper extremity hypertension with diminished and delayed femoral pulses compared to the radial pulse, a difference sometimes called radial-femoral delay. A systolic murmur is often heard over the back between the scapulae. Chest x-ray classically shows rib notching from collateral intercostal vessels and a figure-3 sign of the aortic contour.

#18 · Cardiovascular Disease

Coarctation of the aorta and associated conditions

Answer

Coarctation is strongly associated with a bicuspid aortic valve, found in the majority of patients, and with Turner syndrome. Berry aneurysms of the circle of Willis also occur more often in these patients. Any young patient with new hypertension should have blood pressure checked in both arms and a leg to screen for coarctation.

#19 · Cardiovascular Disease

Acute pericarditis: classic ECG and exam findings

Answer

ECG shows diffuse concave upward ST elevation and PR depression across most leads, with PR elevation and ST depression reciprocally in aVR. Chest pain is sharp, pleuritic, and improves with sitting forward while worsening when lying supine. A pericardial friction rub, best heard with the patient leaning forward at end expiration, is a specific finding.

#20 · Cardiovascular Disease

First-line treatment for acute idiopathic or viral pericarditis

Answer

NSAIDs or aspirin plus colchicine are first-line, and colchicine is continued for about three months to reduce the risk of recurrence. Corticosteroids are avoided as first-line therapy because they increase the rate of recurrent pericarditis. Exercise restriction is recommended until symptoms resolve and inflammatory markers normalize.

#21 · Cardiovascular Disease

Cardiac tamponade: key physical findings

Answer

Beck's triad is hypotension, distended neck veins, and muffled heart sounds. Pulsus paradoxus, an exaggerated drop in systolic blood pressure of more than 10 mmHg during inspiration, is a sensitive finding. Echocardiography shows diastolic collapse of the right ventricle and right atrium along with a large pericardial effusion and a plethoric inferior vena cava that does not collapse with inspiration.

#22 · Cardiovascular Disease

Constrictive pericarditis versus restrictive cardiomyopathy

Answer

Both cause impaired diastolic filling with elevated jugular venous pressure, but constrictive pericarditis shows a thickened or calcified pericardium and prominent respiratory ventricular interdependence, meaning septal shift with inspiration. Restrictive cardiomyopathy usually shows biatrial enlargement and abnormal myocardial tissue on imaging without pericardial thickening. Cardiac catheterization showing discordance between right and left ventricular pressures during respiration favors constriction, while concordance favors restriction.

#23 · Cardiovascular Disease

Kussmaul sign

Answer

Kussmaul sign is a paradoxical rise, or failure to fall, in jugular venous pressure during inspiration. It occurs when the right heart cannot accommodate the increased venous return that normally happens with inspiration. It is classically seen in constrictive pericarditis, restrictive cardiomyopathy, and right ventricular infarction, and it is notably absent in typical cardiac tamponade.

#24 · Cardiovascular Disease

STEMI versus NSTEMI versus unstable angina

Answer

STEMI shows ST elevation on ECG with elevated cardiac troponin and requires emergent reperfusion, ideally primary PCI within 90 minutes of first medical contact. NSTEMI has elevated troponin without ST elevation, though ST depression or T wave inversion may be present, and is managed with an early invasive or ischemia-guided strategy depending on risk. Unstable angina has ischemic symptoms without troponin elevation, reflecting supply-demand mismatch without frank myocardial necrosis.

#25 · Cardiovascular Disease

Initial medical therapy for ACS

Answer

Give aspirin, a P2Y12 inhibitor such as clopidogrel or ticagrelor, anticoagulation with heparin, a high-intensity statin, and a beta blocker if there is no contraindication. Nitroglycerin and morphine help with symptom control but do not change mortality outcomes. Beta blockers should be held if there is evidence of cardiogenic shock or severe decompensated heart failure.

#26 · Cardiovascular Disease

ECG leads for inferior wall MI and the associated coronary artery

Answer

Inferior MI shows ST elevation in leads II, III, and aVF, and is most often due to occlusion of the right coronary artery. Right ventricular involvement should be checked with right-sided leads, especially V4R, since these patients are preload dependent and nitrates can cause profound hypotension. Inferior MI also carries higher risk of bradyarrhythmias and heart block due to AV nodal ischemia.

#27 · Cardiovascular Disease

Wellens syndrome

Answer

Wellens syndrome is a pattern of deeply inverted or biphasic T waves in leads V2 and V3 in a patient who is pain free at the time of the ECG. It indicates critical stenosis of the proximal left anterior descending artery and a high risk of anterior wall MI if untreated. These patients need urgent coronary angiography and should not undergo stress testing, which can precipitate infarction.

#28 · Cardiovascular Disease

Mechanical complications after MI and their timing

Answer

Free wall rupture typically occurs within the first week and presents with sudden hypotension and tamponade, often fatal. Papillary muscle rupture, usually of the posteromedial muscle after inferior MI, causes acute severe mitral regurgitation with a new murmur and pulmonary edema. Ventricular septal rupture causes a new holosystolic murmur with a step-up in oxygen saturation from right atrium to right ventricle, and both septal and papillary muscle rupture usually occur three to five days post-MI.

#29 · Cardiovascular Disease

Approach to stable chest pain with intermediate pretest probability of CAD

Answer

Coronary CT angiography or a stress test with imaging are both reasonable first steps in patients with intermediate pretest probability. Exercise ECG alone is preferred in patients who can exercise and have an interpretable baseline ECG without left bundle branch block, paced rhythm, or baseline ST changes. Pharmacologic stress testing with imaging is used when patients cannot exercise adequately or have an uninterpretable ECG.

#30 · Cardiovascular Disease

Atrial fibrillation: rate versus rhythm control

Answer

Rate control with a beta blocker or non-dihydropyridine calcium channel blocker is reasonable first-line therapy for most patients, especially older patients with minimal symptoms. Rhythm control with antiarrhythmics or catheter ablation is favored in younger patients, those with persistent symptoms despite rate control, or heart failure with reduced ejection fraction where AF may be worsening cardiac function. Regardless of strategy, stroke risk must be assessed independently using the CHA2DS2-VASc score to decide on anticoagulation.

#31 · Cardiovascular Disease

CHA2DS2-VASc score components

Answer

The score awards points for congestive heart failure, hypertension, age 75 or older (2 points), diabetes, prior stroke or TIA (2 points), vascular disease, age 65 to 74, and female sex. A score of 2 or more in men or 3 or more in women generally warrants anticoagulation for atrial fibrillation. A score of 0 in men or 1 in women (from sex alone) usually means anticoagulation can be safely withheld.

#32 · Cardiovascular Disease

Mobitz type I versus Mobitz type II second-degree AV block

Answer

Mobitz I, or Wenckebach, shows progressive PR interval lengthening until a QRS is dropped, and it is usually benign, located at the AV node, and often improves with atropine. Mobitz II shows a constant PR interval with sudden dropped beats, reflects infranodal disease below the AV node, and carries a high risk of progression to complete heart block. Mobitz II generally requires a pacemaker even if asymptomatic.

#33 · Cardiovascular Disease

Torsades de pointes: cause and treatment

Answer

Torsades de pointes is a polymorphic ventricular tachycardia associated with a prolonged QT interval, from causes like hypokalemia, hypomagnesemia, congenital long QT syndrome, or QT-prolonging drugs. First-line treatment is IV magnesium sulfate regardless of the magnesium level. Unstable patients need immediate defibrillation, and any offending drug should be stopped.

#34 · Cardiovascular Disease

Wolff-Parkinson-White syndrome: ECG and treatment pitfall

Answer

WPW shows a short PR interval and a delta wave, a slurred upstroke of the QRS, from an accessory pathway that bypasses the AV node. During atrial fibrillation with WPW, AV nodal blocking agents like beta blockers, calcium channel blockers, digoxin, and adenosine are contraindicated because they can promote conduction down the accessory pathway and precipitate ventricular fibrillation. Procainamide or ibutilide are safer choices, and catheter ablation of the accessory pathway is the definitive treatment.

#35 · Cardiovascular Disease

Indications for permanent pacemaker

Answer

Symptomatic sinus node dysfunction, symptomatic or high-grade Mobitz II second-degree block, and third-degree (complete) heart block are standard indications. Asymptomatic Mobitz I block generally does not require a pacemaker. Post-MI complete heart block, alternating bundle branch block, and symptomatic bradycardia from necessary drug therapy are other common indications.

#36 · Cardiovascular Disease

Most common adult congenital heart defect and its physiology

Answer

Bicuspid aortic valve is the most common congenital cardiac abnormality, though atrial septal defect is the most common congenital lesion that presents in adulthood. ASD causes a left-to-right shunt with right heart volume overload, a fixed split S2, and a systolic ejection murmur at the left upper sternal border from increased flow across the pulmonic valve. Over decades, uncorrected large shunts can reverse to right-to-left flow, called Eisenmenger syndrome, causing cyanosis.

#37 · Cardiovascular Disease

Eisenmenger syndrome

Answer

Eisenmenger syndrome develops when a long-standing left-to-right shunt, such as from a VSD, ASD, or PDA, causes progressive pulmonary vascular resistance to rise until the shunt reverses to right-to-left. This produces cyanosis, clubbing, and erythrocytosis. Once Eisenmenger physiology is established, surgical closure of the defect is contraindicated because it removes the pop-off valve the failing right heart depends on.

#38 · Cardiovascular Disease

Aortic stenosis: classic triad of symptoms

Answer

Severe symptomatic aortic stenosis presents with angina, syncope, and heart failure symptoms such as exertional dyspnea. Onset of symptoms marks a turning point where survival drops sharply without valve replacement, roughly 2 years average survival once heart failure symptoms develop, the worst prognosis, versus about 3 years with syncope and 5 years with angina if untreated. The murmur is a harsh crescendo-decrescendo systolic murmur at the right upper sternal border radiating to the carotids, with a delayed and diminished carotid upstroke called pulsus parvus et tardus.

#39 · Cardiovascular Disease

Mitral regurgitation murmur and key physiology

Answer

Mitral regurgitation produces a holosystolic murmur best heard at the apex radiating to the axilla. Acute severe MR, such as from papillary muscle rupture, causes sudden pulmonary edema and may have a surprisingly soft murmur because pressures equalize quickly between ventricle and atrium. Chronic MR allows the left atrium and ventricle to dilate and compensate, so symptoms may be delayed for years despite significant regurgitant volume.

#40 · Cardiovascular Disease

Mitral stenosis: cause and auscultation

Answer

Rheumatic heart disease is the most common cause of mitral stenosis worldwide. The murmur is a low-pitched diastolic rumble heard best at the apex in the left lateral decubitus position, often preceded by an opening snap; a shorter interval between S2 and the opening snap indicates more severe stenosis. Mitral stenosis predisposes to atrial fibrillation and left atrial thrombus formation because of left atrial enlargement and stasis.

#41 · Cardiovascular Disease

Aortic regurgitation: murmur and peripheral signs

Answer

Chronic AR produces a high-pitched decrescendo diastolic murmur at the left sternal border, best heard with the patient sitting forward in held expiration. Chronic severe AR causes a wide pulse pressure and bounding pulses, giving rise to signs like Corrigan pulse (water-hammer pulse) and head bobbing (de Musset sign). Acute AR, such as from aortic dissection or endocarditis, is a surgical emergency because the noncompliant left ventricle cannot tolerate the sudden volume overload.

#42 · Cardiovascular Disease

Hypertrophic cardiomyopathy: murmur behavior with maneuvers

Answer

HCM causes a systolic murmur from left ventricular outflow tract obstruction that increases in intensity with maneuvers that decrease preload, such as standing up or Valsalva strain, and decreases with maneuvers that increase preload or afterload, such as squatting or handgrip. This is the opposite behavior of most other systolic murmurs, which get louder with increased venous return. HCM is a leading cause of sudden cardiac death in young athletes and is usually caused by sarcomere gene mutations.

#43 · Cardiovascular Disease

HFpEF versus HFrEF: definition and typical patient

Answer

Heart failure with preserved ejection fraction (HFpEF) has an ejection fraction of 50 percent or higher with diastolic dysfunction, and typically affects older patients with hypertension, obesity, diabetes, and atrial fibrillation. HFrEF has an ejection fraction of 40 percent or less due to impaired systolic contraction, often from prior MI or dilated cardiomyopathy. Guideline-directed therapies proven to reduce mortality, like ACE inhibitors, beta blockers, and mineralocorticoid antagonists, benefit HFrEF much more clearly than HFpEF.

#44 · Cardiovascular Disease

SGLT2 inhibitors in heart failure

Answer

SGLT2 inhibitors such as dapagliflozin and empagliflozin reduce hospitalization for heart failure and cardiovascular death across the full range of ejection fraction, including both HFrEF and HFpEF. Benefit occurs independent of diabetes status. They are now part of standard four-pillar guideline-directed therapy for HFrEF alongside beta blockers, ACE inhibitor/ARB/ARNI, and mineralocorticoid receptor antagonists.

#45 · Cardiovascular Disease

Restrictive cardiomyopathy: common causes

Answer

Amyloidosis is a leading cause of restrictive cardiomyopathy, along with sarcoidosis, hemochromatosis, and prior radiation therapy. Echocardiography classically shows a small or normal-sized ventricle with thickened walls, biatrial enlargement, and a granular sparkling myocardial appearance in amyloid. Cardiac MRI with late gadolinium enhancement and, in suspected amyloid, a technetium pyrophosphate scan help confirm the diagnosis noninvasively.

#46 · Cardiovascular Disease

Peripartum cardiomyopathy

Answer

Peripartum cardiomyopathy presents with new systolic heart failure in the last month of pregnancy through five months postpartum, without another identifiable cause. Risk factors include multiparity, multiple gestation, preeclampsia, and older maternal age. Management follows standard heart failure therapy adjusted for pregnancy or lactation safety, and many patients recover left ventricular function within six months, though some progress to chronic cardiomyopathy.

#47 · Cardiovascular Disease

Cor pulmonale: definition and most common cause

Answer

Cor pulmonale is right ventricular enlargement and dysfunction caused by a primary disorder of the lung, pulmonary vasculature, or chest wall that leads to pulmonary hypertension. COPD is the most common underlying cause in adults. Findings include a right ventricular heave, loud P2, jugular venous distension, hepatomegaly, and peripheral edema, and treatment focuses on managing the underlying lung disease and hypoxia rather than the heart directly.

#48 · Cardiovascular Disease

Duke criteria for infective endocarditis: overview

Answer

Diagnosis uses major criteria, which are positive blood cultures with typical organisms or persistent bacteremia, and evidence of endocardial involvement on echocardiography such as a vegetation, plus minor criteria including fever, vascular phenomena, immunologic phenomena, predisposing heart condition or IV drug use, and atypical microbiologic evidence. Definite endocarditis requires 2 major, or 1 major plus 3 minor, or 5 minor criteria. Transesophageal echocardiography is more sensitive than transthoracic and should be pursued if suspicion remains high despite a negative transthoracic study.

#49 · Cardiovascular Disease

Peripheral stigmata of infective endocarditis

Answer

Janeway lesions are painless, flat, erythematous macules on the palms and soles from septic microemboli. Osler nodes are painful, tender nodules on the finger and toe pads from immune complex deposition. Roth spots are retinal hemorrhages with pale centers, and splinter hemorrhages appear as small linear streaks under the nail beds.

#50 · Cardiovascular Disease

Endocarditis prophylaxis: who needs it

Answer

Antibiotic prophylaxis before dental procedures is reserved for the highest-risk patients: those with a prosthetic valve or prosthetic material used in valve repair, prior infective endocarditis, unrepaired cyanotic congenital heart disease, and cardiac transplant recipients with valvulopathy. Most patients with native valve disease, including mitral valve prolapse, do not need prophylaxis. When indicated, a single dose of amoxicillin is given before the procedure.

#51 · Cardiovascular Disease

Organisms typically responsible for infective endocarditis

Answer

Staphylococcus aureus is now the most common cause overall and is especially associated with IV drug use and acute, rapidly destructive disease often involving the tricuspid valve. Viridans group streptococci cause a more subacute course typically on previously damaged or prosthetic valves. Enterococcus and the HACEK group of fastidious gram-negative organisms are other important causes, and Streptococcus bovis (gallolyticus) bacteremia should prompt colonoscopy to look for colon cancer.

#52 · Cardiovascular Disease

Carotid artery stenosis: when to intervene

Answer

Symptomatic carotid stenosis of 70 to 99 percent, meaning a recent TIA or stroke referable to that carotid, warrants carotid endarterectomy or stenting to reduce recurrent stroke risk. Asymptomatic stenosis is managed more selectively, generally considering intervention around 70 percent or greater stenosis in good surgical candidates with reasonable life expectancy, alongside optimal medical therapy. All patients with carotid disease need aggressive risk factor management including statin therapy and antiplatelet agents.

#53 · Cardiovascular Disease

Abdominal aortic aneurysm: screening recommendations

Answer

A one-time abdominal ultrasound is recommended for men aged 65 to 75 who have ever smoked. Selective screening based on individual risk is reasonable for men in that age range who never smoked. Once an AAA is found, size determines follow-up interval and repair threshold, with elective repair generally recommended around 5.5 cm in men or when growth exceeds 0.5 cm in six months.

#54 · Cardiovascular Disease

Aortic dissection: Stanford classification and management

Answer

Stanford type A involves the ascending aorta and is a surgical emergency due to risk of pericardial tamponade, aortic regurgitation, and stroke. Stanford type B is confined to the descending aorta distal to the left subclavian artery and is usually managed medically with blood pressure and heart rate control unless there is malperfusion, rupture, or refractory pain. Initial imaging of choice in a stable patient is CT angiography of the chest, abdomen, and pelvis.

#55 · Cardiovascular Disease

Peripheral artery disease: diagnosis and ankle-brachial index

Answer

The ankle-brachial index is the first-line noninvasive test, with a value of 0.90 or lower diagnostic of PAD; values between 0.91 and 0.99 are borderline and above 1.40 suggest noncompressible calcified vessels requiring further testing. Classic presentation is intermittent claudication, reproducible exertional leg pain that resolves with rest. Management includes supervised exercise therapy, smoking cessation, statin therapy, and an antiplatelet agent, with cilostazol considered for symptomatic relief if not contraindicated by heart failure.

#56 · Cardiovascular Disease

Acute limb ischemia: the six Ps

Answer

The six Ps are pain, pallor, pulselessness, paresthesia, paralysis, and poikilothermia (coldness). Paresthesia and paralysis signal advancing nerve and muscle ischemia and indicate a limb-threatening emergency requiring urgent revascularization. Causes include embolism, often cardiac in origin such as from atrial fibrillation, or in situ thrombosis of a previously diseased atherosclerotic segment.

#57 · Cardiovascular Disease

Deep vein thrombosis: diagnostic approach

Answer

Use a validated clinical prediction rule such as the Wells score to estimate pretest probability. Low-probability patients can be evaluated with a D-dimer, and a negative result reasonably excludes DVT, while higher-probability patients or those with a positive D-dimer should go straight to compression ultrasound. Ultrasound showing lack of full compressibility of the vein is diagnostic.

#58 · Cardiovascular Disease

Chronic venous insufficiency versus arterial insufficiency ulcers

Answer

Venous ulcers occur over the medial malleolus, are shallow with irregular borders and surrounding hyperpigmentation or stasis dermatitis, and are relatively less painful. Arterial ulcers occur on the distal toes or pressure points, are deep with a punched-out appearance, and are quite painful, often worse with leg elevation. Venous disease is treated with compression therapy while arterial disease requires revascularization workup and avoiding compression.

#59 · Cardiovascular Disease

Superior vena cava syndrome

Answer

SVC syndrome presents with facial and upper extremity swelling, distended neck and chest wall veins, and sometimes headache or visual changes from venous congestion. Malignancy, especially lung cancer and lymphoma, is the most common cause, though central venous catheters and pacemaker leads are increasingly common causes too. Diagnosis is confirmed with contrast CT of the chest, and management targets the underlying cause, with stenting used for rapid symptom relief in severe cases.

#60 · Cardiovascular Disease

Syncope: red flags suggesting a cardiac cause

Answer

Syncope during exertion, syncope while supine, absence of prodromal symptoms, a family history of sudden cardiac death, or an abnormal ECG all suggest a cardiac cause and warrant further cardiac workup and often admission. Cardiac syncope carries a higher mortality risk than vasovagal or situational syncope. Structural heart disease, such as severe aortic stenosis or hypertrophic cardiomyopathy, and arrhythmias are the key concerning etiologies to rule out.

#61 · Cardiovascular Disease

Vasovagal syncope: typical features

Answer

Vasovagal syncope is usually triggered by prolonged standing, pain, fear, or emotional stress, and is preceded by a prodrome of nausea, warmth, diaphoresis, and tunnel vision. It results from an exaggerated autonomic reflex causing sudden bradycardia and vasodilation. It is benign and managed with reassurance, hydration, avoiding triggers, and counterpressure maneuvers at the first sign of prodrome.

#62 · Cardiovascular Disease

Revised Cardiac Risk Index for preoperative risk

Answer

The RCRI includes six factors: high-risk surgery, history of ischemic heart disease, history of congestive heart failure, history of cerebrovascular disease, insulin-dependent diabetes, and preoperative creatinine above 2.0 mg/dL. Each factor present increases predicted risk of major cardiac complications, and 0 to 1 factors indicates low risk generally not requiring further cardiac testing before surgery. More than 1 factor with poor functional capacity may prompt further evaluation or optimization before elective surgery.

#63 · Cardiovascular Disease

Preoperative cardiac clearance: functional capacity approach

Answer

If a patient can achieve 4 metabolic equivalents of activity, such as climbing two flights of stairs or walking briskly, without symptoms, further cardiac testing is usually unnecessary regardless of risk factors, and surgery can proceed. If functional capacity is poor or unknown and the patient has elevated risk plus the surgery is high risk, further stress testing may be warranted only if it would change management. Emergency surgery should never be delayed for cardiac workup.

#64 · Cardiovascular Disease

Perioperative beta blockade: current guidance

Answer

Continuing a beta blocker in patients already taking one chronically is recommended through the perioperative period. Starting a beta blocker for the first time just before surgery is not recommended, since trials showed it reduces MI but increases stroke and overall mortality from hypotension and bradycardia. If beta blockade is initiated for a new indication, it should be started well in advance of surgery, not on the day of.

#65 · Cardiovascular Disease

LDL cholesterol goals and statin intensity

Answer

High-intensity statin therapy, aiming for at least a 50 percent LDL reduction, is recommended for patients with established atherosclerotic cardiovascular disease and for most patients with LDL 190 mg/dL or higher. Moderate-intensity statins aiming for 30 to 49 percent reduction are used for primary prevention in patients aged 40 to 75 with diabetes or elevated 10-year ASCVD risk based on the pooled cohort equations. Ezetimibe or a PCSK9 inhibitor can be added if LDL goals are not met on maximally tolerated statin therapy.

#66 · Cardiovascular Disease

Statin-associated muscle symptoms: approach

Answer

First confirm symptoms are truly related to the statin by checking creatine kinase and considering a trial off the drug, since many reported symptoms are not actually caused by statins. If confirmed, options include switching to a different statin, using a lower dose, alternate-day dosing, or trying a non-statin agent like ezetimibe or a PCSK9 inhibitor. Statin therapy should not be abandoned entirely in high-risk patients without exhausting these alternatives, since the cardiovascular benefit is substantial.

#67 · Cardiovascular Disease

Severe hypertriglyceridemia: risk and management

Answer

Triglycerides above 500 mg/dL, and especially above 1000, carry a significant risk of acute pancreatitis, distinct from the cardiovascular risk of moderate elevations. Management focuses on fibrates, omega-3 fatty acids, and strict dietary fat restriction, with severe cases sometimes requiring insulin infusion or apheresis during an acute pancreatitis episode. Secondary causes like poorly controlled diabetes, alcohol use, and certain medications should always be addressed.

#68 · Cardiovascular Disease

Dual antiplatelet therapy after PCI with drug-eluting stent

Answer

Standard duration is dual antiplatelet therapy with aspirin plus a P2Y12 inhibitor for at least 6 to 12 months after drug-eluting stent placement, with the exact duration individualized based on bleeding versus ischemic risk. In patients with ACS, at least 12 months is generally preferred if bleeding risk is not prohibitive. After the DAPT period, most patients continue lifelong aspirin monotherapy.

#69 · Cardiovascular Disease

Anticoagulation choice in atrial fibrillation: DOAC versus warfarin

Answer

Direct oral anticoagulants such as apixaban, rivaroxaban, dabigatran, and edoxaban are preferred over warfarin for most patients with nonvalvular atrial fibrillation due to comparable or better efficacy, lower intracranial bleeding risk, and no need for routine monitoring. Warfarin remains necessary for patients with moderate to severe mitral stenosis or a mechanical heart valve, situations termed valvular atrial fibrillation, where DOACs have shown worse outcomes. Renal function should be checked periodically since DOAC dosing depends on creatinine clearance.

#70 · Cardiovascular Disease

Triple therapy versus dual therapy after PCI in a patient with atrial fibrillation

Answer

Patients with atrial fibrillation who undergo PCI generally need anticoagulation plus antiplatelet therapy together, but current guidance favors a shorter course of triple therapy (anticoagulant plus aspirin plus a P2Y12 inhibitor), often only during the hospitalization, followed by dual therapy with an anticoagulant plus a single antiplatelet, usually clopidogrel. This reduces bleeding risk substantially compared to prolonged triple therapy without a significant increase in ischemic events. The specific duration is individualized based on bleeding and thrombotic risk.

#71 · Cardiovascular Disease

Innocent versus pathologic murmurs in a young patient

Answer

Innocent murmurs, like a Still's murmur, are soft, early systolic, grade 2 or less, heard best at the left lower sternal border, and occur in an asymptomatic patient with a normal exam otherwise. Features suggesting a pathologic murmur include diastolic timing, grade 3 or higher intensity, a harsh or holosystolic quality, radiation to the back or carotids, or association with symptoms like poor growth or cyanosis. Any diastolic murmur warrants echocardiographic evaluation regardless of other findings.

#72 · Cardiovascular Disease

Effect of Valsalva maneuver on heart murmurs

Answer

The strain phase of Valsalva decreases venous return and therefore decreases the intensity of most murmurs, including aortic stenosis and mitral regurgitation. The two exceptions that get louder with Valsalva strain are hypertrophic cardiomyopathy, because reduced preload worsens outflow obstruction, and mitral valve prolapse, because reduced ventricular volume causes the prolapse and its click-murmur to occur earlier in systole. Recognizing these two exceptions is a classic exam distinction.

#73 · Cardiovascular Disease

S3 versus S4 heart sound: meaning

Answer

An S3 occurs early in diastole during rapid ventricular filling and suggests volume overload states like systolic heart failure, mitral regurgitation, or high-output states; it can be normal in young healthy people and pregnant women. An S4 occurs late in diastole during atrial contraction against a stiff, noncompliant ventricle, as seen in longstanding hypertension, aortic stenosis, or hypertrophic cardiomyopathy, and is never normal in older adults. An S4 is absent in atrial fibrillation because it requires organized atrial contraction.

#74 · Cardiovascular Disease

Fixed split S2 versus paradoxical splitting

Answer

A fixed split S2, meaning the split does not vary with respiration, is classic for atrial septal defect because the left-to-right shunt equalizes right and left atrial filling regardless of the respiratory cycle. Paradoxical splitting, where the split narrows with inspiration and widens with expiration, occurs when aortic closure is delayed, as in left bundle branch block or severe aortic stenosis. Normal physiologic splitting widens with inspiration due to increased venous return delaying pulmonic closure.

#75 · Cardiovascular Disease

Continuous machinery murmur in a newborn

Answer

A continuous, machine-like murmur heard throughout systole and diastole at the left upper sternal border or infraclavicular area is classic for patent ductus arteriosus. It results from continuous flow from the higher-pressure aorta into the lower-pressure pulmonary artery throughout the cardiac cycle. Indomethacin or ibuprofen can pharmacologically close a PDA in premature infants, while prostaglandin E1 is given to keep a PDA open in duct-dependent congenital lesions.

#76 · Cardiovascular Disease

Tetralogy of Fallot: components and hallmark event

Answer

The four components are pulmonic stenosis, right ventricular hypertrophy, an overriding aorta, and a ventricular septal defect. Pulmonic stenosis is the key determinant of right-to-left shunting and cyanosis severity. A hypercyanotic tet spell is treated by placing the infant in a knee-to-chest position, which increases systemic vascular resistance and reduces the right-to-left shunt, along with supplemental oxygen and sometimes morphine or phenylephrine.

#77 · Cardiovascular Disease

Approach to a new murmur discovered incidentally on exam

Answer

Any new diastolic murmur, any murmur grade 3 or higher, or a murmur accompanied by symptoms such as dyspnea, chest pain, or syncope warrants an echocardiogram. Grade 1 to 2 systolic murmurs in an asymptomatic patient with an otherwise normal exam can often be observed without imaging, particularly if features of an innocent murmur are present. Echocardiography remains the initial imaging test of choice for essentially all suspected structural or valvular heart disease.

#78 · Cardiovascular Disease

Cardiac auscultation location for the mitral valve versus the aortic valve

Answer

The mitral valve is best heard at the cardiac apex, roughly the fifth intercostal space at the midclavicular line, and is enhanced by placing the patient in the left lateral decubitus position. The aortic valve is best heard at the right second intercostal space near the sternal border, with radiation toward the carotids, and is enhanced by having the patient sit up, lean forward, and exhale. Systematic auscultation across all four classic areas, aortic, pulmonic, tricuspid, and mitral, helps avoid missing murmurs localized to one region.

#79 · Cardiovascular Disease

Mitral valve prolapse: auscultation finding

Answer

MVP produces a midsystolic click, sometimes followed by a late systolic murmur if regurgitation is present, best heard at the apex. Maneuvers that decrease left ventricular volume, like standing or Valsalva, move the click and murmur earlier in systole, while maneuvers that increase volume, like squatting, move them later. Most patients are asymptomatic, but MVP can rarely be complicated by progressive mitral regurgitation, arrhythmia, or, very rarely, endocarditis.

#80 · Cardiovascular Disease

Timed intervals: what a widely split S2 without respiratory variation but wide fixed split suggests versus a widely split S2 that does vary

Answer

A widely split S2 that still varies normally with respiration, just with a larger gap than usual, is seen in conditions delaying right ventricular emptying such as right bundle branch block or pulmonic stenosis. A fixed wide split that does not change with respiration points to atrial septal defect, since the shunt keeps right and left sided volumes equalized throughout the respiratory cycle. Distinguishing fixed versus respiratory-variable splitting narrows the differential significantly on exam.

#81 · Dermatology

Chronic itchy flexural rash in a patient with a history of asthma and allergic rhinitis. Diagnosis?

Answer

Atopic dermatitis. It classically involves the antecubital and popliteal fossae in adults and the face and extensor surfaces in infants. Treatment is emollients, avoidance of triggers, and topical corticosteroids or calcineurin inhibitors for flares. Severe refractory disease may need dupilumab.

#82 · Dermatology

How do you distinguish urticarial vasculitis from ordinary urticaria?

Answer

Urticarial vasculitis produces wheals that last longer than 24 hours, are painful or burning rather than itchy, and leave residual bruising or hyperpigmentation as they resolve. Ordinary urticaria lesions are pruritic and each one fades within 24 hours without a mark. Biopsy of urticarial vasculitis shows leukocytoclastic vasculitis, and it can be associated with hypocomplementemia and underlying connective tissue disease.

#83 · Dermatology

First-line treatment approach for moderate inflammatory acne vulgaris?

Answer

Topical retinoid plus a topical or oral antibiotic such as doxycycline for its anti-inflammatory effect, often combined with benzoyl peroxide to limit resistance. Oral antibiotics are used for a limited course, not indefinitely. Isotretinoin is reserved for severe nodulocystic or scarring acne that fails standard therapy.

#84 · Dermatology

Middle-aged patient with facial flushing, telangiectasias, and papulopustules but no comedones. Diagnosis?

Answer

Rosacea. The absence of comedones distinguishes it from acne vulgaris. Triggers include sun exposure, heat, alcohol, and spicy foods, and treatment includes topical metronidazole or azelaic acid, oral doxycycline for inflammatory flares, and trigger avoidance. Rhinophyma is a late complication from sebaceous gland hypertrophy.

#85 · Dermatology

Silvery scaly plaques on the extensor surfaces of the elbows and knees, with nail pitting. Diagnosis and a key associated systemic finding?

Answer

Psoriasis. Look for the Auspitz sign, pinpoint bleeding when scale is removed, and nail changes like pitting, onycholysis, and oil spots. Up to 30 percent of patients develop psoriatic arthritis, so ask about joint pain and stiffness. Treatment ranges from topical steroids and vitamin D analogs for limited disease to biologics for severe or joint-involving disease.

#86 · Dermatology

What are the classic 6 P's of lichen planus?

Answer

Purple, polygonal, planar, pruritic, papules, and plaques, often with fine white lines on the surface called Wickham striae. Lesions favor the wrists, ankles, and can involve oral mucosa as lacy white patches. It has been linked to hepatitis C infection, so consider screening.

#87 · Dermatology

Diabetic patient with a rapidly spreading, painful, warm area of erythema with poorly demarcated borders on the leg. Likely diagnosis and organism?

Answer

Cellulitis, most commonly caused by group A Streptococcus or Staphylococcus aureus. Poorly demarcated borders and deeper tissue involvement distinguish it from erysipelas, which has sharply demarcated raised borders and is almost always streptococcal. Treatment is empiric antibiotics covering strep and staph, with MRSA coverage if purulence or risk factors are present.

#88 · Dermatology

Flaccid bullae with a positive Nikolsky sign and oral mucosal erosions. Diagnosis and mechanism?

Answer

Pemphigus vulgaris, caused by IgG autoantibodies against desmoglein 3, a protein that holds keratinocytes together in the epidermis. This produces intraepidermal blister formation, so the blisters are flaccid and rupture easily, unlike the tense bullae of bullous pemphigoid. Diagnosis is confirmed by biopsy with direct immunofluorescence showing intercellular IgG deposits, and treatment is systemic corticosteroids plus a steroid-sparing agent like rituximab.

#89 · Dermatology

How do you distinguish vitiligo from tinea versicolor as a cause of hypopigmented patches?

Answer

Vitiligo causes completely depigmented, sharply demarcated white patches from autoimmune destruction of melanocytes, often symmetric and worsening over time with no scale. Tinea versicolor causes finely scaling hypopigmented or hyperpigmented patches from a Malassezia yeast infection, and potassium hydroxide prep shows a spaghetti and meatballs pattern of hyphae and spores. Vitiligo is treated with topical steroids or phototherapy, while tinea versicolor responds to topical or oral antifungals.

#90 · Dermatology

How do you differentiate androgenetic alopecia from alopecia areata on exam?

Answer

Androgenetic alopecia causes gradual, patterned thinning, frontotemporal recession in men and diffuse crown thinning in women, without inflammation or complete bald patches. Alopecia areata causes sudden, well-circumscribed round patches of complete hair loss with normal-appearing scalp skin and characteristic exclamation point hairs at the margins. Androgenetic alopecia is treated with minoxidil or finasteride, while alopecia areata may respond to intralesional or topical corticosteroids.

#91 · Dermatology

What features on a pigmented lesion raise concern for melanoma?

Answer

Use the ABCDE rule: Asymmetry, Border irregularity, Color variation, Diameter greater than 6 millimeters, and Evolution or change over time. Any lesion with these features needs an excisional biopsy with narrow margins to determine Breslow depth, which drives prognosis and further management. A history of blistering sunburns and a changing mole are important red flags to elicit.

#92 · Dermatology

How do you distinguish basal cell carcinoma from squamous cell carcinoma clinically?

Answer

Basal cell carcinoma classically presents as a pearly papule with rolled borders and telangiectasias, grows slowly, and rarely metastasizes. Squamous cell carcinoma presents as a scaly, crusted, or ulcerated hyperkeratotic papule or plaque, often arising from actinic keratosis, and carries a higher risk of metastasis, especially on the lip or ear. Both occur on sun-exposed skin and are treated with surgical excision or Mohs surgery depending on location and size.

#93 · Dermatology

Yellow, thickened, and separating nail plate in an elderly patient. Diagnosis and how to confirm?

Answer

Onychomycosis, a fungal nail infection usually caused by dermatophytes. Confirm with KOH prep or fungal culture before starting treatment, since a normal-appearing nail can mimic other conditions like psoriasis. Oral terbinafine is more effective than topical therapy for most cases, but check liver function before starting it.

#94 · Dermatology

Elderly bedbound patient with a stage 3 pressure injury. What defines this stage and what is the key management principle?

Answer

A stage 3 pressure injury involves full-thickness skin loss with visible subcutaneous fat but no exposed bone, tendon, or muscle. Management centers on offloading pressure with frequent repositioning, moist wound care, and nutritional optimization including adequate protein intake. Debridement of necrotic tissue is needed when present, and infection should be assessed if there is surrounding erythema, warmth, or drainage.

#95 · Dermatology

Patient on a new sulfonamide antibiotic develops fever, mucosal erosions, and a spreading rash with a positive Nikolsky sign. Diagnosis and how is SJS distinguished from TEN?

Answer

This is Stevens-Johnson syndrome or toxic epidermal necrolysis, a drug-induced dermatologic emergency involving keratinocyte apoptosis and epidermal detachment. SJS involves less than 10 percent of body surface area, TEN involves more than 30 percent, and 10 to 30 percent is considered SJS-TEN overlap. Management requires immediate discontinuation of the offending drug and transfer to a burn unit or intensive care for supportive wound care and fluid management.

#96 · Endocrinology Diabetes and Metabolism

Patient with fatigue, hyperpigmentation, hypotension, hyponatremia, and hyperkalemia. Diagnosis and next test?

Answer

This is primary adrenal insufficiency (Addison disease). Hyperpigmentation occurs because low cortisol drives high ACTH, and ACTH shares a precursor with melanocyte stimulating hormone. Confirm with a cosyntropin (ACTH) stimulation test showing an inadequate cortisol rise. Autoimmune adrenalitis is the most common cause in developed countries.

#97 · Endocrinology Diabetes and Metabolism

How does secondary adrenal insufficiency differ from primary on labs and exam?

Answer

Secondary adrenal insufficiency results from pituitary ACTH deficiency, often from chronic steroid use suppressing the axis. Because ACTH is low, there is no hyperpigmentation, and aldosterone is preserved since it is mainly regulated by the renin-angiotensin system, so hyperkalemia is usually absent. Hyponatremia can still occur from impaired free water clearance.

#98 · Endocrinology Diabetes and Metabolism

Best initial treatment for suspected adrenal crisis before confirmatory testing is back?

Answer

Give IV hydrocortisone immediately along with aggressive IV normal saline resuscitation, do not wait for cortisol or ACTH stimulation test results. Hydrocortisone is preferred at stress doses because at high doses it also has mineralocorticoid activity. Check and correct hypoglycemia and hyperkalemia as needed.

#99 · Endocrinology Diabetes and Metabolism

Young woman with hypertension, hypokalemia, and metabolic alkalosis. What is the likely diagnosis and confirmatory test?

Answer

Think primary hyperaldosteronism (Conn syndrome). Screen with a plasma aldosterone to renin ratio, which is elevated because aldosterone is high and renin is suppressed. Confirm with a saline suppression test, then use adrenal CT and adrenal vein sampling to distinguish unilateral adenoma from bilateral hyperplasia.

#100 · Endocrinology Diabetes and Metabolism

Patient with episodic headache, palpitations, and diaphoresis plus labile hypertension. What tumor and workup?

Answer

This triad suggests pheochromocytoma. Screen with plasma free metanephrines or 24 hour urine fractionated metanephrines, which have high sensitivity. Once biochemically confirmed, localize with CT or MRI of the adrenals, and give alpha blockade before beta blockade or surgery to avoid unopposed alpha stimulation causing a hypertensive crisis.

#101 · Endocrinology Diabetes and Metabolism

What clinical features suggest Cushing syndrome, and what is the initial screening test?

Answer

Look for central obesity, purple striae, proximal muscle weakness, easy bruising, moon facies, and a buffalo hump. Initial screening options include a low dose overnight dexamethasone suppression test, 24 hour urine free cortisol, or late night salivary cortisol. Once hypercortisolism is confirmed, measure ACTH to determine if the cause is pituitary, ectopic, or adrenal.

#102 · Endocrinology Diabetes and Metabolism

How do you differentiate ACTH dependent from ACTH independent Cushing syndrome?

Answer

If plasma ACTH is low or suppressed, the source is adrenal, such as an adenoma or carcinoma, and adrenal imaging is next. If ACTH is normal or elevated, the source is pituitary (Cushing disease) or an ectopic ACTH secreting tumor, and a high dose dexamethasone suppression test or pituitary MRI helps distinguish them, since pituitary tumors usually still show partial suppression while ectopic sources typically do not.

#103 · Endocrinology Diabetes and Metabolism

What are the classic signs and symptoms of Graves disease beyond generic hyperthyroidism?

Answer

Graves disease is caused by TSH receptor stimulating antibodies and classically presents with diffuse goiter, exophthalmos, periorbital edema, and pretibial myxedema, none of which occur in other causes of thyrotoxicosis. Patients also have weight loss despite increased appetite, heat intolerance, tremor, and tachycardia or atrial fibrillation. Radioactive iodine uptake scan shows diffusely increased uptake.

#104 · Endocrinology Diabetes and Metabolism

How do you distinguish subacute (de Quervain) thyroiditis from Graves disease when both cause thyrotoxicosis?

Answer

Subacute thyroiditis presents with a painful, tender goiter often after a viral illness, with elevated ESR, and radioactive iodine uptake is low because inflamed follicles are leaking preformed hormone rather than actively producing it. Graves disease has a nontender goiter with high uptake. Subacute thyroiditis is self limited and often followed by a hypothyroid phase before recovery.

#105 · Endocrinology Diabetes and Metabolism

Patient with fever, altered mental status, tachycardia, and vomiting in the setting of known hyperthyroidism. Diagnosis and treatment?

Answer

This is thyroid storm, a life threatening endocrine emergency often precipitated by infection, surgery, or trauma in a patient with underlying thyrotoxicosis. Treat with beta blockers for symptom control, propylthiouracil or high dose methimazole to block hormone synthesis, iodine solution given at least one hour after the antithyroid drug to block release, and glucocorticoids to reduce peripheral T4 to T3 conversion and support possible relative adrenal insufficiency.

#106 · Endocrinology Diabetes and Metabolism

Elderly patient with hypothermia, altered mental status, bradycardia, and hyponatremia. What is this and how is it treated?

Answer

This is myxedema coma, a severe decompensated hypothyroidism, usually precipitated by infection, cold exposure, or sedative use in an elderly patient with untreated hypothyroidism. Treat urgently with IV levothyroxine, IV hydrocortisone (since coexisting adrenal insufficiency may be masked), passive rewarming, and supportive care in an ICU setting, since mortality is high without prompt treatment.

#107 · Endocrinology Diabetes and Metabolism

What is the most common cause of hypothyroidism in iodine sufficient regions, and what antibodies are associated?

Answer

Hashimoto thyroiditis, an autoimmune lymphocytic infiltration of the thyroid, is the most common cause. It is associated with anti thyroid peroxidase (anti TPO) and anti thyroglobulin antibodies. Patients often have a firm, nontender goiter early on, which may shrink as the disease progresses to atrophic thyroiditis.

#108 · Endocrinology Diabetes and Metabolism

How should subclinical hypothyroidism be managed?

Answer

Subclinical hypothyroidism is defined as an elevated TSH with normal free T4. Treatment with levothyroxine is generally recommended if TSH is above 10, if the patient is symptomatic, pregnant, or trying to conceive, or if there are positive anti TPO antibodies with TSH between 7 and 10. Mildly elevated TSH under 10 in an asymptomatic patient without antibodies can often just be monitored.

#109 · Endocrinology Diabetes and Metabolism

What is the first line lipid lowering therapy for a patient with established atherosclerotic cardiovascular disease?

Answer

High intensity statin therapy (such as atorvastatin 40-80 mg or rosuvastatin 20-40 mg) is first line for secondary prevention. The goal is generally to lower LDL by at least 50 percent, and many guidelines target an LDL below 70 mg/dL in very high risk patients. If LDL remains above goal on maximally tolerated statin, ezetimibe or a PCSK9 inhibitor can be added.

#110 · Endocrinology Diabetes and Metabolism

Patient with recurrent pancreatitis and triglycerides over 1000 mg/dL. What is the concern and initial management?

Answer

Severe hypertriglyceridemia above 1000 mg/dL is a well established cause of acute pancreatitis. Initial management includes fasting, IV fluids, and insulin infusion (even without hyperglycemia) to accelerate triglyceride clearance through lipoprotein lipase activation, sometimes with apheresis in severe cases. Long term, fibrates are the most effective agents for lowering triglycerides in this range.

#111 · Endocrinology Diabetes and Metabolism

Woman with irregular menses, hirsutism, acne, and infertility. What diagnostic criteria confirm PCOS?

Answer

Polycystic ovary syndrome is diagnosed using the Rotterdam criteria, requiring at least two of three: oligo or anovulation, clinical or biochemical hyperandrogenism, and polycystic ovaries on ultrasound, after excluding other causes such as thyroid disease, hyperprolactinemia, and nonclassic congenital adrenal hyperplasia. Many patients also have insulin resistance and are at increased risk for type 2 diabetes and endometrial hyperplasia from unopposed estrogen.

#112 · Endocrinology Diabetes and Metabolism

What is first line treatment for PCOS in a woman not currently seeking pregnancy?

Answer

Combined oral contraceptives are first line, since they regulate menstrual cycles, reduce androgen levels, and protect the endometrium from unopposed estrogen exposure. Metformin can be added for patients with insulin resistance or metabolic features. For women desiring pregnancy, letrozole is now preferred over clomiphene for ovulation induction.

#113 · Endocrinology Diabetes and Metabolism

Older man with low testosterone symptoms. What two morning measurements confirm hypogonadism, and what should be checked next?

Answer

Diagnosis requires low total testosterone on at least two separate early morning measurements, since testosterone follows diurnal variation and peaks in the morning. If confirmed, check LH and FSH to distinguish primary (testicular) from secondary (hypothalamic-pituitary) hypogonadism, and consider checking prolactin and pituitary imaging if secondary hypogonadism is found without a clear cause.

#114 · Endocrinology Diabetes and Metabolism

What are contraindications to starting testosterone replacement therapy?

Answer

Testosterone therapy should be avoided in men with untreated prostate cancer or breast cancer, severe untreated obstructive sleep apnea, uncontrolled heart failure, or a markedly elevated hematocrit. It can worsen erythrocytosis, so hematocrit and PSA should be monitored during treatment, along with symptom response.

#115 · Endocrinology Diabetes and Metabolism

What distinguishes type 1 from type 2 diabetes mellitus at diagnosis?

Answer

Type 1 diabetes results from autoimmune beta cell destruction, typically presents in younger patients often with diabetic ketoacidosis at diagnosis, and is associated with positive autoantibodies such as anti GAD65 and low or absent C peptide. Type 2 diabetes is driven by insulin resistance with relative insulin deficiency, usually presents in older or overweight patients, and C peptide is typically normal to elevated early in the disease.

#116 · Endocrinology Diabetes and Metabolism

How do you distinguish diabetic ketoacidosis from hyperosmolar hyperglycemic state?

Answer

DKA typically occurs in type 1 diabetes with glucose usually under 800, significant anion gap metabolic acidosis, and positive ketones, and it develops rapidly over hours. HHS occurs in type 2 diabetes with markedly higher glucose often over 600-1000, minimal ketosis and acidosis, and much higher serum osmolality, developing more gradually over days, often with more severe dehydration and altered mental status.

#117 · Endocrinology Diabetes and Metabolism

What is the cornerstone of DKA treatment and a key pitfall regarding potassium?

Answer

Treatment involves IV fluids, an insulin infusion, and correction of electrolytes. Even though total body potassium is depleted, serum potassium may appear normal or high initially due to acidosis driving potassium out of cells, so insulin should be withheld or potassium repleted first if serum potassium is below 3.3 mEq/L, since insulin will shift potassium into cells and can precipitate dangerous hypokalemia and arrhythmia.

#118 · Endocrinology Diabetes and Metabolism

What first line oral agent for type 2 diabetes has cardiovascular and renal benefits, and what is a key contraindication?

Answer

Metformin is first line and has a favorable safety profile plus modest cardiovascular benefit. It should be avoided or used cautiously in patients with an eGFR below 30 mL/min/1.73m2 due to risk of lactic acidosis, and it should be held before procedures using iodinated contrast in patients with reduced kidney function or acute illness.

#119 · Endocrinology Diabetes and Metabolism

Which diabetes drug classes have proven cardiovascular and renal benefit independent of glucose lowering?

Answer

SGLT2 inhibitors (such as empagliflozin) reduce heart failure hospitalization and slow progression of diabetic kidney disease, and GLP-1 receptor agonists (such as semaglutide and liraglutide) reduce major adverse cardiovascular events in patients with established atherosclerotic disease. Both are now recommended in high risk type 2 diabetics regardless of A1c control.

#120 · Endocrinology Diabetes and Metabolism

How does diabetic nephropathy typically progress, and what is the first sign?

Answer

The earliest detectable sign is microalbuminuria, defined as a urine albumin to creatinine ratio between 30 and 300 mg/g, which should be screened annually starting at diagnosis in type 2 diabetes and after five years in type 1 diabetes. ACE inhibitors or ARBs are used to slow progression by reducing intraglomerular pressure, and SGLT2 inhibitors provide additional renoprotection.

#121 · Endocrinology Diabetes and Metabolism

Patient with diabetes and a painless foot ulcer with loss of protective sensation. What is the underlying process and screening tool?

Answer

This reflects diabetic peripheral neuropathy, a distal symmetric sensorimotor polyneuropathy from chronic hyperglycemia damaging small nerve fibers. Screen annually with a 10 gram monofilament test on the plantar foot surfaces, and educate patients on daily foot inspection since loss of protective sensation predisposes to unnoticed injury and ulceration.

#122 · Endocrinology Diabetes and Metabolism

What are the classic symptoms and workup for primary hyperparathyroidism found incidentally on labs?

Answer

Primary hyperparathyroidism classically presents with hypercalcemia and an inappropriately normal or elevated PTH, often found incidentally as asymptomatic hypercalcemia on routine labs. Remember the mnemonic "bones, stones, groans, and psychiatric overtones" for symptomatic disease: bone pain or osteoporosis, kidney stones, abdominal pain or constipation, and depression or confusion. The most common cause is a solitary parathyroid adenoma.

#123 · Endocrinology Diabetes and Metabolism

How do you distinguish primary hyperparathyroidism from malignancy associated hypercalcemia using PTH?

Answer

In primary hyperparathyroidism, PTH is normal or elevated despite high calcium. In malignancy associated hypercalcemia (from PTHrP secretion or bony metastases), PTH is appropriately suppressed because the high calcium is coming from another source. Checking PTHrP can help confirm humoral hypercalcemia of malignancy when PTH is low.

#124 · Endocrinology Diabetes and Metabolism

Patient with perioral numbness, muscle cramps, and a positive Chvostek sign. What is the diagnosis and common cause?

Answer

These findings indicate hypocalcemia, and Chvostek sign (facial twitching when tapping the facial nerve) along with Trousseau sign (carpal spasm with blood pressure cuff inflation) are classic exam findings. A common cause is post surgical hypoparathyroidism after thyroidectomy or parathyroidectomy from inadvertent removal or devascularization of the parathyroid glands. Treatment involves calcium and calcitriol supplementation.

#125 · Endocrinology Diabetes and Metabolism

What DEXA T-score values define osteopenia and osteoporosis?

Answer

A T-score between -1.0 and -2.5 defines osteopenia (low bone mass), and a T-score of -2.5 or lower defines osteoporosis. A T-score compares bone density to a healthy young adult, while a Z-score compares to age matched peers and is more useful in younger patients or when searching for secondary causes of bone loss.

#126 · Endocrinology Diabetes and Metabolism

What is first line pharmacologic treatment for postmenopausal osteoporosis, and what is a key safety concern?

Answer

Oral bisphosphonates such as alendronate are first line, reducing fracture risk by inhibiting osteoclast mediated bone resorption. Key concerns include esophagitis (patients should remain upright after dosing), and with long term use, rare risks of osteonecrosis of the jaw and atypical femur fractures, which sometimes prompts a drug holiday after several years of therapy.

#127 · Endocrinology Diabetes and Metabolism

How do you interpret vitamin D deficiency and what levels define insufficiency versus deficiency?

Answer

Vitamin D status is assessed with 25-hydroxyvitamin D, since it is the major circulating and storage form. A level below 20 ng/mL defines deficiency, and 20 to 29 ng/mL defines insufficiency, while levels of 30 ng/mL or above are considered sufficient. Severe deficiency can cause secondary hyperparathyroidism, osteomalacia in adults, and rickets in children.

#128 · Endocrinology Diabetes and Metabolism

Patient with headache, bitemporal hemianopsia, and galactorrhea. What is the likely diagnosis and first line treatment if prolactin is very high?

Answer

This presentation suggests a prolactinoma, a pituitary adenoma causing mass effect on the optic chiasm and hyperprolactinemia. Unlike other pituitary tumors, prolactinomas are treated medically first with a dopamine agonist such as cabergoline or bromocriptine, which shrinks the tumor and normalizes prolactin, reserving surgery for medication resistant or intolerant cases.

#129 · Endocrinology Diabetes and Metabolism

Besides prolactinoma, what other conditions can cause a mildly elevated prolactin level?

Answer

Common causes of mild hyperprolactinemia include pregnancy, hypothyroidism (elevated TRH stimulates prolactin release), medications such as antipsychotics and metoclopramide, chronic kidney disease, and stalk compression from a nonfunctioning pituitary mass that blocks dopamine's normal inhibitory effect on prolactin. These should be excluded before attributing hyperprolactinemia to a prolactinoma, especially when levels are only modestly elevated.

#130 · Endocrinology Diabetes and Metabolism

Patient with coarsened facial features, enlarged hands and feet, and new onset diabetes. What is the diagnosis and screening test?

Answer

This presentation suggests acromegaly from excess growth hormone, typically due to a pituitary adenoma. Screen with an IGF-1 level, which reflects average GH secretion over time and is more reliable than a random GH level, then confirm with an oral glucose tolerance test showing failure to suppress GH. Pituitary MRI localizes the tumor, and transsphenoidal surgery is first line treatment.

#131 · Endocrinology Diabetes and Metabolism

Postpartum woman with failure to lactate, fatigue, and hypotension after a complicated delivery with significant blood loss. What is the diagnosis?

Answer

This is Sheehan syndrome, pituitary infarction and necrosis following severe postpartum hemorrhage causing hypoperfusion of the enlarged pregnancy pituitary gland. It presents with varying degrees of hypopituitarism, classically failure to lactate from low prolactin, along with possible deficiencies in ACTH, TSH, and gonadotropins requiring hormone replacement.

#132 · Endocrinology Diabetes and Metabolism

How do you distinguish SIADH from other causes of hyponatremia?

Answer

SIADH causes euvolemic hyponatremia with inappropriately concentrated urine (urine osmolality above 100 mOsm/kg) despite low serum osmolality, along with urine sodium usually above 40 mEq/L and normal thyroid and adrenal function. Common causes include CNS disease, lung disease, malignancy (especially small cell lung cancer), and medications such as SSRIs and carbamazepine. It is a diagnosis of exclusion after ruling out adrenal insufficiency and hypothyroidism.

#133 · Endocrinology Diabetes and Metabolism

Why must chronic severe hyponatremia be corrected slowly, and what is the maximum safe rate?

Answer

Rapid correction of chronic hyponatremia risks osmotic demyelination syndrome (central pontine myelinolysis), which can cause irreversible neurologic damage including quadriparesis and locked in syndrome. Correction should generally not exceed 8 mEq/L in 24 hours, and even slower limits are used in high risk patients such as those with alcoholism, malnutrition, or liver disease.

#134 · Endocrinology Diabetes and Metabolism

Patient with polyuria, polydipsia, and hypernatremia that fails to concentrate urine after desmopressin. What is the diagnosis?

Answer

This pattern indicates nephrogenic diabetes insipidus, where the kidney fails to respond to ADH even though ADH is present or given exogenously. Common causes include lithium use, hypercalcemia, and hypokalemia. This contrasts with central diabetes insipidus, where urine osmolality rises significantly after desmopressin because the kidney can respond once ADH is supplied.

#135 · Endocrinology Diabetes and Metabolism

Patient with episodic confusion, diaphoresis, and palpitations that resolve with eating. What is the likely diagnosis and confirmatory test?

Answer

This presentation with Whipple triad (symptoms of hypoglycemia, documented low glucose, and relief with glucose administration) suggests an insulinoma, an insulin secreting pancreatic neuroendocrine tumor. Confirm with a supervised 72 hour fast showing inappropriately elevated insulin and C peptide during documented hypoglycemia; elevated C peptide helps distinguish endogenous insulin excess from exogenous insulin administration, where C peptide would be suppressed.

#136 · Endocrinology Diabetes and Metabolism

Patient with flushing, diarrhea, and wheezing, found to have a small bowel mass. What is the diagnosis and confirmatory test?

Answer

This triad suggests carcinoid syndrome, caused by a neuroendocrine tumor secreting serotonin and other vasoactive substances, typically with liver metastases since hepatic metabolism otherwise clears serotonin before it reaches systemic circulation. Confirm with elevated 24 hour urine 5-hydroxyindoleacetic acid (5-HIAA), the main serotonin metabolite. Right sided heart valve lesions (carcinoid heart disease) can also develop from chronic serotonin exposure.

#137 · Endocrinology Diabetes and Metabolism

What is autoimmune polyglandular syndrome type 1, and what is its characteristic triad?

Answer

APS type 1 (also called APECED) is caused by mutations in the AIRE gene and usually presents in childhood. Its classic triad is chronic mucocutaneous candidiasis, hypoparathyroidism, and primary adrenal insufficiency, with candidiasis often appearing first. It is inherited in an autosomal recessive pattern, unlike type 2.

#138 · Endocrinology Diabetes and Metabolism

What conditions make up autoimmune polyglandular syndrome type 2, and how does inheritance differ from type 1?

Answer

APS type 2 (Schmidt syndrome) most commonly includes autoimmune adrenal insufficiency plus autoimmune thyroid disease and/or type 1 diabetes mellitus. It typically presents in adulthood and has a polygenic, HLA associated inheritance pattern rather than the single gene autosomal recessive pattern seen in APS type 1. Any patient with one autoimmune endocrinopathy should be screened for others given this clustering tendency.

#139 · Endocrinology Diabetes and Metabolism

What are the diagnostic criteria for metabolic syndrome?

Answer

Metabolic syndrome is diagnosed when at least three of the following are present: abdominal obesity (waist circumference over 40 inches in men or 35 inches in women), triglycerides of 150 mg/dL or higher, HDL below 40 mg/dL in men or below 50 mg/dL in women, blood pressure of 130/85 or higher, and fasting glucose of 100 mg/dL or higher. It identifies patients at increased risk for type 2 diabetes and cardiovascular disease.

#140 · Endocrinology Diabetes and Metabolism

How is central versus primary hypothyroidism distinguished on labs, and why does TSH alone mislead in central disease?

Answer

Primary hypothyroidism shows a low free T4 with an appropriately elevated TSH, since the pituitary is responding correctly to low thyroid hormone. Central (secondary) hypothyroidism from pituitary or hypothalamic disease shows a low free T4 with a low or inappropriately normal TSH, because the pituitary itself cannot mount a proper response, so relying on TSH alone would miss the diagnosis and free T4 must always be checked when central disease is suspected.

#141 · Gastroenterology

Patient with heartburn and regurgitation for years, endoscopy shows salmon colored mucosa extending into the distal esophagus. What is this and what should be done?

Answer

This is Barrett esophagus, meaning normal squamous epithelium has been replaced by intestinal columnar metaplasia from chronic acid exposure. Biopsies are needed to grade dysplasia. Surveillance endoscopy intervals depend on the dysplasia grade, and high grade dysplasia or early cancer is treated with endoscopic ablation or resection rather than watchful waiting.

#142 · Gastroenterology

First line treatment approach for uncomplicated GERD symptoms?

Answer

Start with lifestyle changes such as weight loss, avoiding late meals, and elevating the head of the bed, plus a trial of a proton pump inhibitor once daily before breakfast. If symptoms persist after 4 to 8 weeks or alarm features are present, move to upper endoscopy. Long term PPI use should be reassessed periodically for the lowest effective dose.

#143 · Gastroenterology

Alarm features that mandate endoscopy in a patient with dyspepsia or reflux, rather than empiric therapy?

Answer

Dysphagia, odynophagia, unintentional weight loss, GI bleeding or anemia, persistent vomiting, and a palpable mass or lymphadenopathy are all alarm features. New onset symptoms after age 60 also warrant endoscopy. Any of these should prompt prompt upper endoscopy instead of an empiric medication trial.

#144 · Gastroenterology

Cirrhotic patient with a large volume hematemesis. What is the immediate management sequence?

Answer

Resuscitate with IV fluids and blood products while protecting the airway, then start octreotide and an IV proton pump inhibitor plus prophylactic antibiotics such as ceftriaxone. Upper endoscopy should follow within 12 hours for band ligation of the bleeding varices. Antibiotics reduce mortality and rebleeding risk in cirrhotic patients with GI bleeding regardless of the bleeding source.

#145 · Gastroenterology

How is primary prophylaxis against variceal bleeding chosen in a cirrhotic patient with medium or large varices found on screening endoscopy?

Answer

Options are a nonselective beta blocker such as propranolol or nadolol, or endoscopic variceal band ligation. The choice depends on patient factors like blood pressure tolerance and access to endoscopy. Beta blockers work by lowering portal pressure, while banding directly obliterates the varices.

#146 · Gastroenterology

Patient with dysphagia to solids only, progressive over months, with a history of longstanding GERD. Most likely diagnosis?

Answer

This presentation is classic for a peptic esophageal stricture from chronic acid injury. Solid food dysphagia that progresses gradually points to a mechanical narrowing rather than a motility disorder. Endoscopy with biopsy is needed to confirm benign stricture and exclude esophageal adenocarcinoma, then dilation is performed if benign.

#147 · Gastroenterology

Young patient with dysphagia to both solids and liquids from the start, plus regurgitation of undigested food. What test confirms the diagnosis and what is it?

Answer

This history suggests achalasia, a motility disorder from loss of inhibitory neurons in the lower esophageal sphincter. Esophageal manometry confirms the diagnosis by showing absent peristalsis and incomplete LES relaxation, and barium swallow classically shows a bird beak tapering. Before diagnosing primary achalasia, endoscopy should exclude a pseudoachalasia from a malignancy at the gastroesophageal junction.

#148 · Gastroenterology

How is H. pylori infection tested for and confirmed cured?

Answer

Initial testing can use a stool antigen test, urea breath test, or endoscopic biopsy with rapid urease testing. Serology only shows past exposure and cannot confirm active infection or cure. Test of cure with stool antigen or urea breath test should be done at least 4 weeks after completing therapy and after stopping PPIs for 1 to 2 weeks.

#149 · Gastroenterology

Standard first line regimen for H. pylori eradication in a region without high clarithromycin resistance?

Answer

Clarithromycin triple therapy consists of a PPI, clarithromycin, and amoxicillin (or metronidazole if penicillin allergic) for 14 days. In areas of high clarithromycin resistance or after a prior macrolide exposure, bismuth quadruple therapy with a PPI, bismuth, metronidazole, and tetracycline is preferred instead. Eradication should always be confirmed with a test of cure.

#150 · Gastroenterology

Patient on chronic NSAIDs develops epigastric pain and is found to have a duodenal ulcer. Key next steps?

Answer

Test for H. pylori and treat if positive, and stop the NSAID if possible or switch to a COX-2 selective agent with a PPI if NSAIDs must continue. Duodenal ulcers are rarely malignant so routine biopsy is not required, unlike gastric ulcers which should always be biopsied and rechecked with repeat endoscopy to confirm healing. Add a PPI for ulcer healing regardless of H. pylori status.

#151 · Gastroenterology

Why must all gastric ulcers be biopsied, unlike duodenal ulcers?

Answer

Gastric ulcers carry a meaningful risk of being malignant, whereas duodenal ulcers are almost always benign. Multiple biopsies of the ulcer margin should be taken at the time of diagnosis. Follow up endoscopy in 6 to 8 weeks is recommended to confirm complete healing, and any ulcer that fails to heal should raise concern for malignancy or refractory acid hypersecretion.

#152 · Gastroenterology

Patient with refractory peptic ulcers, diarrhea, and markedly elevated fasting gastrin off PPI. Diagnosis and confirmatory test?

Answer

This suggests Zollinger-Ellison syndrome from a gastrin secreting tumor, often in the duodenum or pancreas. A secretin stimulation test confirms the diagnosis by showing a paradoxical rise in gastrin. Patients should also be screened for MEN1 syndrome given the association with gastrinomas.

#153 · Gastroenterology

Young adult with iron deficiency anemia, bloating, and diarrhea. Which antibody test screens for celiac disease and what confirms it?

Answer

IgA tissue transglutaminase antibody (tTG-IgA) is the preferred screening test, but a total IgA level should be checked simultaneously since selective IgA deficiency causes false negatives. Duodenal biopsy showing villous atrophy, crypt hyperplasia, and intraepithelial lymphocytosis while the patient is still eating gluten confirms the diagnosis. Treatment is strict lifelong avoidance of gluten.

#154 · Gastroenterology

Why must a patient continue eating gluten before celiac serology or biopsy testing?

Answer

Both the antibody tests and the duodenal biopsy findings depend on ongoing gluten exposure to trigger the immune mediated damage. Starting a gluten free diet before testing can normalize antibodies and heal the villi, causing a false negative result. Patients should be counseled not to start dietary changes until testing is complete.

#155 · Gastroenterology

Distinguishing features of Crohn disease versus ulcerative colitis on endoscopy and pathology?

Answer

Crohn disease can affect any part of the GI tract from mouth to anus in a skip pattern, is transmural, and shows granulomas with cobblestoning and strictures or fistulas. Ulcerative colitis is limited to the colon, starts in the rectum and extends proximally in a continuous pattern, and is limited to the mucosa without granulomas. Perianal disease and fistulas strongly favor Crohn disease.

#156 · Gastroenterology

Extraintestinal manifestation of IBD that tracks with bowel disease activity versus one that does not?

Answer

Peripheral arthritis, episcleritis, and erythema nodosum tend to flare along with intestinal disease activity. Primary sclerosing cholangitis, ankylosing spondylitis, uveitis, and pyoderma gangrenosum run independently of bowel activity and do not necessarily improve when the bowel disease is controlled. PSC is strongly associated with ulcerative colitis specifically.

#157 · Gastroenterology

Ulcerative colitis patient with 10 years of disease duration. What screening is recommended going forward?

Answer

Surveillance colonoscopy with random biopsies throughout the colon should begin about 8 years after diagnosis of extensive colitis and then be repeated every 1 to 3 years due to increased colorectal cancer risk. The risk is proportional to disease duration and extent of colonic involvement. Concurrent primary sclerosing cholangitis raises the cancer risk further and warrants earlier and more frequent surveillance.

#158 · Gastroenterology

Average risk adult, when should colorectal cancer screening begin and what are acceptable modalities?

Answer

Screening should begin at age 45 for average risk individuals. Options include colonoscopy every 10 years, FIT testing annually, FIT-DNA testing every 1 to 3 years, or CT colonography every 5 years. Any positive noninvasive test should be followed by a diagnostic colonoscopy.

#159 · Gastroenterology

Patient has a first degree relative diagnosed with colon cancer at age 45. When should this patient start colonoscopy screening?

Answer

Screening should start at age 40, or 10 years before the relative's age at diagnosis, whichever is earlier. This patient should therefore start at age 35. Screening interval is generally every 5 years rather than the standard 10 year interval for average risk patients.

#160 · Gastroenterology

Elderly patient with left lower quadrant pain, fever, and leukocytosis. Best initial imaging and typical management?

Answer

This presentation suggests acute diverticulitis, and CT scan of the abdomen and pelvis with contrast is the best initial imaging to assess severity and look for abscess or perforation. Uncomplicated diverticulitis can often be managed with oral antibiotics as an outpatient in otherwise healthy patients, while complicated disease with abscess, perforation, or obstruction requires hospitalization and possibly drainage or surgery. Colonoscopy should be delayed 6 to 8 weeks after resolution to exclude malignancy, not performed during the acute episode.

#161 · Gastroenterology

Young woman with chronic abdominal pain and altered bowel habits, normal labs and colonoscopy. How is IBS diagnosed and treated?

Answer

IBS is a clinical diagnosis using the Rome criteria, requiring recurrent abdominal pain associated with a change in stool frequency or form, with normal workup to exclude alarm features. Treatment is tailored to the predominant symptom, such as fiber and antispasmodics for pain, loperamide or rifaximin for diarrhea predominant disease, and osmotic laxatives for constipation predominant disease. Alarm features like weight loss, GI bleeding, or anemia should prompt further evaluation rather than an IBS diagnosis.

#162 · Gastroenterology

Painless hematochezia in an otherwise healthy patient with hemorrhoids on exam. What is the key caution?

Answer

Hemorrhoids are a diagnosis of exclusion for rectal bleeding, especially in patients over 45 or with any risk factors for colorectal cancer. A full colonoscopy should still be performed to exclude a more proximal or serious source of bleeding before attributing symptoms to hemorrhoids. Attributing bleeding to hemorrhoids without adequate workup is a classic pitfall that can delay a cancer diagnosis.

#163 · Gastroenterology

Patient with severe epigastric pain radiating to the back, nausea, and lipase three times the upper limit of normal. Diagnosis and initial management?

Answer

This is acute pancreatitis, diagnosed when 2 of 3 criteria are met: characteristic pain, lipase or amylase elevated at least 3 times normal, or characteristic imaging findings. Initial management is aggressive IV fluid resuscitation, pain control, and early oral feeding as tolerated rather than prolonged bowel rest. Gallstones and alcohol are the two most common causes and should be investigated with history and right upper quadrant ultrasound.

#164 · Gastroenterology

Why is early enteral feeding preferred over bowel rest in acute pancreatitis?

Answer

Early oral or enteral feeding, as tolerated within 24 to 72 hours, helps maintain gut mucosal integrity and reduces bacterial translocation, infectious complications, and length of stay compared to prolonged fasting. Parenteral nutrition is reserved for patients who cannot tolerate enteral feeding. This is a shift from older teaching that emphasized bowel rest until pain resolved.

#165 · Gastroenterology

Patient with recurrent episodes of acute pancreatitis over years, now with steatorrhea and new diabetes. Diagnosis and confirmatory findings?

Answer

This presentation suggests chronic pancreatitis with exocrine and endocrine insufficiency. CT or MRI may show pancreatic calcifications, ductal dilation, or atrophy, and fecal elastase is typically low reflecting exocrine insufficiency. Alcohol use is the most common cause, and management includes pancreatic enzyme replacement, pain control, and diabetes management.

#166 · Gastroenterology

Patient with jaundice, right upper quadrant pain, and fever. What is this triad concerning for and what is the treatment?

Answer

This is Charcot triad, concerning for acute cholangitis from an obstructing common bile duct stone or stricture with superimposed infection. Treatment requires IV antibiotics and urgent biliary decompression, typically with ERCP. Adding hypotension and altered mental status to the triad creates Reynolds pentad, signaling more severe suppurative cholangitis needing emergent intervention.

#167 · Gastroenterology

Asymptomatic gallstones found incidentally on imaging. How should this be managed?

Answer

Asymptomatic cholelithiasis generally does not require treatment or cholecystectomy since most patients never develop symptoms. Prophylactic cholecystectomy is reserved for specific situations such as a porcelain gallbladder, very large stones, or gallbladder polyps with concerning features. Patients should be counseled to seek care if biliary colic symptoms develop.

#168 · Gastroenterology

Right upper quadrant pain after fatty meals with a positive sonographic Murphy sign and gallbladder wall thickening. Diagnosis and management?

Answer

This is acute cholecystitis, typically from a cystic duct obstruction by a gallstone with secondary inflammation. Right upper quadrant ultrasound is the first line imaging test, and treatment is IV antibiotics with early laparoscopic cholecystectomy, ideally within 72 hours of symptom onset. HIDA scan can be used when ultrasound findings are equivocal.

#169 · Gastroenterology

Interpretation of hepatitis B serologies: HBsAg negative, anti-HBs positive, anti-HBc negative.

Answer

This pattern indicates immunity from vaccination rather than natural infection, since anti-HBc is absent. Natural immunity from prior resolved infection would show both anti-HBs and anti-HBc positive. No further workup or treatment is needed in this vaccinated pattern.

#170 · Gastroenterology

Interpretation of hepatitis B serologies: HBsAg positive for more than 6 months, elevated HBV DNA, elevated ALT.

Answer

This defines chronic hepatitis B infection with active viral replication and liver inflammation. HBeAg status and HBV DNA level help determine the phase of infection and need for antiviral therapy such as tenofovir or entecavir. These patients need ongoing monitoring for cirrhosis and hepatocellular carcinoma with periodic imaging.

#171 · Gastroenterology

Patient exposed to acute hepatitis A. What is the natural history and does it become chronic?

Answer

Hepatitis A causes acute illness with jaundice, malaise, and elevated transaminases, transmitted via the fecal-oral route, but it never causes chronic infection. Management is supportive since it is typically self-limited. Postexposure prophylaxis with vaccine or immune globulin can be given to close contacts depending on timing and risk.

#172 · Gastroenterology

Which viral hepatitis is most likely to become chronic, and what is the standard treatment approach today?

Answer

Hepatitis C has a high rate of chronicity, with the majority of acutely infected patients failing to clear the virus spontaneously. Direct acting antivirals, taken for 8 to 12 weeks, now cure over 95 percent of chronic hepatitis C infections regardless of genotype in most regimens. All patients with chronic hepatitis C should be evaluated for the degree of liver fibrosis before and monitored after treatment.

#173 · Gastroenterology

Middle aged woman with fatigue, elevated ALT and AST, positive ANA and anti-smooth muscle antibody, elevated IgG. Diagnosis and treatment?

Answer

This is autoimmune hepatitis, more common in women, characterized by these autoantibodies and hypergammaglobulinemia with interface hepatitis on biopsy. Treatment is immunosuppression, typically corticosteroids often combined with azathioprine. Liver biopsy is important to confirm the diagnosis and assess severity before starting treatment.

#174 · Gastroenterology

Patient with bronze skin, diabetes, and elevated ferritin and transferrin saturation. Diagnosis and confirmatory testing?

Answer

This suggests hereditary hemochromatosis, an autosomal recessive disorder of iron overload most often from HFE gene mutations. Elevated transferrin saturation above 45 percent is the best initial screening test, and HFE genetic testing confirms the diagnosis. Treatment is regular therapeutic phlebotomy to reduce iron stores and prevent cirrhosis, cardiomyopathy, and diabetes.

#175 · Gastroenterology

Young patient with liver disease, neuropsychiatric symptoms, and a Kayser-Fleischer ring on exam. Diagnosis and initial test?

Answer

This is Wilson disease, an autosomal recessive disorder of impaired copper excretion causing copper accumulation in the liver and brain. A low serum ceruloplasmin level supports the diagnosis, and 24 hour urinary copper is typically elevated. Treatment involves copper chelation with agents like penicillamine or trientine, or zinc to reduce copper absorption.

#176 · Gastroenterology

Obese patient with type 2 diabetes and incidentally elevated liver enzymes, ultrasound shows hepatic steatosis, and alcohol use is denied. Diagnosis and first line treatment?

Answer

This is nonalcoholic fatty liver disease, now often termed metabolic dysfunction associated steatotic liver disease. The first line treatment is weight loss through diet and exercise, since even a 7 to 10 percent body weight reduction can significantly improve steatosis and inflammation. Patients with risk factors for fibrosis should be assessed with noninvasive tools like the FIB-4 index or elastography to identify advanced disease.

#177 · Gastroenterology

Cirrhotic patient develops confusion, asterixis, and elevated ammonia. Diagnosis and treatment?

Answer

This is hepatic encephalopathy, caused by accumulation of gut derived toxins like ammonia due to impaired liver clearance and portosystemic shunting. First line treatment is lactulose to increase ammonia excretion through the stool, with rifaximin added for recurrent or refractory episodes. Precipitating factors such as infection, GI bleeding, constipation, or medication noncompliance should always be sought and corrected.

#178 · Gastroenterology

Cirrhotic patient presents with fever, abdominal pain, and worsening ascites. What test should be performed and what defines the diagnosis?

Answer

Diagnostic paracentesis should be performed to evaluate for spontaneous bacterial peritonitis. An ascitic fluid absolute neutrophil count of 250 cells per microliter or higher confirms the diagnosis. Treatment is empiric IV antibiotics such as a third generation cephalosporin, along with IV albumin in patients with significant renal dysfunction to reduce the risk of hepatorenal syndrome.

#179 · Gastroenterology

What does a serum-ascites albumin gradient (SAAG) of 1.1 g/dL or higher indicate about the cause of ascites?

Answer

A SAAG of 1.1 g/dL or greater indicates portal hypertension as the cause, as seen in cirrhosis, heart failure, or Budd-Chiari syndrome. A SAAG below 1.1 g/dL suggests a non-portal hypertensive cause such as malignancy, tuberculosis, or pancreatic disease. This gradient is far more useful than the older total protein based transudate versus exudate classification for ascites.

#180 · Gastroenterology

Cirrhotic patient develops rapidly rising creatinine without improvement despite fluid challenge and no other identifiable cause. Diagnosis and management principle?

Answer

This suggests hepatorenal syndrome, a functional renal failure from severe splanchnic vasodilation and renal vasoconstriction in advanced liver disease. Other causes of kidney injury such as hypovolemia, nephrotoxic drugs, and obstruction must be excluded first. Treatment involves albumin plus vasoconstrictors such as terlipressin or midodrine and octreotide, with liver transplantation being the definitive cure.

#181 · Gastroenterology

How should a patient with a suspected upper GI bleed be risk stratified before endoscopy?

Answer

The Glasgow-Blatchford score uses clinical and laboratory data such as blood urea nitrogen, hemoglobin, blood pressure, and heart rate to help decide whether a patient can be managed as an outpatient or needs hospitalization and urgent endoscopy. A very low score identifies patients at low risk who may not need admission. All patients with active bleeding, hemodynamic instability, or high risk features should be admitted for urgent endoscopy.

#182 · Gastroenterology

Patient with painless massive hematochezia, hemodynamically stable, in their 70s. Most likely cause and how does this differ from diverticulitis?

Answer

Diverticular bleeding is the most common cause of significant lower GI bleeding in older adults, arising from a vessel eroding at the neck of a diverticulum, and it is characteristically painless. This differs from diverticulitis, which presents with pain, fever, and inflammation without significant bleeding. Colonoscopy is used both diagnostically and often therapeutically once the patient is stabilized.

#183 · Gastroenterology

Alcoholic patient with forceful vomiting followed by hematemesis. What is the likely diagnosis and how is it distinguished from a Boerhaave tear?

Answer

This suggests a Mallory-Weiss tear, a mucosal laceration at the gastroesophageal junction from forceful retching that usually causes self-limited bleeding managed supportively or with endoscopic therapy. Boerhaave syndrome is a full thickness esophageal rupture presenting with severe chest pain, subcutaneous emphysema, and signs of mediastinitis, which is a surgical emergency. The key distinction is that Boerhaave involves transmural perforation while a Mallory-Weiss tear is only mucosal.

#184 · Gastroenterology

Patient on long term PPI therapy for GERD asks about risks of continued use. What should be discussed?

Answer

Long term PPI use has been associated with increased risk of Clostridioides difficile infection, community acquired pneumonia, bone fractures from reduced calcium absorption, and possible vitamin B12 or magnesium deficiency. These associations are generally modest and the benefit of treating significant reflux disease often outweighs the risk. The lowest effective dose should be used and periodic reassessment of ongoing need is reasonable.

#185 · Gastroenterology

Middle aged patient found to have a single small hyperplastic polyp versus a tubular adenoma on screening colonoscopy. Why does this distinction matter for surveillance timing?

Answer

Hyperplastic polyps, especially small ones in the rectosigmoid colon, carry minimal malignant potential and warrant standard interval screening similar to an average risk patient. Tubular adenomas are precancerous and their number, size, and degree of dysplasia determine how soon the next colonoscopy should occur, typically ranging from 3 to 10 years. Sessile serrated polyps and large or high grade adenomas require shorter surveillance intervals due to higher cancer risk.

#186 · Hematology

Microcytic anemia with low ferritin, high TIBC, and low iron saturation. Diagnosis and next step?

Answer

This is iron deficiency anemia. In a man or postmenopausal woman, the next step is colonoscopy and often upper endoscopy to look for a source of GI blood loss, since occult malignancy must be excluded. Ferritin is the most useful test but remember it is an acute phase reactant and can be falsely normal or elevated with inflammation.

#187 · Hematology

How do you distinguish anemia of chronic disease from iron deficiency anemia when ferritin is equivocal?

Answer

Check TIBC and transferrin saturation, or get a soluble transferrin receptor level. In anemia of chronic disease, ferritin is normal or high, TIBC is low or normal, and iron is low with low saturation because iron is sequestered, not absent. In iron deficiency, TIBC is high and transferrin saturation is very low.

#188 · Hematology

Young man develops hemolysis a few days after starting dapsone or primaquine, or after eating fava beans. Diagnosis?

Answer

This is G6PD deficiency, an X-linked enzyme defect that impairs the ability to handle oxidative stress. Peripheral smear shows bite cells and Heinz bodies. Enzyme levels should be checked after the acute hemolytic episode resolves, since levels can be falsely normal during hemolysis when older deficient cells have already been destroyed.

#189 · Hematology

Warm versus cold autoimmune hemolytic anemia: key differences?

Answer

Warm AIHA is IgG mediated, causes extravascular hemolysis in the spleen, and is often idiopathic or associated with lupus, CLL, or drugs like methyldopa. Cold agglutinin disease is IgM mediated, causes complement driven intravascular and extravascular hemolysis, and is linked to Mycoplasma pneumonia or infectious mononucleosis. Both show a positive direct antiglobulin test, but the type of antibody detected differs.

#190 · Hematology

Patient with hemolytic anemia, thrombocytopenia, and schistocytes on smear. What is the differential and initial workup?

Answer

This picture defines a microangiopathic hemolytic anemia, seen in TTP, HUS, DIC, malignant hypertension, and preeclampsia or HELLP syndrome. Check a coagulation panel, fibrinogen, LDH, haptoglobin, creatinine, and ADAMTS13 activity. TTP classically has normal coagulation studies with severely low ADAMTS13, while DIC shows prolonged PT and PTT with low fibrinogen.

#191 · Hematology

Pentad of TTP and what is the emergency treatment?

Answer

The classic pentad is microangiopathic hemolytic anemia, thrombocytopenia, renal dysfunction, neurologic symptoms, and fever, though not all features need be present to diagnose it. Treatment is emergent plasma exchange, not platelet transfusion, since giving platelets can worsen microthrombosis and precipitate fatal outcomes.

#192 · Hematology

How do you distinguish alpha thalassemia trait from iron deficiency anemia?

Answer

Both cause microcytic anemia, but thalassemia trait has a normal or high red cell count relative to the degree of microcytosis, normal iron studies, and a normal RDW. Iron deficiency has a low red cell count, low ferritin, and elevated RDW. Hemoglobin electrophoresis is usually normal in alpha thalassemia trait, so genetic testing may be needed to confirm.

#193 · Hematology

Beta thalassemia minor versus major: how do they present?

Answer

Beta thalassemia minor (trait) causes mild microcytic anemia with elevated hemoglobin A2 on electrophoresis, and patients are usually asymptomatic. Beta thalassemia major presents in infancy with severe anemia, failure to thrive, and transfusion dependence, leading to iron overload if untransfused patients survive or if chelation is inadequate.

#194 · Hematology

Sickle cell patient with sudden anemia and low reticulocyte count. What is the likely cause?

Answer

This suggests aplastic crisis, most commonly triggered by parvovirus B19 infection, which transiently shuts down erythropoiesis. Because sickle cells have a shortened lifespan, even brief marrow suppression causes rapid hemoglobin drop. This differs from splenic sequestration or hemolytic crisis, which show an elevated reticulocyte response.

#195 · Hematology

What triggers acute chest syndrome in sickle cell disease and how is it managed?

Answer

Acute chest syndrome is triggered by infection, fat embolism from bone marrow infarction, or pulmonary infarction, and presents with fever, chest pain, hypoxia, and a new infiltrate on chest x-ray. Management includes oxygen, empiric antibiotics, incentive spirometry, and exchange transfusion for severe or worsening hypoxia. It is a leading cause of death in sickle cell disease and requires prompt recognition.

#196 · Hematology

Isolated thrombocytopenia in an otherwise healthy adult with normal smear. What is the diagnosis and first-line treatment?

Answer

This is immune thrombocytopenic purpura, a diagnosis of exclusion once other causes like medications, infection, and pseudothrombocytopenia from EDTA clumping are ruled out. First-line treatment for symptomatic or severe thrombocytopenia is corticosteroids, with IVIG reserved for rapid platelet rise when bleeding is significant or surgery is urgent.

#197 · Hematology

Platelet count drops 5 to 10 days after starting heparin. What is the next step?

Answer

Suspect heparin induced thrombocytopenia, especially if the platelet count falls more than 50 percent from baseline. Stop all heparin products immediately, including flushes, and start a non-heparin anticoagulant such as argatroban or bivalirudin, since HIT is a prothrombotic state despite the low platelet count. Confirm with a PF4 antibody assay and, if positive, a serotonin release assay.

#198 · Hematology

Why should platelets not be transfused in HIT unless there is active severe bleeding?

Answer

Platelet transfusion in HIT can theoretically fuel further thrombosis since the pathology is antibody-mediated platelet activation, not simple platelet deficiency. The priority is anticoagulation with a non-heparin agent to prevent life-threatening arterial or venous thrombosis, which occurs in a substantial proportion of untreated patients.

#199 · Hematology

Young woman with recurrent miscarriages and a history of unprovoked DVT. What should you test for?

Answer

This presentation is classic for antiphospholipid syndrome. Test for lupus anticoagulant, anticardiolipin antibodies, and anti-beta-2 glycoprotein I antibodies, with two positive results at least 12 weeks apart required for diagnosis. Patients often have a prolonged PTT that does not correct with mixing studies, reflecting the lupus anticoagulant's paradoxical in vitro versus in vivo effect.

#200 · Hematology

How is antiphospholipid syndrome treated long term after a confirmed thrombotic event?

Answer

Standard treatment is lifelong warfarin, not a direct oral anticoagulant, since DOACs have shown inferior outcomes in antiphospholipid syndrome, particularly in triple-positive patients. Target INR is typically 2 to 3 for venous events, and some experts use a higher target for arterial or recurrent events.

#201 · Hematology

Elevated hemoglobin, low erythropoietin level, and splenomegaly. Likely diagnosis and confirmatory test?

Answer

This suggests polycythemia vera, a myeloproliferative neoplasm. Confirm with JAK2 V617F mutation testing, which is positive in the large majority of cases. Treatment includes therapeutic phlebotomy to keep hematocrit below 45 percent and low dose aspirin, with cytoreductive therapy like hydroxyurea for high risk patients.

#202 · Hematology

Massive splenomegaly, leukocytosis with a full spectrum of granulocyte precursors, and a specific chromosomal translocation. Diagnosis?

Answer

This describes chronic myeloid leukemia, driven by the BCR-ABL fusion gene from the t(9;22) Philadelphia chromosome. The peripheral smear shows the full myeloid maturation spectrum with basophilia. First-line treatment is a tyrosine kinase inhibitor like imatinib, which has transformed prognosis dramatically.

#203 · Hematology

Older adult with unexplained cytopenias, a hypercellular marrow, and dysplastic changes on smear. Diagnosis?

Answer

This is myelodysplastic syndrome, a clonal bone marrow disorder causing ineffective hematopoiesis despite a cellular or hypercellular marrow. Smear findings include hypogranular neutrophils, pseudo-Pelger-Huet cells, and macrocytosis. Risk of transformation to acute myeloid leukemia is stratified using the revised IPSS score based on cytogenetics, cytopenias, and blast percentage.

#204 · Hematology

Auer rods on peripheral smear point to what diagnosis?

Answer

Auer rods are needle-like cytoplasmic inclusions seen in acute myeloid leukemia, particularly abundant in acute promyelocytic leukemia. APL is associated with the t(15;17) translocation and carries a high risk of disseminated intravascular coagulation at presentation, so treatment with all-trans retinoic acid should be started urgently once suspected.

#205 · Hematology

Older adult with painless lymphadenopathy, fatigue, and smudge cells on peripheral smear. Diagnosis?

Answer

Smudge cells are characteristic of chronic lymphocytic leukemia, a clonal proliferation of mature but dysfunctional B lymphocytes. Many patients are asymptomatic at diagnosis and found incidentally on routine labs. Flow cytometry confirms the diagnosis by showing CD5, CD19, CD20, and CD23 positive B cells, and treatment is often deferred until symptomatic progression.

#206 · Hematology

Elderly patient with bone pain, anemia, renal insufficiency, and hypercalcemia. What is the diagnosis and initial workup?

Answer

This CRAB constellation, calcium elevation, renal failure, anemia, and bone lesions, is classic for multiple myeloma. Workup includes serum and urine protein electrophoresis with immunofixation, serum free light chains, and a skeletal survey or whole body MRI. Bone marrow biopsy showing more than 10 percent clonal plasma cells confirms the diagnosis.

#207 · Hematology

How is monoclonal gammopathy of undetermined significance distinguished from multiple myeloma?

Answer

MGUS has a monoclonal protein under 3 g/dL, less than 10 percent clonal plasma cells in the marrow, and no CRAB features of end organ damage. It requires no treatment but lifelong monitoring, since it progresses to myeloma at roughly 1 percent per year. Multiple myeloma requires meeting higher protein thresholds, more marrow involvement, or evidence of end organ damage.

#208 · Hematology

Young adult with painless cervical lymphadenopathy, night sweats, and Reed-Sternberg cells on biopsy. Diagnosis and typical spread pattern?

Answer

This is classic Hodgkin lymphoma. Reed-Sternberg cells are large binucleated cells with an owl-eye appearance and are typically CD15 and CD30 positive. Hodgkin lymphoma classically spreads in a contiguous fashion from one lymph node group to adjacent ones, unlike non-Hodgkin lymphoma, which spreads noncontiguously.

#209 · Hematology

What are B symptoms and why do they matter in lymphoma staging?

Answer

B symptoms are fever above 38 degrees Celsius, drenching night sweats, and unintentional weight loss exceeding 10 percent of body weight over 6 months. Their presence upgrades the Ann Arbor stage designation from A to B and generally indicates more aggressive disease, influencing prognosis and treatment intensity.

#210 · Hematology

A patient develops fever and chills within an hour of starting a red blood cell transfusion, without hemolysis on workup. Diagnosis and management?

Answer

This is a febrile nonhemolytic transfusion reaction, caused by cytokines released from donor white cells during storage. Stop or slow the transfusion, rule out hemolysis and bacterial contamination, and give antipyretics. It is the most common transfusion reaction and can be reduced in future transfusions with leukoreduced blood products.

#211 · Hematology

Patient develops fever, flank pain, and dark urine minutes after starting a blood transfusion. What is happening and what should be done immediately?

Answer

This is an acute hemolytic transfusion reaction, usually caused by ABO incompatibility from a clerical error. Stop the transfusion immediately, keep the IV line open with normal saline, and send the unit and patient blood back to the blood bank to recheck the crossmatch and direct antiglobulin test. This reaction can rapidly progress to DIC, shock, and acute kidney injury if not stopped promptly.

#212 · Hematology

Transfusion patient develops sudden dyspnea, hypoxia, and bilateral pulmonary infiltrates within 6 hours, with normal cardiac function. Diagnosis?

Answer

This is transfusion-related acute lung injury, thought to result from donor antileukocyte antibodies causing neutrophil activation in the lungs. It is a leading cause of transfusion-related death, distinguished from transfusion-associated circulatory overload by the absence of volume overload signs and a normal or low B-type natriuretic peptide. Management is supportive, similar to ARDS.

#213 · Hematology

How do you differentiate TRALI from TACO in a patient who develops respiratory distress after transfusion?

Answer

TACO presents with hypertension, signs of volume overload like jugular venous distension and an S3, and elevated BNP, occurring more often in patients with heart or renal failure who receive large or rapid transfusion volumes. TRALI presents with hypoxia and infiltrates without volume overload signs, normal or mildly elevated BNP, and often fever or hypotension. Treatment for TACO is diuresis, while TRALI is treated supportively without diuretics.

#214 · Hematology

Which patients require irradiated blood products and why?

Answer

Irradiated blood products are needed for severely immunocompromised patients, such as those with hematologic malignancies, stem cell transplant recipients, and patients receiving purine analog chemotherapy, to prevent transfusion-associated graft versus host disease. Irradiation inactivates donor lymphocytes that could otherwise engraft and attack the recipient's tissues, a nearly always fatal complication.

#215 · Hematology

Patient with IgA deficiency develops anaphylaxis with a blood transfusion. What is the mechanism and prevention strategy?

Answer

Patients with selective IgA deficiency can form anti-IgA antibodies, which react against IgA in donor plasma and trigger anaphylactic transfusion reactions even with small volumes. Prevention requires washed red blood cells or blood products from IgA deficient donors for future transfusions, and the patient should be tested and flagged in their medical record.

#216 · Infectious Disease

Diabetic patient with a rapidly spreading, exquisitely painful leg wound, bullae, and pain out of proportion to exam. What is the diagnosis and next step?

Answer

This is necrotizing fasciitis until proven otherwise. Pain out of proportion, crepitus, skin necrosis, and systemic toxicity are red flags. Emergent surgical debridement is the definitive treatment, along with broad spectrum antibiotics covering gram positive, gram negative, and anaerobic organisms. Imaging should never delay surgery in a clearly toxic patient.

#217 · Infectious Disease

How do you distinguish cellulitis from erysipelas?

Answer

Erysipelas involves the upper dermis and superficial lymphatics, producing a sharply demarcated, raised, bright red plaque, most classically on the face or lower leg. Cellulitis involves the deeper dermis and subcutaneous fat, so borders are less distinct and the area is less raised. Erysipelas is usually caused by Streptococcus pyogenes, while cellulitis has a broader range of causes including Staphylococcus aureus.

#218 · Infectious Disease

What organism should you suspect in cellulitis with purulent drainage or abscess formation, and how does treatment differ from nonpurulent cellulitis?

Answer

Purulent cellulitis and abscess are most often caused by Staphylococcus aureus, including community acquired MRSA. Treatment should include incision and drainage of any abscess plus empiric MRSA coverage such as trimethoprim sulfamethoxazole, doxycycline, or clindamycin. Nonpurulent cellulitis is more often streptococcal and can typically be treated with a beta lactam like cephalexin.

#219 · Infectious Disease

A patient stepped on a rusty nail through a tennis shoe and now has a deep foot puncture wound with cellulitis and osteomyelitis. What organism is classically implicated?

Answer

Pseudomonas aeruginosa is classically associated with puncture wounds through the sole of a shoe, because the warm moist environment inside the shoe favors this organism. Empiric antibiotic coverage should include an antipseudomonal agent, and imaging or bone biopsy should be considered to evaluate for osteomyelitis.

#220 · Infectious Disease

What is the classic triad of bacterial meningitis, and how sensitive is it?

Answer

The classic triad is fever, nuchal rigidity, and altered mental status. Unfortunately the full triad is present in only about two thirds of cases, so its absence does not rule out meningitis. Headache and photophobia are also common, and Kernig and Brudzinski signs support the diagnosis but have low sensitivity.

#221 · Infectious Disease

When should a head CT be obtained before lumbar puncture in suspected bacterial meningitis?

Answer

CT before LP is recommended if there are risk factors for mass effect or herniation, including immunocompromise, history of CNS disease, new seizure, papilledema, altered consciousness, or a focal neurologic deficit. If none of these are present, LP can proceed without CT. Regardless of CT timing, empiric antibiotics and dexamethasone should be started immediately and not delayed for imaging.

#222 · Infectious Disease

What is the empiric antibiotic regimen for suspected bacterial meningitis in an immunocompetent adult under 50?

Answer

Vancomycin plus a third generation cephalosporin such as ceftriaxone covers the most likely pathogens, Streptococcus pneumoniae and Neisseria meningitidis. Dexamethasone should be given with or just before the first antibiotic dose, since it reduces neurologic complications and mortality, particularly in pneumococcal meningitis. In patients over 50 or with impaired cellular immunity, ampicillin is added to cover Listeria monocytogenes.

#223 · Infectious Disease

What CSF findings distinguish bacterial from viral meningitis?

Answer

Bacterial meningitis typically shows a high neutrophil predominant white count, low glucose relative to serum, and elevated protein. Viral meningitis usually shows a lymphocyte predominant white count, normal glucose, and mildly elevated protein. CSF glucose that is very low compared to serum glucose is a strong clue for bacterial infection.

#224 · Infectious Disease

A patient with fever, altered mental status, and temporal lobe changes on MRI. What is the most likely diagnosis and treatment?

Answer

This presentation is classic for herpes simplex virus encephalitis, which has a predilection for the temporal lobes. Empiric intravenous acyclovir should be started immediately while awaiting CSF HSV PCR, since delayed treatment significantly worsens outcomes. Untreated HSV encephalitis carries very high mortality and morbidity.

#225 · Infectious Disease

What is the typical CURB-65 score interpretation for community acquired pneumonia?

Answer

CURB-65 scores one point each for confusion, urea (BUN) over 19, respiratory rate 30 or higher, low blood pressure, and age 65 or older. A score of 0 to 1 suggests outpatient treatment is appropriate, 2 suggests consideration of admission, and 3 or higher suggests severe pneumonia often warranting ICU level care. It helps guide site of care decisions alongside clinical judgment.

#226 · Infectious Disease

What is the standard empiric outpatient treatment for community acquired pneumonia in a previously healthy adult with no comorbidities?

Answer

Amoxicillin is now preferred first line for healthy outpatients without comorbidities, per updated guidelines, since it targets Streptococcus pneumoniae well and preserves macrolides. A macrolide like azithromycin or doxycycline can be used as an alternative if local pneumococcal resistance is low. Patients with comorbidities need combination therapy with a beta lactam plus a macrolide, or monotherapy with a respiratory fluoroquinolone.

#227 · Infectious Disease

What defines a positive PPD (tuberculin skin test) at 5 mm, 10 mm, and 15 mm induration cutoffs?

Answer

A 5 mm cutoff applies to high risk groups including HIV positive patients, recent close contacts of active TB, patients with fibrotic changes on chest x-ray consistent with prior TB, organ transplant recipients, and other immunosuppressed patients. A 10 mm cutoff applies to recent immigrants from high prevalence countries, IV drug users, residents or employees of high risk congregate settings, and those with certain comorbidities like diabetes or silicosis. A 15 mm cutoff is used for people with no known risk factors.

#228 · Infectious Disease

A patient has a positive PPD or IGRA but a normal chest x-ray and no symptoms. What is the diagnosis and treatment?

Answer

This represents latent tuberculosis infection, meaning the patient is infected but not contagious and has no active disease. Treatment options include four months of daily rifampin, three months of weekly isoniazid plus rifapentine, or nine months of daily isoniazid with pyridoxine. Active TB must be excluded clinically and radiographically before starting latent TB treatment.

#229 · Infectious Disease

What is the standard four drug regimen for active pulmonary tuberculosis and its typical duration?

Answer

The standard regimen is RIPE: rifampin, isoniazid, pyrazinamide, and ethambutol, given daily for two months. This is followed by a continuation phase of rifampin and isoniazid for an additional four months, for a total of six months in drug susceptible disease. Pyridoxine (vitamin B6) is given with isoniazid to prevent peripheral neuropathy, and liver function should be monitored.

#230 · Infectious Disease

What are the modified Duke criteria used for, and what are the major criteria?

Answer

The modified Duke criteria are used to diagnose infective endocarditis based on clinical, microbiologic, and echocardiographic findings. The two major criteria are positive blood cultures with a typical endocarditis organism from separate cultures, and evidence of endocardial involvement on echocardiography such as vegetation, abscess, or new valvular regurgitation. Diagnosis requires either two major criteria, one major plus three minor, or five minor criteria.

#231 · Infectious Disease

An IV drug user presents with fever and a new murmur. What valve is most commonly affected and what organism is most likely?

Answer

In IV drug users, the tricuspid valve is most commonly affected because injected organisms first reach the right side of the heart. Staphylococcus aureus is the most common causative organism in this population, often presenting with septic pulmonary emboli causing cavitary lung lesions. Right sided endocarditis can have a softer or absent murmur despite significant valve involvement.

#232 · Infectious Disease

A patient with endocarditis has tender nodules on finger and toe pads, painless hemorrhagic macules on palms and soles, and splinter hemorrhages. What are these findings called?

Answer

Tender nodules on the pads of fingers and toes are Osler nodes, thought to be immune complex mediated. Painless hemorrhagic macules on the palms and soles are Janeway lesions, caused by septic microemboli. Splinter hemorrhages under the nails and Roth spots on fundoscopic exam are additional classic peripheral stigmata of infective endocarditis.

#233 · Infectious Disease

Streptococcus bovis (Streptococcus gallolyticus) bacteremia or endocarditis should prompt what additional workup?

Answer

Streptococcus bovis bacteremia is strongly associated with colonic neoplasia, particularly colorectal cancer. Any patient with S. bovis bacteremia or endocarditis should undergo colonoscopy to evaluate for colon cancer or polyps. This association is a classic board testing point linking an infectious finding to an underlying malignancy workup.

#234 · Infectious Disease

What CD4 count thresholds are associated with specific opportunistic infections in HIV?

Answer

Below 200, patients are at risk for Pneumocystis jirovecii pneumonia, and prophylaxis with trimethoprim sulfamethoxazole should be started. Below 100, toxoplasmosis and cryptococcal meningitis become concerns, with toxoplasmosis prophylaxis added at this threshold. Below 50, disseminated Mycobacterium avium complex (MAC) becomes a risk and azithromycin prophylaxis is considered, along with CMV retinitis risk.

#235 · Infectious Disease

What is the recommended approach to antiretroviral therapy timing after a new HIV diagnosis?

Answer

Current guidelines recommend starting antiretroviral therapy as soon as possible after diagnosis, ideally the same day or within days, regardless of CD4 count. Rapid initiation improves viral suppression rates and reduces transmission risk. Delaying therapy is no longer standard practice except in specific circumstances like active opportunistic infection requiring treatment sequencing, such as cryptococcal meningitis.

#236 · Infectious Disease

How do you diagnose acute HIV infection versus chronic HIV infection serologically?

Answer

Acute HIV infection often occurs during the window period when antibody tests are still negative but the patient is highly viremic, so a fourth generation antigen antibody test or HIV RNA viral load is needed to detect it. Chronic infection is confirmed with a positive fourth generation screening test followed by an HIV-1/HIV-2 antibody differentiation assay. Acute HIV can present with a mononucleosis like illness, so it should be considered in patients with fever, rash, and lymphadenopathy with recent risk exposure.

#237 · Infectious Disease

What is the treatment of choice for primary and secondary syphilis?

Answer

A single intramuscular dose of benzathine penicillin G is the treatment of choice for primary, secondary, and early latent syphilis. Late latent syphilis or syphilis of unknown duration requires three weekly doses. Doxycycline is an alternative for penicillin allergic patients, but penicillin desensitization is preferred in pregnancy since it is the only proven safe and effective treatment.

#238 · Infectious Disease

What is a Jarisch-Herxheimer reaction and when does it occur?

Answer

The Jarisch-Herxheimer reaction is an acute febrile illness with headache, myalgia, and chills that occurs within hours after starting antibiotic treatment for syphilis or other spirochetal infections. It results from the release of inflammatory cytokines as spirochetes are rapidly killed. It is self-limited and treated with supportive care such as antipyretics, and it should not be mistaken for a drug allergy.

#239 · Infectious Disease

What is the recommended empiric treatment for gonorrhea given rising resistance concerns?

Answer

Current guidelines recommend a single dose of intramuscular ceftriaxone for uncomplicated gonorrhea, with the dose increased to 500 mg (or 1 g in patients over 150 kg) due to emerging resistance. Routine dual therapy with azithromycin is no longer recommended unless chlamydia has not been excluded. Patients should also be tested and treated for chlamydia if coinfection is not ruled out, and sexual partners should be treated.

#240 · Infectious Disease

A young sexually active woman has dysuria, frequency, and urgency without fever or flank pain. What is the likely diagnosis and first line treatment?

Answer

This presentation is consistent with uncomplicated cystitis, most commonly caused by Escherichia coli. First line treatment options include nitrofurantoin for five days, trimethoprim sulfamethoxazole for three days if local resistance is low, or fosfomycin as a single dose. Fluoroquinolones are generally reserved for more complicated cases due to resistance and side effect concerns.

#241 · Infectious Disease

How does pyelonephritis differ clinically from simple cystitis, and how is it treated?

Answer

Pyelonephritis presents with fever, flank pain, and costovertebral angle tenderness in addition to lower urinary tract symptoms, reflecting upper tract or kidney involvement. Outpatient treatment for mild to moderate cases can use oral fluoroquinolones or trimethoprim sulfamethoxazole based on susceptibility, while hospitalized patients often need intravenous ceftriaxone or a fluoroquinolone. Urine culture should always be obtained to guide therapy and rule out complications like abscess in nonresponders.

#242 · Infectious Disease

Should asymptomatic bacteriuria be treated? What are the exceptions?

Answer

Asymptomatic bacteriuria should generally not be treated because treatment does not improve outcomes and promotes resistance. The main exceptions are pregnant women, because untreated bacteriuria increases risk of pyelonephritis and preterm birth, and patients undergoing an invasive urologic procedure that may cause mucosal bleeding. Catheterized patients and elderly patients with bacteriuria but no symptoms should not be treated.

#243 · Infectious Disease

What is the most common cause of septic arthritis in a sexually active young adult, and how does it typically present?

Answer

Neisseria gonorrhoeae is the most common cause of septic arthritis in sexually active young adults and often presents as disseminated gonococcal infection. This can appear as a migratory polyarthralgia with tenosynovitis and pustular skin lesions, or as a purulent monoarticular arthritis. Synovial fluid, blood, and mucosal site cultures (cervix, urethra, rectum, pharynx) should all be obtained since synovial fluid culture is often negative.

#244 · Infectious Disease

A patient has an acutely swollen, warm, painful joint with fever. What is the next best diagnostic step and why?

Answer

Arthrocentesis with synovial fluid analysis, gram stain, and culture is the essential next step, because septic arthritis is a joint threatening emergency that requires prompt diagnosis and drainage. Synovial white blood cell counts above roughly 50,000 with neutrophil predominance strongly suggest bacterial infection. Imaging should not delay arthrocentesis, and empiric antibiotics should be started after cultures are obtained.

#245 · Infectious Disease

What is the gold standard for diagnosing osteomyelitis, and what imaging is most sensitive?

Answer

Bone biopsy with culture and histopathology is the gold standard for diagnosing osteomyelitis and identifying the causative organism. MRI is the most sensitive imaging modality and can detect marrow edema and abscess earlier than plain films or CT. Plain radiographs often appear normal early in the disease course, with changes typically not visible until at least ten to fourteen days after onset.

#246 · Infectious Disease

A diabetic patient has a chronic foot ulcer, and a probe reaches bone. What does this finding suggest and what organism is most common?

Answer

A positive probe to bone test strongly suggests underlying osteomyelitis and warrants further workup with imaging and possibly bone biopsy. Staphylococcus aureus is the most common causative organism overall, though diabetic foot infections are often polymicrobial including gram negative rods and anaerobes. Treatment typically requires prolonged antibiotics, and surgical debridement or resection may be needed depending on severity.

#247 · Infectious Disease

What are the current Sepsis-3 definitions of sepsis and septic shock?

Answer

Sepsis-3 defines sepsis as life threatening organ dysfunction caused by a dysregulated host response to infection, identified by an acute increase in SOFA score of 2 or more points. Septic shock is a subset of sepsis with persistent hypotension requiring vasopressors to maintain a mean arterial pressure of 65 mmHg or higher, plus a serum lactate greater than 2 mmol/L despite adequate fluid resuscitation. The older SIRS criteria (temperature, heart rate, respiratory rate, white count abnormalities) are no longer central to the sepsis definition but are still used clinically to flag possible infection.

#248 · Infectious Disease

What are the key components of the sepsis one hour bundle?

Answer

The bundle includes measuring lactate and remeasuring if initially elevated, obtaining blood cultures before giving antibiotics, administering broad spectrum antibiotics, giving 30 mL per kg of crystalloid fluid for hypotension or lactate 4 or higher, and starting vasopressors if the patient remains hypotensive during or after fluid resuscitation to maintain MAP of at least 65. Rapid initiation of antibiotics within the first hour is associated with improved survival. Source control, such as draining an abscess or removing an infected line, is also essential.

#249 · Infectious Disease

What is the first line vasopressor in septic shock, and what is added if the patient remains hypotensive?

Answer

Norepinephrine is the first line vasopressor in septic shock because it provides strong alpha adrenergic vasoconstriction with less tachycardia than other agents. If blood pressure remains inadequate despite norepinephrine, vasopressin can be added as a second agent, and epinephrine is another option. Corticosteroids like hydrocortisone can be considered in refractory shock despite adequate fluids and vasopressors.

#250 · Infectious Disease

What defines a central line-associated bloodstream infection (CLABSI), and how is it prevented?

Answer

CLABSI is a laboratory confirmed bloodstream infection occurring in a patient with a central line in place for more than two calendar days that is not related to infection at another site. Prevention relies on a bundle of practices including hand hygiene, maximal sterile barrier precautions during insertion, chlorhexidine skin antisepsis, optimal catheter site selection, and daily review of line necessity with prompt removal when no longer needed. These bundles have significantly reduced CLABSI rates in hospitals.

#251 · Infectious Disease

A hospitalized patient develops watery diarrhea after a course of antibiotics. What is the likely cause and first line treatment?

Answer

This presentation suggests Clostridioides difficile infection, which should be confirmed with a stool PCR or toxin assay. First line treatment for an initial episode is oral vancomycin or fidaxomicin; metronidazole is now reserved for situations where these are unavailable. Contact precautions should be used, and alcohol based hand sanitizer is not effective against C. difficile spores, so soap and water hand washing is required.

#252 · Infectious Disease

What isolation precautions are used for suspected tuberculosis, and what personal protective equipment is required?

Answer

Suspected or confirmed pulmonary tuberculosis requires airborne precautions, which include placement in a negative pressure room with the door closed. Healthcare workers must wear a fit tested N95 respirator or higher level respiratory protection when entering the room. Other classic airborne precaution pathogens include measles and varicella.

#253 · Infectious Disease

Which vaccines are live attenuated and generally contraindicated in significantly immunocompromised patients?

Answer

Live attenuated vaccines include MMR (measles mumps rubella), varicella, zoster (Zostavax, the live version), intranasal influenza (LAIV), yellow fever, and oral typhoid. These are generally avoided in patients with significant immunosuppression, including advanced HIV with low CD4 counts, active chemotherapy, or high dose corticosteroids. The recombinant zoster vaccine (Shingrix) is not live and is preferred in immunocompromised patients eligible for shingles vaccination.

#254 · Infectious Disease

What is the recommended pneumococcal vaccination approach for adults 65 and older?

Answer

Adults 65 and older who have not previously received a pneumococcal conjugate vaccine should receive a single dose of PCV15 or PCV20. If PCV15 is used, it should be followed by PPSV23 at least one year later. If PCV20 is used, no further pneumococcal vaccine is needed. Younger adults with certain high risk conditions may also qualify earlier.

#255 · Infectious Disease

Match the organism to the classic exposure: cat scratch, undercooked pork, unpasteurized dairy, and freshwater exposure with a skin wound.

Answer

Cat scratch disease is caused by Bartonella henselae, transmitted through a cat scratch or bite. Undercooked pork is associated with Trichinella spiralis and Taenia solium. Unpasteurized dairy is classically linked to Brucella and Listeria monocytogenes. Freshwater exposure with a skin wound raises concern for Aeromonas hydrophila, while saltwater exposure raises concern for Vibrio vulnificus.

#256 · Infectious Disease

A patient recently returned from camping with fever, headache, and a rash starting on the wrists and ankles that spreads centrally. What is the diagnosis and treatment?

Answer

This presentation is classic for Rocky Mountain spotted fever, caused by Rickettsia rickettsii and transmitted by tick bite. The rash characteristically begins peripherally on wrists and ankles before spreading centrally to the trunk. Doxycycline should be started empirically based on clinical suspicion without waiting for confirmatory testing, since delayed treatment significantly increases mortality, and this holds true even in children and pregnant patients.

#257 · Infectious Disease

What is the definition of fever of unknown origin (FUO), and what are the major diagnostic categories to consider?

Answer

Classic FUO is defined as a temperature above 38.3 C on multiple occasions, lasting more than three weeks, with no diagnosis after at least one week of appropriate inpatient or outpatient evaluation. The major categories to consider are infections, malignancies particularly lymphoma, connective tissue and autoimmune diseases, and miscellaneous causes such as drug fever. A thorough history, physical exam, and repeated basic labs and imaging often uncover the etiology before invasive workup is needed.

#258 · Infectious Disease

What historical clues suggest drug fever as a cause of unexplained fever?

Answer

Drug fever should be suspected when a patient looks relatively well despite a high fever, often described as a relative disconnect between how sick the patient appears and how high the temperature is. It typically develops one to two weeks after starting the offending medication and resolves within days of stopping it. Eosinophilia and rash may be present, and common culprits include beta lactam antibiotics, sulfonamides, and antiepileptics.

#259 · Infectious Disease

What are the key differences in empiric pneumonia coverage between community acquired pneumonia and hospital acquired or ventilator associated pneumonia?

Answer

Community acquired pneumonia empiric therapy targets typical organisms like Streptococcus pneumoniae and atypicals like Mycoplasma, using agents such as amoxicillin, a macrolide, or a respiratory fluoroquinolone. Hospital acquired and ventilator associated pneumonia require broader coverage for resistant gram negative organisms including Pseudomonas, and often MRSA, using agents like piperacillin tazobactam or cefepime plus vancomycin. The choice depends on risk factors for multidrug resistant organisms, including recent hospitalization, prior antibiotic use, and local resistance patterns.

#260 · Infectious Disease

A patient with a prosthetic heart valve needs antibiotic prophylaxis before which type of procedure, and why only that category?

Answer

Antibiotic prophylaxis for endocarditis is now recommended only before certain dental procedures involving manipulation of gingival tissue or the periapical region of teeth, or perforation of oral mucosa, in high risk patients like those with prosthetic valves, prior endocarditis, or certain congenital heart disease. Prophylaxis is no longer routinely recommended for GI or GU procedures because evidence did not show clear benefit. A single dose of amoxicillin before the procedure is the standard regimen, with clindamycin or a cephalosporin used for penicillin allergic patients.

#261 · Miscellaneous

How do you define sensitivity and specificity, and which one helps you rule out versus rule in a disease?

Answer

Sensitivity is the proportion of people with the disease who test positive, and a highly sensitive test with a negative result helps rule out disease (SnNout). Specificity is the proportion of people without the disease who test negative, and a highly specific test with a positive result helps rule in disease (SpPin). Neither value changes with disease prevalence, unlike predictive values.

#262 · Miscellaneous

A test has sensitivity 90% and specificity 80%. Calculate the positive and negative likelihood ratios.

Answer

LR positive equals sensitivity divided by (1 minus specificity), so 0.9 divided by 0.2 equals 4.5. LR negative equals (1 minus sensitivity) divided by specificity, so 0.1 divided by 0.8 equals 0.125. An LR positive above 10 or LR negative below 0.1 usually produces a large, clinically useful shift in disease probability.

#263 · Miscellaneous

A drug reduces a bad outcome from 40% in placebo to 30% in treated patients. What is the number needed to treat?

Answer

The absolute risk reduction is 40% minus 30%, which is 10 percentage points, or 0.10. Number needed to treat is 1 divided by the absolute risk reduction, so 1 divided by 0.10 equals 10. You would need to treat 10 patients with this drug to prevent one additional bad outcome.

#264 · Miscellaneous

What is the difference between a cohort study and a case control study, and which is better for rare diseases?

Answer

A cohort study starts with exposed and unexposed groups and follows them forward to see who develops disease, allowing calculation of relative risk and incidence. A case control study starts with people who already have disease versus controls and looks backward for exposures, calculating an odds ratio instead. Case control studies are more efficient and preferred for rare diseases because you do not need to wait years for enough cases to accumulate.

#265 · Miscellaneous

What is recall bias, and in which type of study is it most problematic?

Answer

Recall bias occurs when people with a disease remember and report past exposures differently, usually more thoroughly, than healthy controls. This is a major limitation of case control and other retrospective studies since they rely on participants recalling exposure history. Prospective cohort studies avoid this problem because exposure is recorded before the outcome occurs.

#266 · Miscellaneous

What are the key elements required for valid informed consent?

Answer

The patient needs disclosure of the diagnosis, the nature and purpose of the proposed treatment, its risks and benefits, and reasonable alternatives including no treatment. The patient must also have decision making capacity and must give consent voluntarily, free from coercion. Consent can be withdrawn at any time even after it was initially given.

#267 · Miscellaneous

How is decision making capacity assessed, and who determines it versus legal competence?

Answer

Capacity requires that a patient can understand the relevant information, appreciate how it applies to their own situation, reason through the options, and communicate a consistent choice. Any treating physician can assess and document capacity, and it is decision specific, meaning a patient can have capacity for simple choices but not complex ones. Competence is a global legal determination made only by a court, not by a physician.

#268 · Miscellaneous

What is the Tarasoff duty, and when can a physician break confidentiality to warn a third party?

Answer

The Tarasoff duty says a clinician must take reasonable steps to protect an identifiable third party when a patient makes a credible, specific threat of serious harm against them. This overrides usual confidentiality because the duty to protect an identifiable victim outweighs the patient's privacy. Other recognized exceptions to confidentiality include suspected child or elder abuse and certain reportable infectious diseases, which must be reported regardless of patient consent.

#269 · Miscellaneous

If a patient loses decision making capacity and has no advance directive or documented surrogate, who typically makes medical decisions, in what order?

Answer

Most states follow a default surrogate hierarchy, usually starting with a court appointed guardian if one exists, then spouse, then adult children, then parents, then adult siblings, then other relatives or close friends. A durable power of attorney for health care or a written advance directive naming a specific proxy always takes priority over this default list. If no surrogate is available and decisions are urgent, the case may need ethics committee or court involvement.

#270 · Miscellaneous

What is the Swiss cheese model of medical error, and what is the recommended approach to disclosing an error to a patient?

Answer

The Swiss cheese model depicts a health system as multiple layers of defense, each with random holes representing weaknesses, and a serious error occurs only when the holes in several layers line up at once. This framework emphasizes that errors usually result from system failures rather than a single person's fault, which is why root cause analysis focuses on process rather than blame. When an error reaches a patient, current best practice is prompt, honest, and transparent disclosure, including an apology and explanation of what happened and what is being done to prevent recurrence.

#271 · Nephrology and Urology

How does KDIGO define acute kidney injury?

Answer

AKI is defined by any of the following: a rise in serum creatinine of at least 0.3 mg/dL within 48 hours, a rise in creatinine to at least 1.5 times baseline known or presumed to have occurred within the prior 7 days, or urine output less than 0.5 mL/kg/hr for 6 hours. The definition is staged 1 to 3 based on how far creatinine rises or how long oliguria persists. Staging matters for prognosis and for deciding when to involve nephrology or consider dialysis.

#272 · Nephrology and Urology

A patient has AKI with a FeNa of 0.4%. What does this suggest and how is FeNa calculated?

Answer

FeNa is (urine Na x plasma creatinine) divided by (plasma Na x urine creatinine), multiplied by 100. A FeNa under 1% suggests a prerenal cause, meaning the tubules are still avidly reabsorbing sodium because renal perfusion is low but tubular function is intact. Common causes include volume depletion, heart failure, and cirrhosis. FeNa loses reliability if the patient is on diuretics, since diuretics force natriuresis regardless of the underlying cause.

#273 · Nephrology and Urology

What steps reduce the risk of contrast-induced nephropathy before a contrast CT in a patient with CKD?

Answer

The main proven strategy is IV isotonic fluid before and after the study, and using the lowest volume of contrast that still gives a diagnostic study. Hold nephrotoxic drugs like NSAIDs around the time of the scan when possible. Oral N-acetylcysteine and sodium bicarbonate infusions have not shown consistent benefit in trials and are not required. If an alternative imaging modality without iodinated contrast can answer the clinical question, that is often preferred in high-risk patients.

#274 · Nephrology and Urology

How is chronic kidney disease staged by GFR?

Answer

CKD is defined as kidney damage or GFR under 60 mL/min/1.73m2 for at least 3 months. G1 is GFR 90 or above with evidence of kidney damage, G2 is 60-89, G3a is 45-59, G3b is 30-44, G4 is 15-29, and G5 is under 15 or dialysis-dependent. Staging is combined with an albuminuria category (A1, A2, A3) to estimate overall risk, since two patients with the same GFR can have very different prognosis depending on proteinuria.

#275 · Nephrology and Urology

What are the key features of CKD-mineral and bone disorder, and how is it managed?

Answer

As GFR falls, phosphate retention rises, activated vitamin D production falls, and calcium tends to drop, all of which drive secondary hyperparathyroidism and renal osteodystrophy. Management includes dietary phosphate restriction, phosphate binders taken with meals, and active vitamin D analogs or calcimimetics like cinacalcet if PTH stays elevated. Labs to trend are phosphate, calcium, PTH, and sometimes alkaline phosphatase, generally starting surveillance around CKD stage 3.

#276 · Nephrology and Urology

A lupus patient has a kidney biopsy showing diffuse proliferative glomerulonephritis. What ISN/RPS class is this and how is it treated?

Answer

Diffuse proliferative lupus nephritis is Class IV, the most aggressive and common severe form, involving 50% or more of glomeruli. It typically presents with nephritic sediment, hypertension, and often nephrotic-range proteinuria, with low complement and high anti-dsDNA titers. Induction therapy is high-dose corticosteroids plus mycophenolate mofetil or cyclophosphamide, followed by a maintenance immunosuppressant to prevent relapse and preserve kidney function.

#277 · Nephrology and Urology

What is the classic clinical association and biopsy finding of focal segmental glomerulosclerosis?

Answer

FSGS is a common cause of nephrotic syndrome in adults, especially Black patients, and is also linked to obesity, heroin use, HIV, and APOL1 gene variants. Biopsy shows segmental sclerosis in some but not all glomeruli, with effacement of podocyte foot processes on electron microscopy. Primary FSGS often responds partially to corticosteroids, while secondary forms from adaptive hyperfiltration respond better to treating the underlying cause and using an ACE inhibitor or ARB to reduce intraglomerular pressure.

#278 · Nephrology and Urology

A middle-aged adult presents with nephrotic syndrome and positive anti-PLA2R antibodies. What is the diagnosis and what should you screen for?

Answer

This is membranous nephropathy, the most common cause of primary nephrotic syndrome in older non-diabetic adults. Anti-PLA2R antibodies are found in most primary cases and correlate with disease activity. When PLA2R is negative, screen for secondary causes including solid tumors, hepatitis B, lupus, and NSAID use, since membranous nephropathy can be a paraneoplastic presentation, especially in patients over 60.

#279 · Nephrology and Urology

A young adult develops gross hematuria 1-2 days after a upper respiratory infection. What is the likely diagnosis and how does it differ from post-streptococcal GN?

Answer

This synpharyngitic pattern of hematuria coinciding closely with a respiratory illness is classic for IgA nephropathy, the most common glomerulonephritis worldwide. In contrast, post-streptococcal GN has a latent period of 1-3 weeks after the infection before hematuria appears, and it typically shows low complement levels that IgA nephropathy does not. IgA nephropathy is confirmed by biopsy showing IgA deposits in the mesangium and is managed with ACE inhibitor or ARB therapy for proteinuria.

#280 · Nephrology and Urology

How do you distinguish nephrotic syndrome from nephritic syndrome clinically?

Answer

Nephrotic syndrome features heavy proteinuria over 3.5 g/day, hypoalbuminemia, edema, and hyperlipidemia, with a relatively bland urine sediment showing few cells. Nephritic syndrome features hematuria with dysmorphic red cells and red cell casts, milder proteinuria, hypertension, and often a reduced GFR, reflecting active glomerular inflammation. Some diseases like lupus nephritis or membranoproliferative GN can show a mixed nephritic-nephrotic picture.

#281 · Nephrology and Urology

What is first-line treatment for uncomplicated cystitis in a healthy non-pregnant woman?

Answer

First-line options are nitrofurantoin for 5 days, TMP-SMX for 3 days if local resistance is under 20%, or fosfomycin as a single dose. Fluoroquinolones are reserved for more complicated cases because of resistance concerns and side effect risk. Nitrofurantoin should be avoided if there is suspicion for early pyelonephritis since it does not achieve adequate tissue levels outside the bladder.

#282 · Nephrology and Urology

When should asymptomatic bacteriuria be treated?

Answer

In general, asymptomatic bacteriuria should not be treated because treatment does not improve outcomes and promotes resistance. The two well-established exceptions are pregnancy, where untreated bacteriuria raises the risk of pyelonephritis and preterm labor, and before an invasive urologic procedure that is expected to cause mucosal bleeding, such as TURP. Diabetes, catheterization, and older age alone are not indications to treat.

#283 · Nephrology and Urology

A woman has 3 or more UTIs per year. What prevention strategies are available?

Answer

Options include low-dose daily prophylactic antibiotics, post-coital antibiotic prophylaxis if UTIs are related to intercourse, and non-antibiotic measures such as vaginal estrogen in postmenopausal women, which restores protective vaginal flora. Cranberry products have weak and inconsistent evidence. Behavioral advice like voiding after intercourse and adequate hydration is reasonable though evidence is limited.

#284 · Nephrology and Urology

How is uncomplicated acute pyelonephritis treated in an outpatient?

Answer

A fluoroquinolone such as ciprofloxacin for 5-7 days is a common oral choice if local resistance is low, or an oral cephalosporin or TMP-SMX with a longer course if susceptibility is confirmed. Patients who are hemodynamically unstable, pregnant, vomiting, or have complicating factors like obstruction need admission for IV antibiotics. Blood and urine cultures should be obtained before starting therapy in anyone sick enough to be considered for admission.

#285 · Nephrology and Urology

What is the most common type of kidney stone and what metabolic workup is done after a first stone?

Answer

Calcium oxalate stones are the most common type overall. After a first uncomplicated stone, a basic evaluation includes urinalysis, basic metabolic panel, and stone analysis if the stone is recovered; a full 24-hour urine metabolic workup for volume, calcium, oxalate, citrate, and uric acid is generally reserved for recurrent stone formers or high-risk patients. Increasing fluid intake to produce over 2.5 liters of urine per day is the single most effective universal prevention measure.

#286 · Nephrology and Urology

What causes struvite kidney stones and why are they concerning?

Answer

Struvite stones form from urease-producing organisms like Proteus, Klebsiella, and some Staphylococcus species, which split urea into ammonia and alkalinize the urine, precipitating magnesium ammonium phosphate crystals. They tend to grow into large staghorn calculi that fill the renal pelvis and predispose to recurrent infection and abscess. Treatment usually requires surgical removal of the stone because antibiotics alone cannot eradicate the infection while the stone remains.

#287 · Nephrology and Urology

A patient with gout and chronic diarrhea develops recurrent radiolucent kidney stones. What type of stone is this and how is it prevented?

Answer

This presentation is classic for uric acid stones, which form in persistently acidic urine and are radiolucent on plain X-ray, though visible on CT and ultrasound. Risk factors include gout, high purine intake, obesity, and conditions causing low urine volume and acidic urine such as chronic diarrhea. Prevention centers on urinary alkalinization with potassium citrate to raise urine pH above 6, along with hydration and allopurinol if uric acid production is high.

#288 · Nephrology and Urology

A patient presents with severe colicky flank pain radiating to the groin. What is the initial approach to pain control and when is urgent urology consultation needed?

Answer

NSAIDs are first-line for renal colic and are as effective as opioids with fewer side effects, and can be combined with opioids for severe pain. A tamsulosin course can help facilitate passage of stones up to about 10 mm, known as medical expulsive therapy. Urgent urology consultation is needed for an obstructing stone with signs of infection, solitary kidney with obstruction, or intractable pain or vomiting, since infected obstruction is a urologic emergency.

#289 · Nephrology and Urology

A 55-year-old smoker has asymptomatic microscopic hematuria on routine urinalysis. What is the next step?

Answer

Microscopic hematuria in an adult, especially with risk factors like smoking, older age, or occupational exposures, warrants evaluation for urologic malignancy since bladder cancer is a key concern. Workup includes upper tract imaging with CT urography and cystoscopy to directly visualize the bladder. A benign cause such as infection or exercise should be excluded first by repeating the urinalysis after resolution of any confounding factor before committing to a full workup.

#290 · Nephrology and Urology

What is the current guidance on PSA screening for prostate cancer?

Answer

USPSTF recommends individualized decision-making for men age 55 to 69, based on shared decision-making about the modest potential benefit versus harms of overdiagnosis and overtreatment. Screening is not recommended for men 70 and older. PSA is not a perfect test since it can be elevated by BPH, prostatitis, and recent ejaculation or instrumentation, not just cancer, which is why shared decision-making is emphasized over routine screening.

#291 · Nephrology and Urology

What are the medical treatment options for benign prostatic hyperplasia?

Answer

Alpha-1 blockers like tamsulosin relax smooth muscle in the prostate and bladder neck, giving fast symptom relief but not shrinking the gland. 5-alpha reductase inhibitors like finasteride shrink the prostate over months by blocking conversion of testosterone to dihydrotestosterone, and are more useful in larger glands. Combination therapy is often used for larger prostates with bothersome symptoms, and both drug classes can be combined with a PDE5 inhibitor like tadalafil if the patient also has erectile dysfunction.

#292 · Nephrology and Urology

How do you distinguish stress incontinence from urge incontinence based on history?

Answer

Stress incontinence involves small volume leakage with increased intra-abdominal pressure, such as coughing, sneezing, or laughing, and results from urethral sphincter weakness often after childbirth or pelvic surgery. Urge incontinence involves a sudden strong urge to void followed by involuntary leakage, driven by detrusor overactivity, and is treated with bladder training and antimuscarinic or beta-3 agonist medications. Stress incontinence is treated with pelvic floor exercises and sometimes surgical sling procedures.

#293 · Nephrology and Urology

An older man with BPH develops constant dribbling and a palpably distended bladder. What type of incontinence is this?

Answer

This is overflow incontinence, caused by chronic bladder outlet obstruction or an underactive detrusor muscle leading to urinary retention with overflow leakage. Common causes include BPH, diabetic neuropathy affecting the bladder, and anticholinergic medications. Post-void residual volume is markedly elevated, and management addresses the underlying obstruction or requires intermittent or indwelling catheterization.

#294 · Nephrology and Urology

How do you work up hyponatremia in a hospitalized patient?

Answer

Start with serum osmolality to confirm true hypotonic hyponatremia and exclude pseudohyponatremia from hyperlipidemia or hyperproteinemia, and exclude hyperglycemia-driven dilutional hyponatremia. Next assess volume status clinically, since hypovolemic, euvolemic, and hypervolemic hyponatremia have different causes and treatments. Urine sodium and urine osmolality help further, for example a euvolemic patient with concentrated urine and urine sodium over 40 suggests SIADH.

#295 · Nephrology and Urology

What are the diagnostic criteria for SIADH?

Answer

SIADH requires hypotonic hyponatremia with serum osmolality low, inappropriately concentrated urine with urine osmolality above 100 mOsm/kg, clinical euvolemia, urine sodium generally above 40 mEq/L, and normal thyroid and adrenal function to exclude hypothyroidism and adrenal insufficiency as mimics. Common causes include small cell lung cancer, CNS disease, pneumonia, and drugs like SSRIs and carbamazepine. Treatment is fluid restriction first line, with salt tablets, loop diuretics, or vaptans for refractory cases.

#296 · Nephrology and Urology

How do you calculate free water deficit in a hypernatremic patient, and why must correction be slow?

Answer

Free water deficit equals total body water times (current sodium divided by 140, minus 1), where total body water is roughly 0.6 times weight in kg for men and 0.5 times weight for women. For example, a 70 kg man with sodium 160 has TBW of 42 L and a deficit of about 42 x (160/140 - 1), roughly 6 liters. Correction should not exceed about 8-10 mEq/L per day because rapid correction of chronic hypernatremia can cause cerebral edema.

#297 · Nephrology and Urology

A patient has a potassium of 6.8 with peaked T waves on EKG. What is the immediate management sequence?

Answer

Give IV calcium gluconate first to stabilize the cardiac membrane, which does not lower potassium but protects the heart within minutes. Then give IV insulin with concurrent dextrose to shift potassium intracellularly, and consider nebulized albuterol as an additional shifting agent. Finally use potassium removal with loop diuretics if the patient makes urine, or sodium zirconium cyclosilicate or patiromer for the gut, and consider emergent dialysis if there is severe AKI, refractory hyperkalemia, or ongoing EKG changes.

#298 · Nephrology and Urology

What are the main causes of hypokalemia and how is severe hypokalemia treated?

Answer

Causes fall into decreased intake, transcellular shifts such as insulin or beta-agonists, and increased losses through GI loss like vomiting or diarrhea, or renal loss from diuretics, hyperaldosteronism, or renal tubular acidosis. Check magnesium, since hypomagnesemia impairs potassium repletion and must be corrected simultaneously. Severe hypokalemia with EKG changes or below 2.5 mEq/L needs IV potassium replacement with cardiac monitoring, since rapid IV infusion itself can be dangerous.

#299 · Nephrology and Urology

A patient has Na 138, Cl 96, HCO3 12. What is the anion gap and what differential does it suggest?

Answer

Anion gap equals Na minus (Cl plus HCO3), so 138 minus (96 plus 12) equals 30, which is markedly elevated above the normal range of about 8-12. An elevated anion gap metabolic acidosis differential is remembered with MUDPILES: methanol, uremia, diabetic ketoacidosis, propylene glycol or paraldehyde, iron or isoniazid, lactic acidosis, ethylene glycol, and salicylates. The specific cause is narrowed with history, osmolar gap, lactate, ketones, and renal function.

#300 · Nephrology and Urology

How do you use Winter's formula to check respiratory compensation for a metabolic acidosis?

Answer

Winter's formula predicts expected PCO2 as 1.5 times the serum HCO3 plus 8, with a margin of plus or minus 2. For example, if HCO3 is 10, expected PCO2 is 1.5 x 10 + 8, which is 23, so an appropriate range is about 21 to 25. If the measured PCO2 is higher than predicted there is a concurrent respiratory acidosis, and if lower there is a concurrent respiratory alkalosis, meaning the patient has a mixed acid-base disorder rather than simple compensation.

#301 · Neurology

First-line treatment for status epilepticus

Answer

Give a benzodiazepine first, usually IV lorazepam or IM midazolam if there is no IV access. If seizures continue, follow with a second-line agent such as IV fosphenytoin, valproate, or levetiracetam. Check glucose right away since hypoglycemia is a reversible cause that mimics or triggers seizures.

#302 · Neurology

Patient with a single unprovoked seizure and a normal brain MRI and EEG. Start antiepileptic drug?

Answer

Not usually. After a first unprovoked seizure with a normal workup, the recurrence risk is low enough that most guidelines favor observation rather than starting long-term medication. Driving restrictions still apply based on state law. Treatment is reconsidered if a second seizure occurs or if imaging or EEG shows an abnormality.

#303 · Neurology

Todd paralysis

Answer

Todd paralysis is temporary focal weakness that follows a seizure, usually affecting the limb where the seizure activity started. It typically resolves within 24 to 48 hours. It is a clinical diagnosis of exclusion, but a new or unusually prolonged deficit should prompt imaging to rule out stroke or structural lesion.

#304 · Neurology

Sudden severe headache described as the worst headache of my life. Next step?

Answer

This is the classic presentation of subarachnoid hemorrhage and needs an emergent noncontrast head CT. If the CT is done within 6 hours of onset and is negative, it is highly sensitive, but if done later or still suspicious, follow with a lumbar puncture looking for xanthochromia. Do not send this patient home without ruling out SAH.

#305 · Neurology

Acute ischemic stroke, patient presents within 3 hours, no contraindications. Treatment?

Answer

Give IV alteplase (tPA) after confirming there is no hemorrhage on noncontrast CT and blood pressure is under 185/110. The window can extend to 4.5 hours in select patients without certain exclusions like age over 80 or diabetes with prior stroke. Mechanical thrombectomy is also considered for large vessel occlusion, with an extended window up to 24 hours in selected imaging-confirmed cases.

#306 · Neurology

TIA definition and next step in workup

Answer

A TIA is a transient episode of neurologic dysfunction from focal ischemia without evidence of acute infarction on imaging, with symptoms typically resolving within an hour. Because TIA carries a high short-term risk of stroke, workup should be urgent and includes brain MRI, carotid imaging, ECG or telemetry, and echocardiogram to find the source. The ABCD2 score helps estimate stroke risk and guide urgency of evaluation.

#307 · Neurology

Migraine with aura, features and first-line abortive therapy

Answer

Migraine often presents with unilateral throbbing head pain, nausea, and photophobia or phonophobia, sometimes preceded by visual aura like scintillating scotoma. First-line abortive treatment is a triptan or NSAID taken early in the attack. Triptans should be avoided in patients with coronary artery disease or uncontrolled hypertension because of vasoconstrictive effects.

#308 · Neurology

Headache that worsens with lying down and improves when sitting up, worse in the morning

Answer

This pattern suggests increased intracranial pressure, often from a mass lesion, hydrocephalus, or edema. Morning worsening happens because lying flat overnight raises intracranial pressure further. This warrants brain imaging before doing a lumbar puncture, since LP in the setting of a mass lesion risks herniation.

#309 · Neurology

Guillain-Barre syndrome, clinical picture and key test

Answer

GBS presents with ascending, symmetric weakness and areflexia, often days to weeks after a gastrointestinal or respiratory infection like Campylobacter jejuni. Lumbar puncture classically shows albuminocytologic dissociation, meaning high protein with a normal white cell count. Watch respiratory function closely with serial vital capacity measurements since respiratory failure can develop quickly and may need ICU support.

#310 · Neurology

Guillain-Barre syndrome treatment

Answer

Treatment is IVIG or plasma exchange, started as early as possible in patients with significant weakness or respiratory compromise. Corticosteroids are not effective and are not used. Most patients recover over weeks to months, though some are left with residual weakness.

#311 · Neurology

Carpal tunnel syndrome, presentation and diagnosis

Answer

Carpal tunnel syndrome causes numbness and tingling in the thumb, index, middle, and part of the ring finger from median nerve compression at the wrist. Symptoms are often worse at night and can be reproduced with Tinel or Phalen maneuvers. Nerve conduction studies confirm the diagnosis when it is unclear, and risk factors include repetitive wrist motion, pregnancy, hypothyroidism, and diabetes.

#312 · Neurology

Diabetic peripheral neuropathy, typical pattern and management

Answer

Diabetic neuropathy usually presents as a symmetric, distal, sensory-predominant neuropathy in a stocking-glove distribution, often with burning pain or numbness starting in the feet. Tight glycemic control helps slow progression. First-line medications for painful symptoms include pregabalin, gabapentin, duloxetine, or tricyclic antidepressants.

#313 · Neurology

Bell palsy, presentation and how to distinguish from stroke

Answer

Bell palsy causes acute unilateral facial weakness affecting both the upper and lower face, including the forehead, because it is a peripheral seventh nerve lesion. A central stroke typically spares the forehead because of bilateral cortical innervation to the upper face. Treatment is early oral corticosteroids, ideally within 72 hours, with antivirals sometimes added for severe cases.

#314 · Neurology

Differentiating delirium from dementia at the bedside

Answer

Delirium has an acute onset over hours to days, fluctuates throughout the day, and features impaired attention and altered level of consciousness. Dementia develops gradually over months to years with a relatively stable level of alertness and preserved attention until late stages. Any acute change in mental status in an older patient should prompt a search for a reversible cause such as infection, medication, or metabolic derangement.

#315 · Neurology

Initial workup for new dementia symptoms

Answer

Basic workup includes TSH, vitamin B12, complete metabolic panel, CBC, and a brain MRI or CT to look for reversible or structural causes like normal pressure hydrocephalus, subdural hematoma, or mass. Depression can also mimic dementia, sometimes called pseudodementia, and should be considered especially if onset is subacute. Cognitive screening tools like the MoCA or MMSE help establish baseline severity.

#316 · Neurology

Parkinson disease, core features and first-line treatment in younger vs older patients

Answer

Core features are resting tremor, bradykinesia, rigidity, and postural instability, often with a shuffling gait and reduced arm swing. In younger patients, dopamine agonists are often preferred first to delay motor complications from levodopa. In older patients or those with more significant symptoms, levodopa-carbidopa is typically used first since it is more effective and better tolerated in that group.

#317 · Neurology

Bacterial meningitis, empiric antibiotics in an adult

Answer

Empiric therapy is vancomycin plus a third-generation cephalosporin like ceftriaxone, covering Streptococcus pneumoniae and Neisseria meningitidis. Add ampicillin in patients over 50 or immunocompromised to cover Listeria monocytogenes. Dexamethasone is given with or just before the first antibiotic dose to reduce neurologic complications, particularly with pneumococcal meningitis.

#318 · Neurology

Myasthenia gravis, hallmark clinical feature and confirmatory testing

Answer

Myasthenia gravis causes fluctuating, fatigable weakness that worsens with repeated activity and improves with rest, often starting with ptosis or diplopia. Acetylcholine receptor antibodies are the most common confirmatory test, and edrophonium testing or ice pack test can support the diagnosis at the bedside. CT chest should be done to look for a thymoma, which is associated with the disease.

#319 · Neurology

Multiple sclerosis, classic presentation and diagnostic imaging finding

Answer

MS often presents with episodes separated in time and space, such as optic neuritis, sensory disturbances, or limb weakness, typically in a young adult. MRI of the brain and spinal cord classically shows periventricular white matter lesions disseminated in space and time. CSF analysis may show oligoclonal bands, supporting the diagnosis when MRI findings are not definitive.

#320 · Neurology

Distinguishing peripheral vertigo (BPPV) from central vertigo (stroke)

Answer

Peripheral vertigo like BPPV has sudden, brief episodes triggered by position change, with a positive Dix-Hallpike test and normal neurologic exam otherwise. Central vertigo from a stroke or lesion is more likely to have direction-changing nystagmus, inability to walk, or other neurologic deficits, and does not fatigue with repeated positioning. The HINTS exam, checking head impulse, nystagmus, and test of skew, helps distinguish the two and is more sensitive than early MRI for detecting stroke in this setting.

#321 · Obstetrics and Gynecology

A 45-year-old woman has a palpable, mobile, well-circumscribed breast mass. What is the next step?

Answer

Any new palpable breast mass in a woman over 30 needs diagnostic imaging, usually diagnostic mammogram plus ultrasound, followed by tissue sampling with core needle biopsy if the mass is solid or suspicious. In women under 30, ultrasound alone is often the first study since breast tissue is dense. A mass should never be dismissed as benign based on exam alone, even if it feels smooth and mobile like a fibroadenoma.

#322 · Obstetrics and Gynecology

What are the diagnostic criteria for polycystic ovary syndrome (PCOS)?

Answer

PCOS is diagnosed using the Rotterdam criteria, requiring at least two of three features: oligo- or anovulation, clinical or biochemical hyperandrogenism, and polycystic ovaries on ultrasound. Other causes of androgen excess like congenital adrenal hyperplasia and androgen-secreting tumors should be excluded first. PCOS also carries long term risks of insulin resistance, type 2 diabetes, and endometrial hyperplasia from unopposed estrogen.

#323 · Obstetrics and Gynecology

How is PCOS managed in a woman not seeking pregnancy?

Answer

Combined oral contraceptives are first line to regulate cycles, protect the endometrium, and reduce hirsutism and acne. Metformin can be added for insulin resistance and metabolic risk reduction. Weight loss through lifestyle changes improves ovulation and metabolic parameters, and spironolactone can be used for hirsutism if oral contraceptives alone are insufficient.

#324 · Obstetrics and Gynecology

What features raise concern for ovarian cancer on imaging or exam?

Answer

Concerning features include a complex adnexal mass with solid components, thick septations, papillary projections, ascites, and irregular borders on ultrasound. Postmenopausal women with a new pelvic mass or elevated CA-125 warrant gynecologic oncology referral. Ovarian cancer often presents late with vague symptoms like bloating, early satiety, and pelvic pressure, which is why it is frequently diagnosed at an advanced stage.

#325 · Obstetrics and Gynecology

A 38-year-old woman has heavy, prolonged menstrual bleeding and an enlarged, irregular uterus on exam. Diagnosis and workup?

Answer

This presentation is classic for uterine fibroids (leiomyomas). Pelvic ultrasound is the initial imaging study to confirm size, number, and location of fibroids. Check a CBC for anemia from chronic blood loss, and consider MRI if surgical planning is needed or the diagnosis is unclear.

#326 · Obstetrics and Gynecology

What are treatment options for symptomatic uterine fibroids?

Answer

Options range from medical to surgical depending on symptom severity and fertility goals. Medical therapy includes hormonal contraceptives, tranexamic acid for bleeding, and GnRH agonists or antagonists for short term shrinkage before surgery. Procedural options include uterine artery embolization, myomectomy for women wanting to preserve fertility, and hysterectomy for definitive treatment when childbearing is complete.

#327 · Obstetrics and Gynecology

A young woman has cyclic pelvic pain, dysmenorrhea, and dyspareunia. What is the likely diagnosis and how is it confirmed?

Answer

This presentation suggests endometriosis, where endometrial-like tissue grows outside the uterus. Laparoscopy with biopsy is the definitive diagnostic method, but empiric medical treatment is often started first based on clinical suspicion alone. NSAIDs and hormonal suppression with combined oral contraceptives or progestins are first line therapy before pursuing surgery.

#328 · Obstetrics and Gynecology

What is the most common presenting symptom of endometrial cancer, and what is the next diagnostic step?

Answer

Postmenopausal bleeding is the most common presenting symptom of endometrial cancer and should never be attributed to atrophy without evaluation. Endometrial biopsy is the initial diagnostic test. Transvaginal ultrasound measuring endometrial stripe thickness can help triage risk, with a stripe greater than 4 mm in a postmenopausal woman warranting biopsy.

#329 · Obstetrics and Gynecology

What are the risk factors for endometrial cancer?

Answer

Risk factors relate to unopposed estrogen exposure: obesity, nulliparity, early menarche, late menopause, tamoxifen use, PCOS, and estrogen-only hormone therapy without a progestin in women with a uterus. Diabetes and hypertension are also associated. Combined oral contraceptives and progestin therapy are protective by opposing estrogen's effect on the endometrium.

#330 · Obstetrics and Gynecology

A pregnant woman at 30 weeks presents with a blood pressure of 155/100, headache, and 2+ proteinuria. Diagnosis and immediate management?

Answer

This is preeclampsia with severe features given the headache and significantly elevated blood pressure. Management includes magnesium sulfate for seizure prophylaxis, antihypertensives such as labetalol or hydralazine for severe range blood pressure, and prompt obstetric consultation regarding timing of delivery. Delivery is the definitive treatment, with timing balanced against fetal maturity and maternal severity.

#331 · Obstetrics and Gynecology

How does peripartum cardiomyopathy typically present, and how is it diagnosed?

Answer

Peripartum cardiomyopathy presents with new heart failure symptoms, such as dyspnea, orthopnea, and edema, occurring in the last month of pregnancy through five months postpartum, without another identifiable cause. Diagnosis requires an echocardiogram showing reduced ejection fraction, typically below 45 percent. It is a diagnosis of exclusion after ruling out other causes of cardiomyopathy, and treatment follows standard heart failure management adapted for pregnancy or lactation status.

#332 · Obstetrics and Gynecology

A woman with a positive pregnancy test presents with unilateral pelvic pain and vaginal spotting. What must be ruled out first?

Answer

Ectopic pregnancy must be ruled out immediately since it is a life-threatening emergency if it ruptures. Transvaginal ultrasound combined with quantitative beta-hCG is used to locate the pregnancy, and if beta-hCG is above the discriminatory zone without an intrauterine pregnancy seen, ectopic pregnancy should be strongly suspected. Hemodynamic instability or signs of rupture require immediate surgical intervention rather than waiting for further testing.

#333 · Obstetrics and Gynecology

How should a stable, unruptured ectopic pregnancy be managed medically?

Answer

Methotrexate is used for medical management in a hemodynamically stable patient with a small ectopic mass, no fetal cardiac activity, and beta-hCG below a defined threshold, typically around 5000. Contraindications include hepatic or renal impairment, breastfeeding, and immunodeficiency. Beta-hCG levels are followed serially after treatment to confirm resolution.

#334 · Obstetrics and Gynecology

A 52-year-old woman reports hot flashes, night sweats, and vaginal dryness with irregular periods. How should menopause management be approached?

Answer

Menopausal hormone therapy with estrogen, combined with a progestin if the uterus is intact, is the most effective treatment for vasomotor symptoms and is generally safe for healthy women under 60 or within 10 years of menopause onset. For women who cannot or prefer not to use hormones, options include SSRIs or SNRIs for hot flashes and vaginal estrogen specifically for genitourinary symptoms. Hormone therapy should be avoided in women with a history of breast cancer, unexplained vaginal bleeding, or venous thromboembolism.

#335 · Obstetrics and Gynecology

How do you choose an appropriate contraceptive method for a 35-year-old smoker with migraines with aura?

Answer

Migraine with aura is an absolute contraindication to estrogen-containing contraceptives due to increased stroke risk, and this risk is compounded by smoking. Progestin-only options are safe alternatives, including the progestin-only pill, implant, injection, or a levonorgestrel IUD. Long-acting reversible contraceptives like the IUD or implant are also excellent choices given their high efficacy and lack of estrogen exposure.

#336 · Medical Oncology

60-year-old smoker with a central lung mass and hypercalcemia. What lung cancer subtype is most likely and why?

Answer

Squamous cell carcinoma is the classic cause of malignancy related hypercalcemia because it secretes PTH related peptide. It also tends to arise centrally near the main bronchi, unlike adenocarcinoma which is usually peripheral. Squamous tumors are strongly linked to smoking and can cavitate.

#337 · Medical Oncology

What driver mutations should be tested for in advanced non-small cell lung cancer before starting treatment?

Answer

Standard molecular testing includes EGFR, ALK, ROS1, BRAF, KRAS G12C, MET exon 14, RET, NTRK, and PD-L1 expression. Finding a targetable mutation like EGFR or ALK allows use of an oral targeted agent instead of chemotherapy, with better response rates and fewer side effects. Testing should happen before first line therapy whenever possible since it changes the treatment plan.

#338 · Medical Oncology

Why is small cell lung cancer treated differently from non-small cell lung cancer?

Answer

Small cell lung cancer grows fast, spreads early, and is almost never resectable at diagnosis, so it is treated primarily with chemotherapy and radiation rather than surgery. It is highly sensitive to chemotherapy initially but relapses quickly and carries a poor long term prognosis. Limited stage disease gets concurrent chemoradiation, while extensive stage gets chemotherapy plus immunotherapy.

#339 · Medical Oncology

A 45-year-old woman has a new breast mass with a BRCA1 mutation. What receptor profile is most likely and what does that mean for treatment?

Answer

BRCA1 mutated breast cancers are frequently triple negative, meaning they lack estrogen receptor, progesterone receptor, and HER2 expression. This rules out hormone therapy and HER2 targeted drugs like trastuzumab, so treatment relies on chemotherapy and increasingly immunotherapy or PARP inhibitors. Triple negative disease tends to be more aggressive with a higher early recurrence risk.

#340 · Medical Oncology

What is the mechanism and major toxicity concern of trastuzumab in HER2 positive breast cancer?

Answer

Trastuzumab is a monoclonal antibody that blocks the HER2 receptor, which is overexpressed in about 15 to 20 percent of breast cancers and drives aggressive growth. Its main toxicity is cardiotoxicity, causing a reversible drop in left ventricular ejection fraction rather than permanent cell death like anthracyclines. Baseline and periodic echocardiograms are required during treatment.

#341 · Medical Oncology

How does tamoxifen work and what are its two major long-term risks?

Answer

Tamoxifen is a selective estrogen receptor modulator that blocks estrogen effects in breast tissue while acting like estrogen in the uterus and bone. Its major risks are endometrial cancer from the uterine estrogenic effect and venous thromboembolism. It is used in both premenopausal and postmenopausal women with hormone receptor positive breast cancer.

#342 · Medical Oncology

Why are aromatase inhibitors like anastrozole only used in postmenopausal women?

Answer

Aromatase inhibitors block the peripheral conversion of androgens to estrogen in fat and other tissues, which is the main source of estrogen after menopause. In premenopausal women the ovaries still make estrogen directly, so blocking aromatase alone triggers a compensatory rise in ovarian estrogen production and does not work. Their main side effects are bone loss and joint pain.

#343 · Medical Oncology

What is the recommended colorectal cancer screening schedule for an average risk adult?

Answer

Average risk adults should start screening at age 45 with either colonoscopy every 10 years, FIT testing every year, or CT colonography every 5 years, among other accepted options. Screening should continue through age 75, with individualized decisions from 76 to 85. A first degree relative with colorectal cancer under age 60 usually triggers earlier and more frequent colonoscopy.

#344 · Medical Oncology

What tumor marker is used to monitor colorectal cancer, and what is it not useful for?

Answer

Carcinoembryonic antigen, or CEA, is used to monitor for recurrence after treatment and to track response to therapy in colorectal cancer. It is not sensitive or specific enough to be used as a screening test since it can be elevated by smoking and other benign conditions and can be normal even with active cancer. A rising CEA after treatment should prompt imaging to look for recurrence.

#345 · Medical Oncology

A patient with new painless jaundice, weight loss, and a palpable nontender gallbladder. What is this finding called and what does it suggest?

Answer

This is Courvoisier sign, a palpable nontender gallbladder in a jaundiced patient, and it strongly suggests malignant obstruction of the common bile duct, most often from pancreatic head cancer. It differs from gallstone disease, where the gallbladder is usually tender or scarred and does not distend easily. Workup should include CT or MRCP and CA 19-9 level.

#346 · Medical Oncology

What is the most important risk factor for hepatocellular carcinoma and how is screening done?

Answer

Cirrhosis from any cause, especially chronic hepatitis B, hepatitis C, and alcohol related liver disease, is the leading risk factor for hepatocellular carcinoma. Screening is done with liver ultrasound every 6 months, sometimes combined with alpha fetoprotein, in patients with cirrhosis or chronic hepatitis B even without cirrhosis. Early detection matters because curative options like resection or transplant only work on small tumors.

#347 · Medical Oncology

A 68-year-old man has an elevated PSA on routine screening. What is the current recommendation for PSA screening?

Answer

Shared decision making is recommended for men aged 55 to 69, weighing the modest mortality benefit against risks of overdiagnosis and overtreatment such as unnecessary biopsies and incontinence or erectile dysfunction from treatment. Routine screening is not recommended after age 70. Men at higher risk, such as Black men or those with a family history, may benefit from starting the conversation earlier.

#348 · Medical Oncology

What does a painless testicular mass in a young man require as the next step, and why should biopsy be avoided?

Answer

A painless testicular mass should be evaluated with scrotal ultrasound and tumor markers including AFP, beta hCG, and LDH, and if suspicious it should go straight to radical inguinal orchiectomy rather than transscrotal biopsy. Biopsy or a transscrotal approach risks seeding the scrotal skin and disrupting normal lymphatic drainage, which changes staging and treatment. Testicular cancer is highly curable even at advanced stages because it responds very well to chemotherapy.

#349 · Medical Oncology

What are the two elevated tumor markers most associated with germ cell testicular tumors, and how do they differ by histology?

Answer

Alpha fetoprotein is elevated in nonseminomatous germ cell tumors such as yolk sac and embryonal carcinoma but is never elevated in pure seminoma. Beta hCG can be elevated in both seminoma and nonseminomatous tumors. A high AFP effectively rules out pure seminoma and points toward mixed or nonseminomatous histology.

#350 · Medical Oncology

What is the classic triad of renal cell carcinoma, and how often does it actually present that way?

Answer

The classic triad is flank pain, hematuria, and a palpable abdominal mass, but this full triad is actually uncommon and usually signals advanced disease. Most renal cell carcinomas today are found incidentally on imaging done for unrelated reasons. Renal cell carcinoma can also cause paraneoplastic effects like erythrocytosis from ectopic erythropoietin production.

#351 · Medical Oncology

A postmenopausal woman presents with vaginal bleeding. What must always be ruled out first, and how?

Answer

Any postmenopausal bleeding must be evaluated for endometrial cancer, since it is the most common gynecologic malignancy and postmenopausal bleeding is its hallmark presenting symptom. Workup starts with transvaginal ultrasound to measure endometrial stripe thickness, followed by endometrial biopsy if the stripe is thickened or bleeding persists. Unopposed estrogen exposure, obesity, and tamoxifen use are major risk factors.

#352 · Medical Oncology

Why is ovarian cancer often called a silent killer, and what marker is used to monitor it?

Answer

Ovarian cancer usually causes vague symptoms like bloating, early satiety, and pelvic discomfort that are easily mistaken for benign conditions, so most cases are diagnosed at an advanced stage. CA 125 is used to monitor treatment response and detect recurrence, but it is not a reliable screening test because it can be elevated by many benign conditions like fibroids and endometriosis. There is currently no recommended screening test for average risk women.

#353 · Medical Oncology

What causes almost all cases of cervical cancer, and how has screening changed over time?

Answer

Persistent infection with high risk strains of human papillomavirus, especially types 16 and 18, causes nearly all cervical cancer. Current guidelines use cytology (Pap) alone every 3 years starting at age 21, then switch at age 30 to either Pap plus HPV co-testing every 5 years or primary HPV testing alone every 5 years. HPV vaccination before sexual debut dramatically lowers lifetime risk.

#354 · Medical Oncology

A patient with newly diagnosed multiple myeloma. What is the CRAB criteria used to diagnose end organ damage?

Answer

CRAB stands for hyperCalcemia, Renal insufficiency, Anemia, and Bone lesions, and any one of these findings in the setting of a clonal plasma cell disorder confirms symptomatic multiple myeloma requiring treatment. Bone lesions are classically lytic punched out lesions seen on skeletal survey or whole body imaging rather than blastic lesions. Monoclonal protein on serum or urine electrophoresis and increased plasma cells on bone marrow biopsy support the diagnosis.

#355 · Medical Oncology

How do you distinguish Hodgkin lymphoma from non-Hodgkin lymphoma on biopsy and clinical behavior?

Answer

Hodgkin lymphoma is defined by the presence of Reed-Sternberg cells, large cells with a classic owl eye binucleated appearance, and it tends to spread in a predictable contiguous pattern from one lymph node group to the next. Non-Hodgkin lymphoma lacks Reed-Sternberg cells and often spreads in a noncontiguous, less predictable pattern. Hodgkin lymphoma has a bimodal age distribution and generally carries a better prognosis with treatment.

#356 · Medical Oncology

What is the Philadelphia chromosome and which leukemia is it associated with?

Answer

The Philadelphia chromosome is a translocation between chromosomes 9 and 22 that creates the BCR-ABL fusion gene, producing a constitutively active tyrosine kinase that drives uncontrolled cell growth. It is the hallmark of chronic myeloid leukemia and is found in most cases. Tyrosine kinase inhibitors like imatinib target this fusion protein directly and have transformed CML into a manageable chronic condition.

#357 · Medical Oncology

A patient with acute promyelocytic leukemia develops disseminated intravascular coagulation. What should be started immediately?

Answer

All-trans retinoic acid, ATRA, should be started as soon as acute promyelocytic leukemia is suspected, even before genetic confirmation, because it directly addresses the life threatening coagulopathy by promoting differentiation of the malignant promyelocytes. Waiting for confirmatory testing risks fatal bleeding from the associated DIC. ATRA combined with arsenic trioxide has made APL one of the most curable forms of acute leukemia.

#358 · Medical Oncology

How do you distinguish superficial spreading melanoma risk using the ABCDE criteria, and what feature matters most for prognosis?

Answer

ABCDE stands for Asymmetry, Border irregularity, Color variation, Diameter greater than 6 millimeters, and Evolving appearance, and any of these features in a pigmented lesion should prompt biopsy. The single most important prognostic factor once melanoma is diagnosed is Breslow depth, the measured thickness of tumor invasion, which determines staging and the need for sentinel lymph node biopsy. Ulceration and mitotic rate also affect prognosis.

#359 · Medical Oncology

A cancer patient presents with facial swelling, distended neck veins, and shortness of breath that worsens when lying flat. What is the diagnosis and first step?

Answer

This presentation is classic for superior vena cava syndrome, usually caused by a mediastinal mass, most often lung cancer or lymphoma, compressing the SVC and obstructing venous return from the head and arms. The first step is CT chest with contrast to confirm the diagnosis and identify the cause. Treatment includes elevating the head of the bed and addressing the underlying tumor with chemotherapy, radiation, or stenting depending on urgency.

#360 · Medical Oncology

What lab findings define tumor lysis syndrome and which patients are at highest risk?

Answer

Tumor lysis syndrome is defined by hyperkalemia, hyperphosphatemia, hyperuricemia, and hypocalcemia from the phosphate binding calcium, resulting from rapid destruction of tumor cells releasing their intracellular contents. Patients with bulky, rapidly dividing tumors like Burkitt lymphoma or acute leukemia are at highest risk, especially right after starting chemotherapy. Prevention includes aggressive IV hydration and allopurinol or rasburicase before treatment starts.

#361 · Medical Oncology

A cancer patient reports new severe back pain, leg weakness, and difficulty urinating. What must be ruled out emergently?

Answer

This presentation should raise immediate concern for malignant spinal cord compression, most often from vertebral metastases compressing the epidural space. MRI of the entire spine should be obtained emergently since delay in treatment can cause permanent paralysis. Treatment includes high dose corticosteroids right away, followed by radiation therapy or surgical decompression.

#362 · Medical Oncology

What is the mechanism behind hypercalcemia of malignancy in solid tumors without bone metastases, and how is it treated?

Answer

Many solid tumors, especially squamous cell carcinomas, secrete PTH related peptide, which mimics parathyroid hormone and raises calcium by increasing bone resorption and renal calcium reabsorption without needing direct bone invasion. Initial treatment is aggressive IV normal saline hydration, followed by bisphosphonates like zoledronic acid or denosumab for more durable calcium lowering. Calcitonin can be used for a rapid but short lived effect while bisphosphonates take full effect.

#363 · Medical Oncology

Which chemotherapy drug class is most associated with irreversible dose dependent cardiotoxicity, and how is risk monitored?

Answer

Anthracyclines like doxorubicin cause cumulative, dose dependent cardiotoxicity from oxidative damage to cardiac myocytes, and the damage is largely irreversible once it occurs, unlike the reversible dysfunction seen with trastuzumab. Risk is monitored with baseline and periodic echocardiograms to track ejection fraction, and lifetime cumulative dose is tracked to avoid exceeding safe thresholds. Combining anthracyclines with trastuzumab significantly increases cardiotoxicity risk.

#364 · Medical Oncology

A patient on immune checkpoint inhibitor therapy develops watery diarrhea and abdominal pain. How should this immune related adverse event be managed?

Answer

This is likely immune checkpoint inhibitor colitis, an immune related adverse event caused by the drug unleashing T cells against normal tissue, in this case the bowel. Mild cases can sometimes continue therapy with close monitoring, but moderate to severe cases require holding the immunotherapy and starting high dose corticosteroids. If steroids fail to control symptoms within a few days, infliximab or vedolizumab is added.

#365 · Medical Oncology

What endocrine immune related adverse events should be checked in a patient on immune checkpoint inhibitors who develops new fatigue?

Answer

New fatigue in a patient on checkpoint inhibitors should prompt checking thyroid function, since immune mediated thyroiditis causing hypothyroidism or less commonly hyperthyroidism is common. Adrenal insufficiency and hypophysitis, especially with ipilimumab, should also be considered and screened with morning cortisol and ACTH if suspected. Unlike most other immune related adverse events, thyroid dysfunction and adrenal insufficiency are often permanent and require lifelong hormone replacement rather than just holding the drug.

#366 · Ophthalmology

Open-angle vs angle-closure glaucoma: how do they differ and how is each treated?

Answer

Open-angle glaucoma is chronic and painless, caused by slow outflow obstruction in the trabecular meshwork, and is usually found on routine eye exam with gradual peripheral vision loss. Angle-closure glaucoma is an acute emergency with severe eye pain, headache, nausea, a fixed mid-dilated pupil, and a red eye with cloudy cornea. Open-angle disease is managed with topical drops like prostaglandin analogs or beta blockers, while acute angle closure needs urgent ophthalmology referral for IOP-lowering drugs and laser iridotomy.

#367 · Ophthalmology

How do you distinguish dry from wet age-related macular degeneration, and why does the distinction matter?

Answer

Dry macular degeneration is more common and progresses slowly, with drusen deposits seen on exam and gradual central vision loss. Wet macular degeneration involves abnormal choroidal neovascularization that can cause sudden, severe central vision distortion or loss, often described as straight lines appearing wavy on an Amsler grid. Wet disease is treated with intravitreal anti-VEGF injections, so any patient with dry AMD who reports new or sudden visual distortion needs urgent ophthalmology evaluation to rule out conversion to the wet form.

#368 · Ophthalmology

What eye findings define diabetic retinopathy, and what is the internist's role in screening?

Answer

Nonproliferative diabetic retinopathy shows microaneurysms, dot-blot hemorrhages, and cotton wool spots, while proliferative disease features neovascularization that can bleed into the vitreous or cause tractional retinal detachment. Diabetic macular edema can cause vision loss at any stage and is a leading cause of blindness in working-age adults. Every patient with type 2 diabetes needs a dilated eye exam at diagnosis and then annually, and type 1 patients need one starting five years after diagnosis, since tight glucose and blood pressure control reduce progression.

#369 · Ophthalmology

A 72-year-old woman has a new severe unilateral headache, jaw pain with chewing, and a brief episode of vision loss in one eye. What is the concern and what must happen immediately?

Answer

This presentation is classic for giant cell (temporal) arteritis, and the transient vision loss is amaurosis fugax, which signals impending permanent blindness from ophthalmic artery involvement. This is a true emergency: start high-dose systemic corticosteroids immediately, before waiting for temporal artery biopsy confirmation, because any delay risks irreversible vision loss. Check an ESR and CRP right away, and arrange biopsy within about a week of starting steroids since the diagnostic yield persists for some time after treatment begins.

#370 · Ophthalmology

A young adult presents with a painful red eye, photophobia, and blurred vision, and exam shows cell and flare in the anterior chamber. What is the diagnosis and what systemic conditions should be considered?

Answer

This is uveitis, inflammation of the uveal tract that can be anterior, intermediate, or posterior, and anterior uveitis is the type most often seen by internists. It is frequently linked to systemic disease, including the HLA-B27 associated spondyloarthropathies, inflammatory bowel disease, sarcoidosis, and infections like syphilis or tuberculosis. Treatment involves topical corticosteroids and cycloplegic drops under ophthalmology guidance, and recurrent or bilateral cases warrant a workup for an underlying systemic inflammatory or infectious cause.

#371 · Otolaryngology and Dental Medicine

A patient reports brief episodes of spinning vertigo triggered by rolling over in bed, lasting under a minute. What is the diagnosis and how do you confirm and treat it?

Answer

This is benign paroxysmal positional vertigo, caused by displaced otoconia in the semicircular canals. Confirm it with the Dix-Hallpike maneuver, which reproduces the vertigo and shows a characteristic torsional, upbeating nystagmus with a brief latency. Treatment is the Epley canalith repositioning maneuver, which is curative in most patients and does not require vestibular suppressant medication.

#372 · Otolaryngology and Dental Medicine

What is the classic symptom tetrad of Meniere disease and how is it managed?

Answer

Meniere disease presents with episodic vertigo lasting minutes to hours, fluctuating low frequency sensorineural hearing loss, tinnitus, and aural fullness in the affected ear, thought to result from endolymphatic hydrops. First line management is a low sodium diet and a diuretic such as hydrochlorothiazide or acetazolamide to reduce endolymph volume. Vestibular suppressants like meclizine can help acute attacks, and refractory cases may need intratympanic steroids or gentamicin.

#373 · Otolaryngology and Dental Medicine

An 80 year old reports gradual, symmetric hearing loss over years, worse for high pitched sounds and understanding speech in noisy rooms. What is the diagnosis and next step?

Answer

This presentation is classic for presbycusis, age related sensorineural hearing loss affecting high frequencies first due to cochlear hair cell degeneration. Audiometry confirms a bilateral, symmetric, high frequency sensorineural pattern and helps exclude other causes. Hearing aids are the mainstay of treatment, and untreated hearing loss in older adults is linked to social isolation and an increased risk of cognitive decline.

#374 · Otolaryngology and Dental Medicine

How do you distinguish acute otitis media from otitis externa on exam, and what is first line treatment for each?

Answer

Acute otitis media shows a bulging, erythematous, poorly mobile tympanic membrane with middle ear effusion, and pulling the pinna does not worsen pain. Otitis externa, or swimmer's ear, causes pain with pinna traction or tragal pressure and an edematous, erythematous ear canal, often with a normal tympanic membrane. Treat otitis media with amoxicillin when antibiotics are indicated, and treat otitis externa with topical fluoroquinolone or aminoglycoside drops rather than oral antibiotics.

#375 · Otolaryngology and Dental Medicine

A patient presents with fever, severe sore throat, drooling, and a muffled 'hot potato' voice. What diagnosis must be excluded urgently and how should it be managed?

Answer

This presentation raises concern for epiglottitis or a deep neck space infection such as a peritonsillar or retropharyngeal abscess, both of which can rapidly compromise the airway. Avoid agitating the patient or examining the oropharynx with a tongue depressor, since this can precipitate complete airway obstruction, and secure definitive airway evaluation with anesthesia or ENT at bedside. Management includes urgent airway assessment, IV antibiotics, and imaging or laryngoscopy once the airway is deemed stable, with abscesses often requiring surgical drainage.

#376 · Psychiatry

What are the core diagnostic criteria for major depressive disorder?

Answer

At least 5 of 9 SIGECAPS symptoms present nearly every day for 2 weeks, including depressed mood or anhedonia. Symptoms are sleep, interest, guilt, energy, concentration, appetite, psychomotor changes, and suicidal ideation. The episode must cause significant distress or functional impairment and not be better explained by substance use or another medical condition.

#377 · Psychiatry

A patient with depression asks about their suicide risk. What history features raise the risk most?

Answer

Prior suicide attempts are the strongest predictor of future attempts. Other high risk factors include a specific plan with access to lethal means, male sex, older age, living alone, recent major loss, active substance use, and a family history of suicide. Always ask directly about ideation, plan, intent, and access to firearms or pills, since asking does not increase risk.

#378 · Psychiatry

First line pharmacotherapy for major depressive disorder and key counseling point?

Answer

SSRIs such as sertraline or escitalopram are first line due to favorable side effect profiles. Counsel patients that full effect takes 4 to 6 weeks and there is a boxed warning for increased suicidal thinking in patients under 25 during the first few weeks. Continue therapy for at least 6 to 12 months after remission of a first episode to prevent relapse.

#379 · Psychiatry

How is bipolar I disorder distinguished from bipolar II disorder?

Answer

Bipolar I requires at least one manic episode lasting 7 days or requiring hospitalization, with or without depressive episodes. Bipolar II requires at least one hypomanic episode, lasting at least 4 days, plus at least one major depressive episode, and never a full manic episode. Mania causes marked impairment or psychosis, while hypomania does not.

#380 · Psychiatry

First line treatment options for acute mania in bipolar I disorder?

Answer

Lithium, valproate, or second generation antipsychotics such as quetiapine or olanzapine are first line, often used together for severe episodes. Antidepressants should be avoided as monotherapy because they can precipitate a manic switch. Lithium also has the strongest evidence for reducing suicide risk in bipolar disorder.

#381 · Psychiatry

What baseline labs and monitoring are needed before and during lithium therapy?

Answer

Check baseline renal function, thyroid function, and an ECG in patients with cardiac risk factors, since lithium can cause nephrogenic diabetes insipidus and hypothyroidism. Monitor lithium levels along with renal and thyroid function periodically during treatment. Warn patients about toxicity risk with dehydration, NSAIDs, ACE inhibitors, and thiazide diuretics, which reduce lithium clearance.

#382 · Psychiatry

How is alcohol use disorder screened for and what tool is commonly used?

Answer

The CAGE questionnaire asks about attempts to Cut down, Annoyance at criticism, Guilt about drinking, and needing an Eye opener, with 2 or more positive answers suggesting a problem. The AUDIT-C is a validated quantitative alternative often used in primary care. Screening should be routine since many patients underreport use unless asked directly.

#383 · Psychiatry

What is the timeline and management of alcohol withdrawal, including delirium tremens?

Answer

Mild symptoms like tremor and anxiety start 6 to 24 hours after the last drink, withdrawal seizures can occur at 12 to 48 hours, and delirium tremens with confusion, autonomic instability, and hallucinations peaks at 48 to 96 hours. Benzodiazepines are the treatment of choice, dosed using a symptom triggered protocol like CIWA-Ar. Thiamine should be given before glucose to prevent Wernicke encephalopathy.

#384 · Psychiatry

First line medications for opioid use disorder maintenance treatment?

Answer

Buprenorphine and methadone are first line for medication assisted treatment and both reduce mortality and relapse compared to abstinence alone. Buprenorphine is a partial agonist that can be prescribed in office settings, while methadone requires dispensing through a certified opioid treatment program. Naltrexone is an alternative for highly motivated patients who have already completed detoxification.

#385 · Psychiatry

What are the core positive and negative symptoms used to diagnose schizophrenia?

Answer

Diagnosis requires at least 2 of the following for a month, with at least one being delusions, hallucinations, or disorganized speech: delusions, hallucinations, disorganized speech, grossly disorganized or catatonic behavior, and negative symptoms like flat affect or avolition. Symptoms must cause functional decline and be present for at least 6 months total including prodromal or residual phases. Onset is typically in the late teens to twenties, earlier in men than women.

#386 · Psychiatry

Why are second generation antipsychotics generally preferred over first generation agents, and what should be monitored?

Answer

Second generation antipsychotics like risperidone and olanzapine have lower rates of extrapyramidal symptoms and tardive dyskinesia than first generation agents like haloperidol. However they carry higher risk of metabolic syndrome, so monitor weight, fasting glucose, and lipids at baseline and periodically. Clozapine is reserved for treatment resistant disease because of agranulocytosis risk requiring regular absolute neutrophil count monitoring.

#387 · Psychiatry

How is generalized anxiety disorder diagnosed and treated?

Answer

GAD involves excessive, difficult to control worry about multiple areas of life for at least 6 months, accompanied by symptoms like restlessness, fatigue, poor concentration, irritability, muscle tension, or sleep disturbance. SSRIs or SNRIs are first line pharmacotherapy, and cognitive behavioral therapy is equally effective. Benzodiazepines can help acutely but should be limited due to dependence risk, especially in older adults.

#388 · Psychiatry

What distinguishes panic disorder from a single panic attack, and what is the first line treatment?

Answer

Panic disorder requires recurrent unexpected panic attacks plus at least a month of persistent worry about future attacks or maladaptive behavior change to avoid them. Attacks themselves involve abrupt surges of fear with symptoms like palpitations, sweating, chest pain, and a sense of doom, peaking within minutes. SSRIs are first line long term treatment, with cognitive behavioral therapy as an equally effective option and benzodiazepines reserved for short term bridging.

#389 · Psychiatry

What are the diagnostic criteria and first line treatment for PTSD?

Answer

PTSD requires exposure to actual or threatened death, serious injury, or sexual violence, followed by intrusion symptoms, avoidance, negative mood or cognition changes, and hyperarousal lasting more than a month. Trauma focused psychotherapies like prolonged exposure or cognitive processing therapy are first line. SSRIs, particularly sertraline and paroxetine, are the first line pharmacologic option.

#390 · Psychiatry

How does obsessive compulsive disorder present and what is first line treatment?

Answer

OCD involves recurrent, intrusive obsessions that cause anxiety, paired with compulsions performed to reduce that anxiety, which are time consuming or impairing. Common themes include contamination fears with washing rituals or symmetry concerns with checking behaviors. First line treatment is exposure and response prevention therapy plus SSRIs, often at higher doses than used for depression.

#391 · Psychiatry

How is somatic symptom disorder diagnosed, and how does it differ from malingering?

Answer

Somatic symptom disorder involves one or more distressing physical symptoms accompanied by excessive thoughts, anxiety, or behaviors related to those symptoms, lasting more than 6 months. The symptoms are real to the patient and not intentionally produced, unlike malingering, where symptoms are consciously fabricated for secondary gain such as disability payments or avoiding work. Management focuses on regular scheduled visits with one primary provider rather than extensive repeat testing.

#392 · Psychiatry

What are the features of restless legs syndrome and its first line treatment?

Answer

Restless legs syndrome causes an urge to move the legs, often with uncomfortable sensations, that worsens at rest in the evening and improves with movement. Iron deficiency is a common secondary cause, so check ferritin and replace iron if low, targeting a ferritin above 75. First line pharmacologic treatment for persistent symptoms is a dopamine agonist like pramipexole or an alpha-2-delta ligand like gabapentin.

#393 · Psychiatry

What is the gold standard test for obstructive sleep apnea and its first line treatment?

Answer

Polysomnography is the gold standard diagnostic test, quantifying the apnea hypopnea index to grade severity. Risk factors include obesity, large neck circumference, and retrognathia, and the STOP-BANG questionnaire helps screen for it. Continuous positive airway pressure is first line treatment and reduces daytime sleepiness and cardiovascular risk.

#394 · Psychiatry

How do borderline and antisocial personality disorders differ in their core features?

Answer

Borderline personality disorder is marked by unstable relationships, identity disturbance, impulsivity, chronic feelings of emptiness, and recurrent self harm or suicidal behavior, often with fear of abandonment. Antisocial personality disorder involves a pervasive pattern of disregard for the rights of others, deceit, impulsivity, and lack of remorse, with conduct disorder typically present before age 15. Dialectical behavior therapy is the most evidence based treatment for borderline personality disorder specifically.

#395 · Psychiatry

What are the key features and treatment of neuroleptic malignant syndrome?

Answer

Neuroleptic malignant syndrome presents with fever, severe muscle rigidity, autonomic instability, and altered mental status, typically starting days to weeks after starting or increasing an antipsychotic. Labs classically show markedly elevated creatine kinase and leukocytosis. Treatment involves immediately stopping the offending agent, supportive cooling and hydration, and dantrolene or bromocriptine for severe cases.

#396 · Pulmonary Disease

How is asthma severity classified before starting treatment?

Answer

Severity is based on symptom frequency, nighttime awakenings, need for rescue inhaler, and lung function. Intermittent asthma has symptoms twice a week or less. Mild, moderate, and severe persistent asthma are defined by increasing daytime symptoms, more frequent nighttime awakenings, and lower FEV1 percent predicted.

#397 · Pulmonary Disease

What spirometry pattern confirms asthma?

Answer

Obstructive pattern with reduced FEV1/FVC ratio that improves by at least 12 percent and 200 mL after bronchodilator. A positive bronchoprovocation test with methacholine also supports the diagnosis when baseline spirometry is normal. Reversibility is the key feature separating asthma from fixed obstruction.

#398 · Pulmonary Disease

What is the preferred step-up therapy for mild persistent asthma per current guidelines?

Answer

Two step-up options exist. Scheduled low dose inhaled corticosteroid with formoterol used as both daily maintenance and reliever, known as SMART or MART therapy, is preferred for persistent disease. For very mild disease, as needed low dose inhaled corticosteroid-formoterol alone with no scheduled dosing is a separate GINA-preferred option, sometimes called anti-inflammatory reliever therapy, and is not itself called SMART. A daily low dose inhaled corticosteroid with a short acting beta agonist for rescue is an older alternative. Formoterol works well for reliever use because it has both rapid onset and long duration.

#399 · Pulmonary Disease

What triad suggests aspirin exacerbated respiratory disease?

Answer

Asthma, chronic rhinosinusitis with nasal polyps, and respiratory reactions to aspirin or other NSAIDs. Patients develop bronchospasm and nasal congestion within minutes to hours of NSAID exposure. Leukotriene modifiers can help control symptoms and aspirin desensitization is an option for select patients.

#400 · Pulmonary Disease

How does COPD differ from asthma on spirometry?

Answer

COPD shows a persistent obstructive pattern with FEV1/FVC below 0.70 that does not fully normalize after bronchodilator. Asthma typically shows more reversibility. COPD is diagnosed only when the ratio remains reduced after bronchodilator testing in a patient with compatible symptoms and exposure history, usually smoking.

#401 · Pulmonary Disease

What is the initial maintenance inhaler choice for a COPD patient with frequent dyspnea but no exacerbations?

Answer

A long acting muscarinic antagonist or long acting beta agonist as monotherapy, or the two combined if symptoms remain bothersome. Inhaled corticosteroids are reserved for patients with frequent exacerbations or elevated blood eosinophils. Bronchodilators are the backbone of COPD symptom control regardless of exacerbation history.

#402 · Pulmonary Disease

Which COPD patients benefit from long term home oxygen therapy?

Answer

Those with resting PaO2 of 55 mmHg or less, or oxygen saturation 88 percent or less on room air. Patients with PaO2 56 to 59 mmHg qualify if they have cor pulmonale, right heart failure, or hematocrit above 55 percent. Long term oxygen improves survival in these hypoxemic patients.

#403 · Pulmonary Disease

What triggers antibiotic use in a COPD exacerbation?

Answer

Increased sputum purulence combined with either increased sputum volume or increased dyspnea, or any exacerbation requiring mechanical ventilation. Common regimens include a macrolide, doxycycline, or amoxicillin-clavulanate. Systemic corticosteroids are also given for most moderate to severe exacerbations to shorten recovery time.

#404 · Pulmonary Disease

What lab finding should prompt testing for alpha-1 antitrypsin deficiency?

Answer

COPD or emphysema developing in a patient younger than 45, especially with minimal or no smoking history, or basilar predominant emphysema on imaging. A family history of early liver or lung disease also warrants testing. Diagnosis is made by checking serum alpha-1 antitrypsin level followed by genotyping if low.

#405 · Pulmonary Disease

What is the classic high resolution CT pattern of idiopathic pulmonary fibrosis?

Answer

Usual interstitial pneumonia pattern showing basal and peripheral predominant reticular opacities, honeycombing, and traction bronchiectasis without significant ground glass. This pattern is often enough to diagnose IPF without biopsy when the clinical picture fits. Honeycombing is the most specific feature.

#406 · Pulmonary Disease

Which two drugs are approved to slow disease progression in idiopathic pulmonary fibrosis?

Answer

Pirfenidone and nintedanib. Both are antifibrotic agents that slow the rate of decline in forced vital capacity but do not reverse existing fibrosis or cure the disease. Common side effects include gastrointestinal upset, and nintedanib also carries a bleeding risk and liver toxicity.

#407 · Pulmonary Disease

A young patient presents with bilateral hilar lymphadenopathy, erythema nodosum, and arthralgias. What is the diagnosis and what is this presentation called?

Answer

This is sarcoidosis presenting as Lofgren syndrome. Lofgren syndrome has an excellent prognosis and often resolves spontaneously without treatment. It does not usually require tissue biopsy because the clinical presentation is considered diagnostic.

#408 · Pulmonary Disease

What is the pathologic hallmark of sarcoidosis on biopsy?

Answer

Noncaseating granulomas, meaning granulomas without central necrosis, found in affected organs such as lymph nodes, lung, skin, or eyes. The diagnosis requires compatible clinical and radiographic findings plus exclusion of other granulomatous diseases like tuberculosis and fungal infection. Elevated ACE level can support but does not confirm the diagnosis.

#409 · Pulmonary Disease

When is systemic treatment indicated in pulmonary sarcoidosis?

Answer

Treatment is indicated for progressive symptoms, declining lung function, or involvement of critical organs such as the heart, eyes, or central nervous system. Oral corticosteroids are first line therapy. Asymptomatic patients with stable disease, especially stage I disease, are often just observed.

#410 · Pulmonary Disease

What is the Wells score used for and what does a low score allow?

Answer

The Wells score estimates pretest probability of pulmonary embolism based on clinical findings like tachycardia, immobilization, prior VTE, and hemoptysis. A low pretest probability combined with a negative D-dimer allows PE to be ruled out without imaging. Higher pretest probability patients should proceed directly to CT pulmonary angiography.

#411 · Pulmonary Disease

What defines a massive versus submassive pulmonary embolism?

Answer

Massive PE causes sustained hypotension, defined as systolic blood pressure below 90 mmHg for at least 15 minutes, from right ventricular failure. Submassive PE has evidence of right ventricular strain on imaging or elevated troponin and BNP but the patient remains hemodynamically stable. Massive PE is an indication for thrombolytic therapy.

#412 · Pulmonary Disease

How long should anticoagulation continue after a first unprovoked pulmonary embolism?

Answer

Indefinite anticoagulation is generally recommended after weighing bleeding risk, since the recurrence risk after stopping is significant. A provoked PE from a reversible risk factor like surgery is usually treated for three months. Ongoing bleeding risk assessment should guide the decision to continue long term.

#413 · Pulmonary Disease

What hemodynamic definition is used for pulmonary hypertension?

Answer

Mean pulmonary artery pressure greater than 20 mmHg measured by right heart catheterization. This replaced the older threshold of 25 mmHg. Right heart catheterization remains the gold standard confirmatory test even though echocardiography is used for initial screening.

#414 · Pulmonary Disease

What are the five WHO groups of pulmonary hypertension?

Answer

Group 1 is pulmonary arterial hypertension, group 2 is due to left heart disease, group 3 is due to lung disease or hypoxia, group 4 is chronic thromboembolic pulmonary hypertension, and group 5 is due to unclear or multifactorial mechanisms. Treatment approach differs greatly by group, and pulmonary vasodilators used in group 1 can worsen group 2 or 3 disease. Correctly identifying the group is essential before starting therapy.

#415 · Pulmonary Disease

Why is chronic thromboembolic pulmonary hypertension important to identify?

Answer

It is a potentially curable cause of pulmonary hypertension through pulmonary thromboendarterectomy surgery. It results from incomplete resolution of prior pulmonary emboli that organize into fibrotic material obstructing the pulmonary arteries. Ventilation-perfusion scanning is more sensitive than CT angiography for detecting this condition.

#416 · Pulmonary Disease

How do you distinguish a transudative from an exudative pleural effusion?

Answer

Light's criteria are used: an exudate is present if pleural fluid to serum protein ratio exceeds 0.5, pleural fluid to serum LDH ratio exceeds 0.6, or pleural fluid LDH exceeds two-thirds the upper limit of normal serum LDH. Meeting any one criterion classifies the effusion as exudative. Transudates are typically caused by heart failure, cirrhosis, or nephrotic syndrome from systemic pressure or protein imbalances.

#417 · Pulmonary Disease

What pleural fluid findings define an empyema requiring chest tube drainage?

Answer

Frank pus on aspiration, a positive Gram stain or culture, or pleural fluid pH below 7.2 with a parapneumonic effusion. Glucose below 60 mg/dL also supports complicated effusion. These findings mean antibiotics alone are unlikely to succeed and tube thoracostomy or surgical drainage is needed.

#418 · Pulmonary Disease

What is the immediate management of a tension pneumothorax?

Answer

Immediate needle decompression, traditionally at the second intercostal space midclavicular line though the fourth or fifth intercostal space at the anterior axillary line is now often preferred, followed by chest tube placement. Do not wait for imaging confirmation if the patient is hemodynamically unstable with signs like tracheal deviation and absent breath sounds. This is a clinical diagnosis in an unstable patient.

#419 · Pulmonary Disease

What scoring system helps decide whether a community acquired pneumonia patient can be treated outpatient?

Answer

CURB-65 assesses confusion, urea greater than 7 mmol/L, respiratory rate 30 or higher, blood pressure below 90/60, and age 65 or older. A score of 0 to 1 suggests outpatient treatment is appropriate, while higher scores suggest hospitalization or ICU care. The Pneumonia Severity Index is a more detailed alternative scoring tool.

#420 · Pulmonary Disease

What organisms should be covered when treating hospital acquired or ventilator associated pneumonia in a patient with risk factors for resistance?

Answer

Coverage should include Pseudomonas aeruginosa and MRSA in patients with risk factors such as prior IV antibiotics within 90 days, high local resistance rates, or septic shock. This typically means an antipseudomonal beta-lactam plus vancomycin or linezolid. Risk factor assessment determines whether broad or narrow empiric therapy is chosen.

#421 · Pulmonary Disease

How is latent tuberculosis infection distinguished from active disease?

Answer

Latent TB means a positive tuberculin skin test or interferon-gamma release assay without symptoms and with a normal or nonspecific chest x-ray, and the patient is not contagious. Active TB has symptoms like cough, fever, night sweats, and weight loss, often with cavitary upper lobe disease on imaging, and the patient can transmit infection. Sputum acid-fast smear and culture confirm active disease.

#422 · Pulmonary Disease

What is the standard initial four drug regimen for active pulmonary tuberculosis?

Answer

Rifampin, isoniazid, pyrazinamide, and ethambutol, often remembered as RIPE, given for two months. This is followed by rifampin and isoniazid alone for an additional four months if the organism is fully susceptible. Ethambutol can be stopped early once susceptibility results confirm the organism is not resistant.

#423 · Pulmonary Disease

What underlying conditions commonly predispose to bronchiectasis?

Answer

Cystic fibrosis, prior severe pneumonia, allergic bronchopulmonary aspergillosis, primary ciliary dyskinesia, and immunodeficiency states like common variable immunodeficiency. Nontuberculous mycobacterial infection is both a cause and a complication. Recurrent infection and inflammation damage the airway walls leading to permanent dilation.

#424 · Pulmonary Disease

What does chest CT show in bronchiectasis and what is the classic sign?

Answer

Bronchial dilation with airway diameter exceeding that of the adjacent pulmonary artery, known as the signet ring sign. Bronchial wall thickening and lack of normal tapering toward the periphery are also seen. High resolution CT is the imaging test of choice since chest x-ray often misses mild disease.

#425 · Pulmonary Disease

What are the Berlin criteria used to diagnose ARDS?

Answer

Onset within one week of a known insult, bilateral opacities on chest imaging not fully explained by effusion or nodules, respiratory failure not fully explained by cardiac failure or fluid overload, and a PaO_2/FiO_2 ratio of 300 or less on at least 5 cmH_2O of PEEP. Severity is graded mild, moderate, or severe based on the PaO_2/FiO_2 ratio. All four criteria must be present together.

#426 · Pulmonary Disease

What ventilator strategy improves survival in ARDS?

Answer

Lung protective ventilation with low tidal volumes of about 6 mL/kg of predicted body weight and plateau pressure kept below 30 cmH_2O. This reduces ventilator induced lung injury from overdistension. Permissive hypercapnia is often accepted as a tradeoff to keep tidal volumes and pressures low.

#427 · Pulmonary Disease

When should prone positioning be used in ARDS?

Answer

In patients with moderate to severe ARDS, generally a PaO2/FiO2 ratio below 150, who remain hypoxemic despite optimized mechanical ventilation. Prone positioning improves oxygenation by recruiting dorsal lung regions and has been shown to reduce mortality when used early and for prolonged sessions of at least 12 to 16 hours per day. It requires a trained team due to the risk of tube dislodgement during turning.

#428 · Pulmonary Disease

What are the four major histologic types of lung cancer and which is most associated with smoking?

Answer

Adenocarcinoma, squamous cell carcinoma, small cell carcinoma, and large cell carcinoma. Small cell and squamous cell carcinoma have the strongest association with smoking and tend to arise centrally. Adenocarcinoma is now the most common type overall and can occur in nonsmokers, often presenting peripherally.

#429 · Pulmonary Disease

Why is small cell lung cancer treated differently from non-small cell lung cancer?

Answer

Small cell lung cancer grows rapidly, spreads early, and is usually treated with chemotherapy and radiation rather than surgery because most cases are already metastatic or unresectable at diagnosis. It is staged simply as limited or extensive stage rather than using the TNM system. It is also strongly associated with paraneoplastic syndromes like SIADH and Lambert-Eaton syndrome.

#430 · Pulmonary Disease

Who qualifies for annual low dose CT lung cancer screening under current guidelines?

Answer

Adults aged 50 to 80 with at least a 20 pack-year smoking history who currently smoke or quit within the past 15 years. Screening should stop once a person has not smoked for 15 years or develops a health problem that substantially limits life expectancy. This screening has been shown to reduce lung cancer mortality in high risk populations.

#431 · Pulmonary Disease

What features on imaging suggest a solitary pulmonary nodule is benign versus malignant?

Answer

Benign features include small size under 6 mm, smooth well-defined margins, and dense central, popcorn, or diffuse calcification, along with stability over two years on prior imaging. Malignant features include larger size, spiculated or irregular margins, upper lobe location, and growth on serial imaging. Growth rate and doubling time on follow-up CT help guide the decision to biopsy versus continue surveillance.

#432 · Pulmonary Disease

What is the standard diagnostic test for obstructive sleep apnea and how is severity defined?

Answer

Polysomnography, either in-lab or home sleep testing in appropriate patients, measures the apnea-hypopnea index, the number of apneas and hypopneas per hour of sleep. Mild is 5 to 15 events per hour, moderate is 15 to 30, and severe is greater than 30 events per hour. An AHI of 5 or more with symptoms, or 15 or more regardless of symptoms, confirms the diagnosis.

#433 · Pulmonary Disease

What is first line treatment for moderate to severe obstructive sleep apnea?

Answer

Continuous positive airway pressure, or CPAP, which splints the airway open during sleep. Weight loss, avoiding alcohol and sedatives before bed, and positional therapy are also important adjuncts. Oral appliances are an alternative for mild disease or patients who cannot tolerate CPAP.

#434 · Pulmonary Disease

What is the initial ventilator setting approach for a patient with severe COPD exacerbation requiring intubation?

Answer

Use a lower respiratory rate and allow adequate expiratory time to prevent breath stacking and auto-PEEP, since air trapping is the main risk. Permissive hypercapnia is generally accepted to avoid worsening hyperinflation. Bronchodilators should be continued aggressively through the ventilator circuit.

#435 · Pulmonary Disease

What does auto-PEEP indicate and how is it detected on the ventilator?

Answer

Auto-PEEP, also called intrinsic PEEP, occurs when a new breath begins before the previous one has fully exhaled, trapping air and raising pressure in the lungs. It is detected by an end-expiratory hold maneuver showing pressure above the set PEEP. It is common in COPD and asthma patients on mechanical ventilation and can cause hypotension and barotrauma if severe.

#436 · Pulmonary Disease

How should a clinician approach a patient with chronic cough lasting more than 8 weeks and a normal chest x-ray?

Answer

The three most common causes are upper airway cough syndrome from postnasal drip, asthma, and gastroesophageal reflux disease. Empiric treatment trials targeting these causes are usually pursued before extensive further workup. ACE inhibitor use should also be reviewed and stopped if present since it is a common medication-induced cause.

#437 · Pulmonary Disease

What is the diagnostic approach to hemoptysis in a patient with a significant smoking history?

Answer

Chest CT and often bronchoscopy are indicated to evaluate for malignancy, since lung cancer is an important cause in this population. Massive hemoptysis, generally defined as greater than 100 to 600 mL in 24 hours depending on the source, is a medical emergency requiring airway protection and localization for possible bronchial artery embolization. Positioning the bleeding lung in the dependent position helps protect the airway while awaiting definitive management.

#438 · Pulmonary Disease

What is the initial diagnostic approach to unexplained chronic dyspnea?

Answer

A stepwise workup includes chest x-ray, spirometry with lung volumes and diffusion capacity, and often an echocardiogram to evaluate for cardiac causes. If these are unrevealing, cardiopulmonary exercise testing can help distinguish cardiac, pulmonary, deconditioning, or psychogenic causes. History and physical exam findings should guide which tests are prioritized first.

#439 · Pulmonary Disease

What does a low diffusing capacity for carbon monoxide, or DLCO, suggest when obstruction is present on spirometry?

Answer

A reduced DLCO with obstruction favors emphysema over chronic bronchitis, since emphysema destroys alveolar-capillary surface area needed for gas transfer. Chronic bronchitis typically has a preserved DLCO despite airflow obstruction. DLCO also helps assess severity and prognosis in interstitial lung disease.

#440 · Pulmonary Disease

What is the role of a D-dimer test in the evaluation of suspected pulmonary embolism?

Answer

D-dimer has high sensitivity but low specificity, so a negative result in a patient with low or moderate pretest probability effectively rules out PE. A positive D-dimer does not confirm PE since it rises with infection, surgery, pregnancy, and many other conditions, and requires further imaging. D-dimer is not useful in high pretest probability patients, who should go straight to CT pulmonary angiography.

#441 · Rheumatology and Orthopedics

A 55-year-old man wakes with a hot, swollen, exquisitely tender first MTP joint. What is the diagnosis and confirmatory test?

Answer

This presentation is classic for acute gout, often called podagra when it involves the first MTP joint. Diagnosis is confirmed by arthrocentesis showing needle-shaped, negatively birefringent monosodium urate crystals under polarized light. Serum uric acid can be normal during an acute flare, so it should not be used to rule out gout.

#442 · Rheumatology and Orthopedics

How do pseudogout crystals differ from gout crystals on joint fluid analysis?

Answer

Pseudogout is caused by calcium pyrophosphate dihydrate crystals, which are rhomboid shaped and weakly positively birefringent under polarized light. Gout crystals are needle shaped and strongly negatively birefringent. Pseudogout most often affects the knee and is associated with chondrocalcinosis on x-ray.

#443 · Rheumatology and Orthopedics

What is first-line treatment for an acute gout flare in a patient with normal renal function?

Answer

NSAIDs such as indomethacin or naproxen are first line for acute gout in patients without contraindications. Colchicine is an alternative, most effective if started within 24 to 36 hours of symptom onset. Oral or intra-articular glucocorticoids are used when NSAIDs and colchicine are contraindicated, such as in renal impairment or peptic ulcer disease.

#444 · Rheumatology and Orthopedics

When should urate-lowering therapy be started in gout, and what is the target uric acid level?

Answer

Urate-lowering therapy such as allopurinol is started after the acute flare has resolved, not during it, since abrupt changes in uric acid can worsen or prolong the attack. It is indicated for patients with recurrent flares, tophi, urate nephropathy, or chronic kidney disease. The target serum uric acid is below 6 mg per dL, and colchicine or NSAID prophylaxis should be given for several months when starting therapy to prevent mobilization flares.

#445 · Rheumatology and Orthopedics

A young man with chronic low back pain and morning stiffness that improves with exercise. What condition should you suspect and what imaging finding confirms it?

Answer

This history is typical for ankylosing spondylitis, a seronegative spondyloarthropathy that causes inflammatory back pain improving with activity and worsening with rest. Sacroiliitis on plain films or MRI is the key diagnostic finding, and advanced disease shows a bamboo spine from vertebral fusion. Most patients are HLA-B27 positive, though this test is not required for diagnosis.

#446 · Rheumatology and Orthopedics

What extra-articular findings are associated with ankylosing spondylitis?

Answer

Ankylosing spondylitis is associated with anterior uveitis, which presents with a painful red eye and photophobia. It can also cause aortic regurgitation, restrictive lung disease from costovertebral involvement, and an increased risk of inflammatory bowel disease. Enthesitis, or inflammation at tendon insertion sites like the Achilles tendon, is also common.

#447 · Rheumatology and Orthopedics

What are the classic skin and nail findings of psoriatic arthritis, and what joint pattern is characteristic?

Answer

Psoriatic arthritis causes nail pitting, onycholysis, and dactylitis, which is diffuse swelling of an entire digit sometimes called sausage digit. It classically involves the distal interphalangeal joints, distinguishing it from rheumatoid arthritis, and can cause a destructive arthritis mutilans in severe cases. Skin psoriasis usually precedes joint disease but can develop afterward.

#448 · Rheumatology and Orthopedics

What is reactive arthritis and what is the classic triad?

Answer

Reactive arthritis is a spondyloarthropathy triggered by a recent gastrointestinal infection, such as Salmonella or Shigella, or a genitourinary infection with Chlamydia trachomatis. The classic triad is arthritis, conjunctivitis, and urethritis, sometimes remembered as cannot see, cannot pee, cannot climb a tree. It is HLA-B27 associated and typically self-limited, treated with NSAIDs.

#449 · Rheumatology and Orthopedics

A patient has symmetric small joint polyarthritis of the hands with morning stiffness lasting over an hour. What labs support the diagnosis of rheumatoid arthritis?

Answer

Rheumatoid factor and anti-cyclic citrullinated peptide antibody are the key serologic markers, with anti-CCP being more specific for rheumatoid arthritis. Inflammatory markers like ESR and CRP are often elevated. The classic joints involved are the MCP and PIP joints while the DIP joints are typically spared, unlike osteoarthritis.

#450 · Rheumatology and Orthopedics

What hand deformities are seen in longstanding rheumatoid arthritis?

Answer

Ulnar deviation at the MCP joints, swan neck deformity with PIP hyperextension and DIP flexion, and boutonniere deformity with PIP flexion and DIP hyperextension are classic. These result from chronic synovitis damaging tendons and joint capsules. Cervical spine involvement, particularly C1-C2 subluxation, is important to assess before intubation or neck manipulation.

#451 · Rheumatology and Orthopedics

What is Felty syndrome?

Answer

Felty syndrome is a triad of rheumatoid arthritis, splenomegaly, and neutropenia. It typically occurs in patients with longstanding, seropositive rheumatoid arthritis. Patients are at increased risk of recurrent bacterial infections due to the neutropenia.

#452 · Rheumatology and Orthopedics

What is the first-line disease-modifying therapy for rheumatoid arthritis and what monitoring is required?

Answer

Methotrexate is the first-line DMARD for rheumatoid arthritis and is often combined with folic acid to reduce toxicity. Baseline and periodic monitoring includes CBC, liver function tests, and renal function, since methotrexate can cause hepatotoxicity, bone marrow suppression, and pneumonitis. Biologic agents like TNF inhibitors are added when methotrexate alone is insufficient, but require screening for latent tuberculosis and hepatitis B before starting.

#453 · Rheumatology and Orthopedics

A young woman presents with malar rash, joint pain, and photosensitivity. What is the most sensitive autoantibody screening test and what is most specific?

Answer

This presentation suggests systemic lupus erythematosus. Antinuclear antibody, or ANA, is highly sensitive and used as a screening test, being positive in over 95 percent of SLE patients. Anti-double-stranded DNA and anti-Smith antibodies are highly specific for SLE, with anti-dsDNA levels also correlating with disease activity and lupus nephritis flares.

#454 · Rheumatology and Orthopedics

What antibody is associated with drug-induced lupus, and which drugs commonly cause it?

Answer

Drug-induced lupus is associated with anti-histone antibodies, which are positive in the large majority of cases. Common culprits include hydralazine, procainamide, isoniazid, and minocycline. Unlike idiopathic SLE, drug-induced lupus typically spares the kidneys and central nervous system and resolves after stopping the offending drug.

#455 · Rheumatology and Orthopedics

How does lupus nephritis present and how is it diagnosed?

Answer

Lupus nephritis presents with proteinuria, hematuria, red cell casts, and sometimes declining renal function, and is a major cause of morbidity in SLE. Diagnosis and classification require a renal biopsy, which determines the histologic class and guides treatment intensity. Class III and IV, which are proliferative forms, typically require induction therapy with high-dose steroids plus mycophenolate mofetil or cyclophosphamide.

#456 · Rheumatology and Orthopedics

What is antiphospholipid syndrome and which antibodies define it?

Answer

Antiphospholipid syndrome causes recurrent arterial or venous thrombosis and pregnancy morbidity such as recurrent miscarriage. It is defined by persistent positivity for lupus anticoagulant, anticardiolipin antibodies, or anti-beta-2-glycoprotein I antibodies on two occasions at least 12 weeks apart. It is often associated with SLE but can occur as a primary condition, and treatment for thrombosis is typically lifelong anticoagulation with warfarin.

#457 · Rheumatology and Orthopedics

A patient has skin thickening of the fingers and face along with Raynaud phenomenon and dysphagia. What condition is this and what antibody is associated with limited disease?

Answer

This presentation is consistent with systemic sclerosis, or scleroderma. Limited cutaneous disease, formerly called CREST syndrome, is associated with anticentromere antibodies and involves calcinosis, Raynaud phenomenon, esophageal dysmotility, sclerodactyly, and telangiectasia. Skin involvement in limited disease stays distal to the elbows and knees, sparing the trunk.

#458 · Rheumatology and Orthopedics

What antibody is associated with diffuse systemic sclerosis and what complication should be screened for?

Answer

Diffuse systemic sclerosis is associated with anti-Scl-70, also called anti-topoisomerase I, antibodies, and involves skin thickening proximal to the elbows and knees plus truncal involvement. Patients are at risk for scleroderma renal crisis, presenting with malignant hypertension and acute kidney injury, which is treated urgently with ACE inhibitors. Interstitial lung disease and pulmonary arterial hypertension are also major causes of mortality and should be screened for with pulmonary function tests and echocardiography.

#459 · Rheumatology and Orthopedics

How is primary Raynaud phenomenon distinguished from secondary Raynaud phenomenon?

Answer

Primary Raynaud phenomenon typically occurs in young women, is symmetric, and is not associated with tissue damage or an underlying disease. Secondary Raynaud phenomenon is more often asymmetric, occurs in older patients, can cause digital ulcers or tissue necrosis, and is linked to underlying connective tissue disease such as systemic sclerosis. Nailfold capillaroscopy showing capillary dropout or dilation suggests secondary Raynaud phenomenon and an underlying rheumatologic disease.

#460 · Rheumatology and Orthopedics

A 70-year-old woman presents with new headache, jaw claudication, and visual disturbance. What is the diagnosis, next step, and how is it confirmed?

Answer

This is classic for giant cell arteritis, a large vessel vasculitis affecting the temporal artery among others. Because vision loss can become permanent within hours, high-dose glucocorticoids should be started immediately without waiting for biopsy confirmation. Temporal artery biopsy remains the diagnostic gold standard but can be done after steroids are started, since pathologic changes persist for one to two weeks.

#461 · Rheumatology and Orthopedics

What condition commonly coexists with giant cell arteritis and how does it present?

Answer

Polymyalgia rheumatica commonly coexists with or precedes giant cell arteritis. It presents with symmetric aching and stiffness in the shoulder and hip girdles, worse in the morning, along with elevated ESR and CRP, typically in patients over age 50. It responds dramatically to low-dose glucocorticoids, and a poor response should prompt reconsideration of the diagnosis.

#462 · Rheumatology and Orthopedics

What are the three major ANCA-associated vasculitides and how are they distinguished?

Answer

Granulomatosis with polyangiitis causes upper and lower respiratory tract disease plus glomerulonephritis and is associated with c-ANCA, or anti-proteinase 3 antibodies. Microscopic polyangiitis causes glomerulonephritis and pulmonary hemorrhage without granulomas and is associated with p-ANCA, or anti-myeloperoxidase antibodies. Eosinophilic granulomatosis with polyangiitis, formerly Churg-Strauss, involves asthma, eosinophilia, and vasculitis, and is p-ANCA/MPO positive in only about half of cases, less consistently than GPA or MPA.

#463 · Rheumatology and Orthopedics

What is the classic presentation of granulomatosis with polyangiitis?

Answer

Granulomatosis with polyangiitis classically presents with sinusitis, saddle nose deformity from nasal septal destruction, hemoptysis from pulmonary nodules or cavities, and rapidly progressive glomerulonephritis. Chest imaging often shows nodules or cavitary lesions. Treatment involves induction with high-dose glucocorticoids plus rituximab or cyclophosphamide.

#464 · Rheumatology and Orthopedics

A patient with palpable purpura on the lower extremities, arthralgias, and abdominal pain after a recent upper respiratory infection. What is the diagnosis and what antibody deposits are found on biopsy?

Answer

This presentation is typical of IgA vasculitis, formerly Henoch-Schonlein purpura, most common in children but also seen in adults. Skin biopsy shows leukocytoclastic vasculitis with IgA immune complex deposition. It can also cause hematuria from IgA nephropathy-like renal involvement and is usually self-limited, though severe renal or GI involvement may require glucocorticoids.

#465 · Rheumatology and Orthopedics

What is polyarteritis nodosa and what is its classic association?

Answer

Polyarteritis nodosa is a medium-vessel vasculitis that spares the lungs and does not involve ANCA antibodies. It causes mononeuritis multiplex, livedo reticularis, renal artery involvement with hypertension, and mesenteric ischemia, often with aneurysms visible on angiography. It has a strong association with chronic hepatitis B infection.

#466 · Rheumatology and Orthopedics

A patient presents with proximal muscle weakness and a heliotrope rash with Gottron papules. What is the diagnosis and what enzyme test supports it?

Answer

This presentation is classic for dermatomyositis, an inflammatory myopathy with characteristic skin findings including a violaceous heliotrope rash on the eyelids and Gottron papules over the knuckles. Creatine kinase and aldolase are typically elevated, reflecting muscle inflammation. Dermatomyositis carries an increased risk of underlying malignancy, so age-appropriate cancer screening is warranted, especially in adults over 40.

#467 · Rheumatology and Orthopedics

How does polymyositis differ from dermatomyositis, and what is the gold standard diagnostic test for both?

Answer

Polymyositis causes symmetric proximal muscle weakness with elevated CK, similar to dermatomyositis, but lacks the characteristic skin findings. Muscle biopsy is the gold standard diagnostic test and helps distinguish the two histologically, along with electromyography showing myopathic changes. Both conditions are treated with glucocorticoids as first-line therapy, often with a steroid-sparing agent added.

#468 · Rheumatology and Orthopedics

What autoantibody is associated with antisynthetase syndrome and what clinical features define it?

Answer

Antisynthetase syndrome is associated with anti-Jo-1 antibodies, which target histidyl-tRNA synthetase. It presents with a combination of myositis, interstitial lung disease, inflammatory arthritis, fever, Raynaud phenomenon, and mechanic hands, which are cracked, thickened skin on the finger pads. Interstitial lung disease is often the most significant driver of morbidity and mortality in this syndrome.

#469 · Rheumatology and Orthopedics

A patient reports dry eyes and dry mouth. What condition is suspected and what antibodies and test confirm it?

Answer

This presentation suggests Sjogren syndrome, an autoimmune exocrinopathy. Anti-Ro, also called SSA, and anti-La, also called SSB, antibodies are commonly positive, and a lip or salivary gland biopsy showing lymphocytic infiltration can confirm the diagnosis when serology is negative. The Schirmer test can objectively measure decreased tear production.

#470 · Rheumatology and Orthopedics

What is the long-term malignancy risk associated with Sjogren syndrome?

Answer

Sjogren syndrome carries an increased risk of B-cell non-Hodgkin lymphoma, particularly mucosa-associated lymphoid tissue, or MALT, lymphoma of the salivary glands. Persistent parotid gland enlargement, palpable purpura, or low complement levels should raise suspicion for lymphoma development. Patients with these risk features warrant closer monitoring over time.

#471 · Rheumatology and Orthopedics

A woman with anti-Ro antibodies is pregnant. What fetal complication is she at risk for?

Answer

Anti-Ro and anti-La antibodies can cross the placenta and cause neonatal lupus, most seriously manifesting as congenital heart block in the fetus. This is typically detected by fetal echocardiography monitoring during pregnancy. Other neonatal lupus features include a transient skin rash and cytopenias that resolve as maternal antibodies clear from the infant's circulation.

#472 · Rheumatology and Orthopedics

A patient has widespread pain, fatigue, and multiple tender points, with normal labs and imaging. What is the diagnosis and how is it managed?

Answer

This presentation is consistent with fibromyalgia, a chronic pain syndrome characterized by widespread musculoskeletal pain, fatigue, sleep disturbance, and cognitive difficulty often called fibro fog. Diagnosis is clinical, based on widespread pain criteria, and laboratory and imaging studies are typically normal, used mainly to exclude other conditions. Management includes patient education, exercise, cognitive behavioral therapy, and medications like duloxetine, pregabalin, or milnacipran.

#473 · Rheumatology and Orthopedics

How is fibromyalgia distinguished from an inflammatory rheumatologic disease?

Answer

Fibromyalgia does not cause objective synovitis, joint swelling, or elevated inflammatory markers like ESR and CRP, unlike inflammatory arthritides. Morning stiffness in fibromyalgia is typically brief compared to the prolonged stiffness of rheumatoid arthritis. Overlap can occur since fibromyalgia is common in patients with underlying autoimmune disease, so a thorough exam is needed to avoid missing concurrent pathology.

#474 · Rheumatology and Orthopedics

A patient presents with an acutely hot, swollen single joint and fever. What is the most urgent diagnosis to exclude and how?

Answer

Septic arthritis must be excluded urgently because untreated infection can rapidly destroy cartilage and joint function. Arthrocentesis with synovial fluid analysis, Gram stain, and culture is required, with synovial white blood cell counts typically above 50,000 per microliter and predominantly neutrophilic suggesting infection. Empiric antibiotics should be started promptly after cultures are obtained, without waiting for final results, given the risk of rapid joint destruction.

#475 · Rheumatology and Orthopedics

What is the most common organism causing septic arthritis in sexually active young adults, and how does it typically present?

Answer

Disseminated gonococcal infection from Neisseria gonorrhoeae is a common cause of septic arthritis in sexually active young adults. It can present as a triad of migratory polyarthralgia, tenosynovitis, and a pustular or vesicular skin rash, sometimes without a frankly purulent monoarthritis. Blood, joint fluid, and genital cultures may be needed since joint fluid cultures are often negative in disseminated gonococcal disease.

#476 · Rheumatology and Orthopedics

What is Lyme arthritis and how does it typically present?

Answer

Lyme arthritis is a late manifestation of Borrelia burgdorferi infection, occurring weeks to months after the initial tick bite, often after the erythema migrans rash has resolved. It typically causes intermittent or persistent monoarthritis or oligoarthritis, most commonly affecting the knee. Diagnosis is made with serology using an enzyme immunoassay followed by confirmatory Western blot, and treatment is with oral doxycycline.

#477 · Rheumatology and Orthopedics

An elderly patient has an isolated elevated alkaline phosphatase with normal calcium, phosphate, and liver enzymes. What condition should be suspected and what imaging finding is characteristic?

Answer

This pattern suggests Paget disease of bone, a disorder of increased and disorganized bone remodeling. Alkaline phosphatase is elevated from increased osteoblastic activity while calcium and phosphate remain normal. Plain films classically show thickened, sclerotic bone with cortical thickening, and the skull, pelvis, spine, and long bones are commonly affected.

#478 · Rheumatology and Orthopedics

What are the complications of Paget disease of bone?

Answer

Complications of Paget disease include bone pain, deformity, pathologic fractures, and hearing loss from skull involvement compressing cranial nerves. High-output heart failure can occur from increased vascularity of pagetic bone in extensive disease. Rarely, osteosarcoma can develop within pagetic bone, presenting as new severe pain or a mass at a previously affected site.

#479 · Rheumatology and Orthopedics

What is first-line treatment for symptomatic Paget disease of bone?

Answer

Bisphosphonates, such as zoledronic acid, are first-line treatment for symptomatic Paget disease, as they suppress the excessive osteoclastic bone turnover. Treatment is indicated for symptomatic patients or those with disease at high-risk sites like the skull or weight-bearing bones. Alkaline phosphatase levels are used to monitor response to therapy.

#480 · Rheumatology and Orthopedics

What are the major risk factors for avascular necrosis of bone, and what is the imaging test of choice?

Answer

Major risk factors for avascular necrosis include chronic glucocorticoid use, alcohol abuse, sickle cell disease, trauma, and SLE. The femoral head is the most commonly affected site. MRI is the imaging test of choice since it can detect avascular necrosis before changes appear on plain radiographs, which only become abnormal in later stages.

#481 · Rheumatology and Orthopedics

How does osteoarthritis typically present and what joints are classically involved?

Answer

Osteoarthritis presents with joint pain that worsens with activity and improves with rest, along with brief morning stiffness usually lasting less than 30 minutes. It classically involves the DIP joints causing Heberden nodes, the PIP joints causing Bouchard nodes, the first carpometacarpal joint, hips, and knees. Plain films show joint space narrowing, osteophytes, subchondral sclerosis, and subchondral cysts.

#482 · Rheumatology and Orthopedics

How is osteoarthritis managed in a patient with mild to moderate knee pain?

Answer

Initial management includes weight loss, low-impact exercise such as swimming or cycling, and physical therapy to strengthen surrounding muscles. Acetaminophen and topical or oral NSAIDs are used for pain control, with NSAIDs used cautiously in older adults due to gastrointestinal and renal risks. Intra-articular glucocorticoid injections can be used for flares, and total joint replacement is reserved for severe, refractory disease.

#483 · Rheumatology and Orthopedics

What is the significance of anti-CCP antibodies compared to rheumatoid factor?

Answer

Anti-cyclic citrullinated peptide antibodies are more specific for rheumatoid arthritis than rheumatoid factor, which can be positive in other conditions like chronic infections, Sjogren syndrome, and even in healthy older adults. Anti-CCP can also be present years before clinical symptoms of rheumatoid arthritis develop. Its presence is associated with more erosive, aggressive joint disease.

#484 · Rheumatology and Orthopedics

What is the significance of anti-U1-RNP antibodies?

Answer

Anti-U1-RNP antibodies are associated with mixed connective tissue disease, a condition that combines features of SLE, systemic sclerosis, and polymyositis. High-titer anti-RNP in the absence of other SLE-specific antibodies like anti-Smith is a defining serologic feature. Patients often present with Raynaud phenomenon, swollen hands, and myositis along with overlapping symptoms of the component diseases.

#485 · Rheumatology and Orthopedics

What autoantibody pattern is most specific for limited cutaneous systemic sclerosis versus diffuse systemic sclerosis, and what about anti-RNA polymerase III?

Answer

Anticentromere antibodies are most specific for limited cutaneous systemic sclerosis, while anti-Scl-70, or anti-topoisomerase I, is more associated with diffuse disease and pulmonary fibrosis. Anti-RNA polymerase III antibodies are associated with diffuse skin disease and a markedly increased risk of scleroderma renal crisis. Testing this antibody panel helps predict disease subtype and complication risk in newly diagnosed systemic sclerosis.

#486 · Geriatric Syndromes

What are the main risk factors for pressure injuries in hospitalized elderly patients?

Answer

Immobility, incontinence, poor nutrition, and impaired sensation are the biggest drivers. Shear and friction from repositioning also contribute. The Braden scale is commonly used to screen risk, and prevention centers on frequent repositioning, pressure-redistributing surfaces, and keeping skin clean and dry.

#487 · Geriatric Syndromes

How do you stage a pressure injury with intact skin but non-blanchable redness?

Answer

This is stage 1. The skin is intact but shows persistent redness that does not blanch with pressure. There may be changes in temperature, firmness, or sensation compared to surrounding tissue. It is a warning sign to offload pressure immediately before skin breakdown progresses.

#488 · Geriatric Syndromes

What four elements must a patient demonstrate to have decisional capacity for a medical decision?

Answer

The patient must be able to understand the relevant information, appreciate how it applies to their own situation, reason through the options and consequences, and communicate a clear, consistent choice. Capacity is decision specific and can fluctuate, so a patient may have capacity for a simple decision but not a complex one.

#489 · Geriatric Syndromes

Who determines whether a patient lacks decisional capacity, and what happens next?

Answer

A physician, not a judge, makes this determination as part of clinical care. Capacity is distinct from legal competence, which only a court can decide. If a patient lacks capacity, the team turns to an advance directive if one exists, or to a surrogate decision maker following the state's hierarchy, usually starting with a spouse.

#490 · Geriatric Syndromes

An elderly patient develops acute confusion and fluctuating attention over two days during a hospital stay. Dementia or delirium?

Answer

This is delirium. It has an acute onset, fluctuates over hours to days, and features inattention as the hallmark finding, often with disorganized thinking or altered level of consciousness. Dementia develops gradually over months to years and attention is usually preserved until late stages.

#491 · Geriatric Syndromes

What is the first step in evaluating new delirium in an elderly inpatient?

Answer

Look for a reversible medical cause rather than assuming it is baseline cognitive decline. Common culprits include infection, medications, especially anticholinergics, sedatives, and opioids, metabolic derangements, urinary retention, fecal impaction, and pain. Treating the underlying cause is more effective than antipsychotics, which are reserved for severe agitation that is unsafe for the patient or staff.

#492 · Geriatric Syndromes

How do you distinguish urge incontinence from stress incontinence on history?

Answer

Urge incontinence presents with a sudden strong need to void followed by leakage, often from detrusor overactivity, and is treated with bladder training or antimuscarinics like oxybutynin. Stress incontinence causes leakage with coughing, sneezing, or lifting due to weak pelvic floor or sphincter support, and is treated with pelvic floor exercises or surgery.

#493 · Geriatric Syndromes

What does the mnemonic DIAPPERS refer to in urinary incontinence?

Answer

It lists reversible causes of transient incontinence to check before assuming a chronic type: delirium, infection, atrophic vaginitis or urethritis, pharmaceuticals, psychological factors, excess urine output, restricted mobility, and stool impaction. Correcting these often resolves incontinence without further workup.

#494 · Geriatric Syndromes

What are the key components of a fall risk evaluation in an older adult?

Answer

Ask about prior falls, gait and balance using tests like Timed Up and Go, vision, orthostatic blood pressure, foot problems, and home hazards. Review the medication list for sedatives, antihypertensives, and other drugs that increase fall risk. A multifactorial assessment guides targeted interventions rather than a single fix.

#495 · Geriatric Syndromes

What is the difference between hospice and palliative care?

Answer

Palliative care focuses on symptom relief and quality of life and can be provided alongside curative treatment at any stage of a serious illness. Hospice is a specific type of palliative care for patients with a prognosis of six months or less who have chosen to forgo curative treatment. Both emphasize comfort, but hospice requires that disease-directed therapy be stopped.

#496 · Geriatric Syndromes

What is the Beers criteria and why does it matter in older patients?

Answer

The Beers criteria is a list of medications that are potentially inappropriate for older adults because the risks often outweigh the benefits in this population. Examples include benzodiazepines, first-generation antihistamines, and certain anticholinergics, which raise the risk of falls, delirium, and cognitive impairment. It is a tool to guide deprescribing and safer prescribing.

#497 · Geriatric Syndromes

Why is polypharmacy dangerous in the geriatric population beyond just drug interactions?

Answer

Age-related changes in renal and hepatic clearance alter drug metabolism, so standard doses can accumulate and cause toxicity. Polypharmacy also increases the risk of prescribing cascades, where a side effect of one drug is treated with another drug instead of stopping the first. Regular medication reconciliation and deprescribing reduce these risks.

#498 · Geriatric Syndromes

An elderly woman presents with hip pain and inability to bear weight after a fall, with the leg shortened and externally rotated. What is the concern and what is the priority?

Answer

This presentation is classic for hip fracture. Early surgical repair, ideally within 24 to 48 hours, improves outcomes and reduces complications like pneumonia, venous thromboembolism, and further deconditioning. Preoperative optimization should not unnecessarily delay surgery, and postoperative care should include early mobilization and osteoporosis treatment.

#499 · Geriatric Syndromes

What clinical clues raise suspicion for elder mistreatment?

Answer

Look for injuries inconsistent with the stated history, delays in seeking care, poor hygiene or malnutrition without a clear medical cause, and a caregiver who answers for the patient or seems unwilling to leave the room during the exam. Financial exploitation can present as sudden changes in the patient's finances or missing belongings. Suspected cases should be reported to adult protective services.

#500 · Geriatric Syndromes

What domains does a comprehensive geriatric assessment cover?

Answer

It evaluates medical issues, functional status including activities of daily living and instrumental activities of daily living, cognition, mood, mobility and fall risk, nutrition, polypharmacy, and social and financial support. This multidimensional approach identifies problems that a standard visit might miss and guides individualized care planning, including goals of care and end-of-life preferences when appropriate.